IEEE pesgre 2025

CONFERENCE HANDBOOK

POWER ELECTRONICS AND RENWABLE ENERGY FOR SUSTAINABLE DEVELOPMENT

4TH IEEE INTERNATIONAL CONFERENCE ON POWER ELECTRONICS SMART GRID AND RENEWABLE ENERGY

December 18-21, 2025,
IIT Dharwad, Dharwad, Karnataka, India

IIT DHARWAD | IEEE Kerala Section | IEEE IA/IE/PELS Jt. Chapter Kerala | IEEE POWER ELECTRONICS SOCIETY | IEEE PES Power & Energy Society | IEEE Advancing Technology for Humanity

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About PESGRE

IEEE PESGRE 2025 is the 4th IEEE International Conference on Power Electronics, Smart Grid and Renewable Energy, bringing together researchers, industry professionals and educators to discuss advances in power conversion, electric transportation and sustainable energy systems. The theme of this edition of conference is "Power Electronics and Renewable Energy for Sustainable Development", being held during 18th - 21st December 2025 at Indian Institute of Technology Dharwad, Dharwad, Karnataka, India.

The conference covers core areas in power electronics, smart grids, renewable energy, electric vehicles, energy storage, and modern power systems, including policy and standards. It also includes special sessions on emerging topics and dedicated Students and Young Professionals (SYP) activities to promote early-career research and networking in the global power and energy community.

IEEE PESGRE is the biennial flagship Conference of IEEE Kerala Section, IEEE IA/IE/PELS Joint Chapter Kerala and IEEE Industry Applications Society. The technical co-sponsors of the conference include IEEE Power Electronics Society, IEEE Industrial Electronics Society, and IEEE Power & Energy Society.

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Welcome Message

On behalf of the Organizing Committee and the Technical Program Committee, we are delighted to welcome you to the **4th IEEE International Conference on Power Electronics, Smart Grid, and Renewable Energy (IEEE PESGRE 2025)**, hosted at the vibrant campus of Indian Institute of Technology Dharwad, Karnataka, India, from December 18 - 21, 2025.

IEEE PESGRE is the biennial flagship conference of **IEEE Kerala Section, IEEE IA/IE/PELS Joint Chapter Kerala**, and the **IEEE Industry Applications Society**, financial co-sponsorship from these bodies. Technical co-sponsors include the **IEEE Power Electronics Society**, **IEEE Industrial Electronics Society**, and **IEEE Power & Energy Society**.

The theme for this edition is **"Power Electronics and Renewable Energy for Sustainable Development** . The conference will spotlight cutting-edge technologies, strategies, and challenges in power electronics, renewable energy, electric vehicles, grid integration, and smart grid solutions. IEEE PESGRE 2025 serves as a global platform for exchanging ideas, presenting technical innovations, and discussing current trends and future perspectives in areas such as power converter topology, control techniques, grid integration, etc.

This edition conference reflects a truly international scope, participants and presentations from **15 countries**, which includes, Canada, China, Estonia, Germany, India, Israel, Italy, Norway, the Netherlands, New Zealand, Poland, UAE, USA, etc. This shows the increasing popularity of the conference by underscoring the global commitment to advancing energy systems.

Out of **400 submissions, 210 papers** were accepted after a rigorous review process three reviewers per paper involving **20 track chairs** and **350 expert reviewers** from different countries. Finally, a total of **179 camera-ready papers** will be presented across 30 technical oral sessions. The program features:

- 4 Keynote Addresses
- 1 IES SYPA Keynote
- 4 Tutorials
- Two Women in Engineering (WiE) Sessions
- PELS Young Professional Mentorship Program
- Panel Discussion

- Five Industry Sessions
- Three oral presentation sessions covering 180 papers and
- Cultural events

We are proud to note the strong industry presence at PESGRE 2025, highlighting the growing real-world relevance of power electronic technologies and the importance of academia-industry collaboration. Together, we are shaping the future of sustainable energy systems. This belief is strengthened by our sponsors who are from different industrial sectors, related to manufacturers of power supplies and testing equipment, defence systems, renewable energy, battery technology, simulation tools, etc. Our sincere appreciation goes to the sponsors for their invaluable support:

- **Title Sponsor**: JSCGI Private Limited
- **Platinum Sponsors**: Pragna Microdesigns and Defence Research and Development Organisation (DRDO)
- **Gold Sponsors**: Convergent Technologies + Tektronix
- **Silver Sponsors**: Karnataka Renewable Energy Development Limited (KREDL), OPAL-RT Technologies, and AARJAY International
- **Bronze Sponsors**: Charge House, HIOKI, Typhoon HIL, MathWorks, CDAC, and COMSOL
- **Other Sponsors**: Amara Raja Group, IFB, Reliamotive Labs, FLUKE, and Keysight

We extend our heartfelt gratitude to the Technical Program Committee chairs, track chairs, and reviewers for ensuring the high technical quality of this event, and to the authors, whose innovative contributions form the core of this conference. Finally, as we gather to share knowledge and explore innovations, let us remember our common mission: building resilient, efficient, and sustainable energy systems. Power electronic technologies are at the heart of this transformation, and your participation is vital in shaping the future. We encourage you to take full advantage of networking opportunities—during technical sessions, coffee breaks, and social events. We look forward to having you all here and have an inspiring, productive, and memorable week in Dharwad.

Dr. Satish Naik

Dr. Dmitri Vinnikov

Dr. Sheldon Williamson

Co-General Chairs, IEEE PESGRE 2025

Schedule

TIME		DAY				
Start Time	End Time	18th December 2025		19th December 2025	20th December 2025		21st December 2025
08:00	09:00	Registration Venue: CLT		Registration Venue: CLT	Registration Venue: CLT		-
09:00	09:30			Oral presentations [parallel technical sessions] Venue: CLT Classrooms	Oral presentations [parallel technical sessions] Venue: CLT Classrooms		Oral presentations [parallel technical sessions] Venue: CLT Classrooms
09:30	10:00	Tutorial -1 Shan, Aravind, Gourahari Venue: 107	Tutorial -2 Francesca, Calina Venue: 106				
10:00	10:30						
10:30	11:00			Coffee break Venue: CLT EXPO area	Coffee break Venue: CLT EXPO area		Coffee Break Venue: CLT EXPO area
11:00	11:30	Coffee Break Venue: CLT EXPO area		Inauguration + Panel discussion Venue: CLT 500 seater	Keynote - 4 Prof. Drazen+ Industry sessions (Convergent + AARJAY) Venue: CLT 500 seater		Industry sessions (Opal-RT + MathWorks+ Reliamotive) Venue: CLT 500 seater
11:30	12:30	Tutorial -1 Shan, Aravind, Gourahari Venue: 107	Tutorial -2 Francesca, Calina Venue: 106				
12:30	13:00						
13:00	13:30	Lunch break Venue: CLT First Floor Back Foyer		Lunch break Venue: CLT First Floor Back Foyer	Lunch break Venue: CLT First Floor Back Foyer		Closing remarks Venue: 500 seater
13:30	14:30						Lunch break Venue:CLT First Floor Back Foyer
14:30	16:00	Tutorial -3 Riccardo, Mattia, Husev Venue: 107	Tutorial – 4 Sandeep Anand, Abhinav Venue: 106	Keynote - 1 Prof. Mariusz + Industry sessions (JSC group) Venue: CLT 500 seater	Keynote - 5 Prof. Alon + SYP Forum (Prof. Tripura Pidikiti) Venue: CLT 500 seater		
16:00	16:30	Coffee Break Venue: CLT EXPO area		Coffee Break Venue: CLT EXPO area	Coffee Break Venue: CLT EXPO area		
16:30	17:00	Tutorial -3 Riccardo, Mattia, Husev Venue: 107	Tutorial – 4 Sandeep Anand, Abhinav Venue: 106	Keynote - 2 Prof. B G Fernandes + Keynote – 3 Prof. Rajendra Venue: CLT 500 seater	WIE (Venue: CLT-500 seater)	PELs mentorship (Venue: CLT-LG 007)	
17:00	17:30						
17:30	18:00						
18:00	18:30						
19:00	19:30	Cultural event Venue: CLT 500 seater			-		
19:30	20:00						
20:00	22:30	Welcome reception Venue: KRDC			Gala Dinner Venue: Denissons Hotel, Hubli		
22:30	23:00						
23:00	23:30						

Please check website for updated schedule: https://pesgre2025.org/

Keynote Talks

Keynote 1: Interlinking hybrid distribution transformer for hybrid power grids

Prof. Mariusz Malinowski, Professor at the Institute of Control and Industrial Electronics, WUT, Poland.

About the Speaker: Professor Mariusz Malinowski received his Ph.D. and D.Sc. degrees in electrical engineering from the Warsaw University of Technology (WUT), Poland. He has authored over 200 technical papers and seven books. His research focuses on the control and modulation of grid-side converters, multilevel converters, smart grids, and renewable energy systems. He has received numerous awards, including the IEEE IES David Irwin Early Career Award, IEEE IES Bimal Bose Energy Systems Award, and the Polish Prime Minister Award. He is a Fellow of IEEE and a member of the Polish Academy of Sciences.

Keynote 2: Brushless Field-excitation Systems for Motors in Electric Vehicles

Prof. Baylon G. Fernandes, Professor at the Department of Electrical Engineering, IIT Bombay, India.

About the Speaker: Professor Baylon G. Fernandes received the B.Tech. degree from the National Institute of Technology Karnataka, Surathkal, India, in 1984, the M.Tech. degree from the Indian Institute of Technology Kharagpur, India, in 1989, and the Ph.D. degree from the Indian Institute of Technology Bombay (IIT Bombay), India, in 1993, all in electrical engineering. Since 1998, he has been the Department of Electrical Engineering, IIT Bombay, where he is currently a Professor. From 2015 to 2020, he served as the Chairman of the Department. He is a Principal Investigator of the National Centre for Photovoltaic Research and Education and the National Centre of Excellence in Technology for Internal Security at IIT Bombay. His research interests include power electronic converters, motors for electric vehicles, and power electronic interfaces for non-conventional energy sources. Prof. Fernandes is a Senior Member of IEEE and serves as an Associate Editor for IEEE Transactions on Power Electronics.

Keynote 3: Transformative Role of Power Electronics in Accelerating Economic Growth of Bharat

Prof. Rajendra Singh, Houser Banks Distinguished Professor at Clemson University, USA

About the Speaker: Professor Rajendra Singh is the Houser Banks Distinguished Professor in the Holcombe Department of Electrical and Computer Engineering and Automotive Engineering at Clemson University (CU). Inspired by Thomas Edison's vision of solar energy, he pursued his Ph.D. dissertation on Silicon Solar Cells during the 1973 energy crisis. Over the past 50 years, he has significantly contributed to the growth of the photovoltaic and semiconductor industries. expertise in operations, project leadership, R&D, product commercialization, and startups, Dr. Singh is dedicated to mitigating climate challenges through sustainable green electric power. He has authored or co-authored over 500 publications and edited more than fifteen conference proceedings. He has delivered over 60 keynote addresses and invited talks at national and international conferences. Dr. Singh is the founding technical chair of the IEEE DC Microgrid Conference and currently serves as Chair of the IEEE Power and Energy Society Committee on End-to-End DC Power. He holds six patents, technologies from his lab licensed for commercialization. Recognized numerous national and international awards, he was named one of the ten Global Champions of Photovoltaics by Photovoltaics World in 2010. In 2014, he was honored by US President Barack Obama as a White House 'Champion of Change for Solar Deployment.' In 2019, he received the Hind Rattan (Jewel of India) Award. Dr. Singh is a Fellow of IEEE, SPIE, ASM, and AAAS.

Keynote 4: Solid State transformer: A Promising Technology Needing Many Breakthroughs

Prof. Drazen Dujic, Associate Professor and Head of the Power Electronics Laboratory at EPFL, Switzerland.

About the Speaker: Drazen Dujic received the Dipl. Ing. and MSc degrees from the University of Novi Sad, Serbia, and the PhD degree from Liverpool John Moores University, UK. He has worked as a Research Assistant at the University of Novi Sad, a Research Associate at Liverpool John Moores University, and in various roles at ABB Switzerland Ltd. Since 2014, he has been EPFL. His research focuses on high-power electronic systems and high-performance drives. He has received multiple awards, including the Istvan Nagy Award (2024), the EPE Outstanding Service Award (2018), and the Isao Takahashi Power Electronics Award (2014). He is an IEEE Fellow.

Keynote 5: Magnetic energy harvesting from overhead power lines

Alon Kuperman, Ben-Gurion University of the Negev, Israel

About the Speaker: Alon Kuperman (Senior Member, IEEE) is a Full Professor the School of Electrical and Computer Engineering, Ben-Gurion University of the Negev, Beer-Sheva, Israel and the Director of Applied Energy Laboratory. He holds multiple international patents and owns an independent consultancy services company. He is also a co-founder of two start-up companies focusing on wireless power transfer. His research interests include all aspects of energy conversion and applied control.

Tutorial Sessions

Tutorial – 1: From Prototype to Product Readiness: IndustrialDesign Considerations for Dual Active BridgeConverters

Dr. Shan, Dr. Aravind, Dr. Gourahari, Schneider Electric, Bangalore

Abstract: The Dual Active Bridge (DAB) converter is a promising DC-DC converter topology for medium-high power applications, offering bidirectional power transfer, galvanic isolation, and high efficiency. While academic research emphasizes on novel modulation strategies, robust control methods, and efficiency improvement; the transition from a prototype to an industrial product requires a broader and more holistic perspective.

In addition to core power electronics, this tutorial presents the design of a Three-Phase Dual Active Bridge (3-DAB) converter for industrial applications, emphasizing the broader requirements that define product readiness. Critical aspects such as thermal management, mechanical robustness, and electromagnetic compliance (EMC/EMI) significantly influence industrial viability. Equally important are Design for Manufacturing (DFM), Design for Assembly (DFA), and Design for Reliability (DFR), which ensure scalability, cost-effectiveness, and consistency in mass production. Reliability practices, including DMA, DFMEA, further strengthen product robustness and long-term performance. rising labour costs, industries are increasingly dependent on robotic and automated assembly lines. Therefore, converter design must also be automation-friendly, supporting repeatable and reliable processes such as automated soldering, reflow compatibility. Considering all these factors, the converter design must ultimately adhere to competitive market requirements by achieving high efficiency while minimizing cost, weight, and volume. This tutorial intends to give a brief insight to all the above said factors taking the Three Phase DAB as an example.

Tutorial 2: Wireless Charging of Electric Vehicles: Design and Implementation

Dr. Francesca, Dr. Calina, Assistant Professor, Eindhoven University of Technology (TU/e), The Netherlands

Abstract: Let's assume that you want to design and implement a wireless charging system based on initial specifications and requirements. Where do we start the process? How do we choose the coil topology, compensation network, and power conversion stages? How do you model tolerances and evaluate/predict performance ? At the end of this tutorial lecture, the student will be able to: Define boundary conditions and make design choices for the subcomponents of the wireless charger. Optimize the system's parameters. Evaluate the design considering manufacturing tolerances and other non-idealities.

Delivery methods: Lecture slides Guided design task

Tutorial 3: Reliability and circularity in the future highly efficient and high-power density DC power electronic converters

Dr. Riccardo[1], Dr. Mattia[1], Dr. Husev[2], University of Bologna, Italy[1], Gdansk University of Technology, Poland[2]

Abstract: As the global energy landscape evolves toward decarbonization, electrification, and digitalization, the demand for highly efficient and compact DC power electronic converters continues to surge. These converters are pivotal in enabling next-generation applications across renewable energy systems, electric mobility, data centers, and aerospace technologies. However, achieving high power density and efficiency often brings challenges related to thermal management, component stress, and long-term system reliability.

This keynote will explore the interplay between reliability and circularity in the design and operation of future power converters. It will examine recent advances in wide-bandgap semiconductors, intelligent control strategies, and modular topologies that address efficiency and density goals while also focusing on design for longevity, ease of maintenance, reuse, and lifecycle carbon footprint.

Bridging academic insights industrial relevance, the keynote will outline strategies for embedding circular economy principles into the converter lifecycle—spanning design, usage, and lifecycle assessment. Attendees will gain a forward-looking perspective on how engineering innovation and sustainable design practices can together pave the way for resilient, resource-efficient power electronics in a rapidly transforming energy ecosystem.

Tutorial 4: SiC MOSFETs Based Power Electronics: Challengesand Opportunities

Prof. Sandeep Anand$_1$, Dr. Abhinav$_2$, IIT Bombay$_1$ and IIT Bhubaneswar$_2$, India

Abstract: The transition from conventional silicon IGBTs to silicon carbide (SiC) MOSFETs is redefining modern power electronics by enabling faster switching and higher switching frequency operation. This tutorial will explore the fundamental need for higher switching frequencies to achieve reduced passive component size while considering overall system efficiency. We will delve into the I-V characteristics of SiC MOSFETs, their switching behaviour, and the impact of parasitic capacitances on switching losses. A detailed comparison between SiC MOSFETs and Si-IGBTs will highlight the clear performance advantages of wide-bandgap semiconductors, while also addressing critical design challenges. Attendees will gain insights into circuit-level issues such as dv/dt-induced false turn-on, EMI/EMC concerns, gate driver complexities, and the need for fast-acting short-circuit protection. We will also discuss key reliability concerns, including gate oxide and body diode degradation mechanisms. The tutorial will also include the medium-voltage applications of the SiC MOSFETs, where these challenges become even more pronounced. The session will conclude a brief presentation of our recent research findings, offering practical perspectives and design recommendations. This tutorial aims to equip engineers and researchers a clear understanding of both the opportunities and obstacles in harnessing the full potential of SiC MOSFETs in next-generation power electronic systems.

List of Abstracts

Oral Session OP1

Technical Track TR01

[#19] Loss-Aware and Thermally-Constrained Design Strategy for 200 kW SiC based Stack

Uppal Das (Indian Institute of Science)*; Surjakanta Mazumder (Indian Institute of Science, Bengaluru); Manish Mandal (Indian Institute of Science Bangalore); Shamibrota Kishore Roy (Indian Institute of Science); Kaushik Basu (IISc-Bangalore)

the rising adoption of electric vehicles (EVs), the demand for fast and reliable public charging infrastructure is increasing rapidly. In this context, high-power converters are becoming increasingly important, particularly in applications like traction inverters and onboard EV systems, where the power levels involved can reach several hundred kilowatts. Conventional high-power converters based on IGBTs typically operate at low switching frequencies (around 5 kHz), necessitating bulky passive components. To enhance power density, increasing the switching frequency is essential, as it directly contributes to reducing the size of these passive elements. Recent advancements in Silicon Carbide (SiC) technology have enabled converters to operate at significantly higher frequencies (up to approximately 50 kHz), thereby improving power density and overall system compactness. However, at these elevated switching frequencies, parasitic inductances become more prominent, leading to challenges such as increased dv/dt, crosstalk, and elevated switching losses. These issues can result in unintended partial turn-on events, ultimately compromising the reliability and efficiency of the system. To address these challenges, this paper presents a systematic design methodology for a 200 kW SiC-based converter stack, optimised for minimal losses and constrained by thermal limits. The proposed procedure facilitates the optimal selection of the SiC power module based on the desired power rating and switching frequency. The effectiveness of the design is experimentally validated under rated operating conditions using a double-pulse test, which accurately quantifies switching and conduction losses.

[#76] A Data-driven Loss Model of GaN-HEMT-based Synchronous Buck Converter Critical Operating Frequency

Subhradip Mukherjee (Subhradip Mukherjee)*; Mahesh Kumar (IIT Madras); N. Lakshminarasamma (IIT Madras)

Wide bandgap (WBG) semiconductor devices such as gallium nitride (GaN) high electron mobility transistors (HEMTs) have significantly advanced power electronics by enabling higher switching frequencies, compact form factors, greater power density, and improved efficiency. Despite these advantages, accurately modeling losses in GaN-based converters remains a key challenge, especially under varying operating conditions. This paper presents a data-driven loss modeling approach for a GaN HEMT-based buck converter, combining empirical simulation data theoretical analysis to estimate both fixed and variable losses across a range of switching frequencies and load levels. Curve-fitting techniques are employed to develop analytical loss functions for key parameters, enabling precise prediction of total power loss. The model also facilitates analysis of the trade-offs among switching frequency, efficiency, and thermal performance. Based on this analysis, an optimal switching frequency is identified that minimizes total power loss while maintaining thermal and operational reliability.

[#141] Desaturation Protection Design for SiC MOSFETs a Low (1.5 us) Short-Circuit Clearance Time

Ravi Teja Pogulaguntla (IIT Madras)*; Arun Karuppaswamy Balasubramanian (IIT Madras)

Short-circuit protection is vital for ensuring the safe operation of high-power-density power converters. While desaturation (desat) protection is widely used for short-circuit protection of Si IGBTs, the lower short-circuit stand time (< 3 us) requirement makes its design challenging for wide band gap SiC MOSFETs. The desat protection can be adapted for SiC MOSFETs certain modifications to achieve a lower short-circuit clearance time. This paper provides a comprehensive discussion on adjusting the short-circuit clearance time in the desat protection circuit specifically for SiC MOSFETs. It also presents various commercially available gate driver integrated circuits (ICs) that are compatible the modified desat circuit. The behavior of SiC MOSFETs under short-circuit conditions, including that at elevated case temperatures, is thoroughly examined, and significant observations are reported. The effectiveness of the designed protection circuit is validated through experiments conducted on a double pulse test fixture.

[#198] Design and Experimental Validation of a Half-Bridge Gate Driver for SiC MOSFETs

Ravi Teja Pogulaguntla (IIT Madras)*; Arun Karuppaswamy Balasubramanian (IIT Madras)

Silicon Carbide MOSFETs offer the advantage of operating at significantly higher switching frequencies compared to conventional silicon IGBTs, presenting significant advantages for high-performance power electronics. However, realizing reliable operation at these frequencies necessitates a carefully optimized gate driver design. This paper provides a comprehensive design methodology for a gate driver board for SiC MOSFETs, a focus on the design of the gate driver power supply and the selection criteria for an appropriate gate driver integrated circuit. The work incorporates practical insights derived from extensive hardware development and experimental validation—insights that extend beyond conventional datasheet specifications. Furthermore, the paper outlines essential printed circuit board layout guidelines critical for gate driver design. A four-layer gate driver board for half-bridge configuration is designed, fabricated, and validated through a double pulse test setup to demonstrate its effectiveness.

[#322] Selection of Negative Gate Voltage for GaN HEMTs Considering Switching Oscillations and Loss Minimization

Venkata Raghavendra I (IIT BOMBAY)*; Sandeep Anand (IIT BOMBAY)

Wide-bandgap (WBG) devices are revolutionising high-density power electronics. Among them, GaN HEMTs are preferred in low-power, high-frequency applications due to their high switching frequency, low on-state resistance, and reduced power losses, all in a small package size. Nevertheless, rapid switching introduces voltage ringing due to gate and power loop parasitics and can lead to unwanted device triggering by virtue of GaN's low threshold voltage. Applying a negative gate bias limits the ringing below threshold values, but it increases dead time losses. This paper proposes a systematic approach to determining negative gate voltage selection during the design phase, aiming to minimise iterative prototyping, accelerate the development process, and enhance both switching performance and system reliability. The proposed method is experimentally validated on a custom-designed 5kw GaN based motor controller (inverter) prototype. The best negative voltage selected can save about 21\{}% of losses.

[#434] Comparative Analysis of EMI in Multilevel Inverters Si and SiC Power Devices

Sarvvesh R (BMS College of Engineering, Bangalore); Dinakar R (BMS College of Engineering, Bangalore); saniya Ahamad (BMS College of Engineering, Bangalore); saicharan K (BMS College of Engineering, Bangalore); Soniya Agrawal (BMS College of Engineering, Bangalore)*; Prema V (BMS College of Engineering, Bangalore)

Multilevel inverters (MLIs) are widely employed in modern power electronics owing to their high conversion efficiency, low harmonic distortion, and improved voltage waveforms. Advances in semiconductor technology have enhanced MLIs in terms of efficiency, compactness, and reliability. Silicon (Si) devices remain prevalent due to their cost-effectiveness, reliability, and mature manufacturing. Conversely, Silicon Carbide (SiC) devices offer higher thermal performance, greater breakdown voltage, and enable higher switching frequencies , reducing losses and system size. However, the fast switching of SiC devices introduces increased electromagnetic interference (EMI) and electromagnetic compatibility (EMC) concerns due to higher dv/dt rates. This paper presents a comparative EMI analysis of Cascaded-H bridge (CHB) multilevel inverter using Si and SiC Power devices. LT-spice Open source software has been utilized to detect the EMI in CHB inverter for a frequency of 50 kHz.

Technical Track TR04

[#72] Multi-Port Fast EV Charging Architecture Renewable Integration and Adaptive Filtering for Power Quality Enhancement

PAVAN SINGH (INSTITUTE OF ENGINEERING & TECHNOLOGY LUCKNOW)*; Divyank srivastava (INSTITUTE OF ENGINEERING & TECHNOLOGY LUCKNOW); Bhim Singh (IIT DELHI); Arunima Verma (INSTITUTE OF ENGINEERING & TECHNOLOGY LUCKNOW); Saurabh Mani Tripathi (Kamla Nehru Institute of Technology, Sultanpur)

This paper presents a robust and intelligent fast DC electric vehicle (EV) charging architecture integrating a photovoltaic (PV) array and a battery energy storage system (BESS) through a bidirectional DC-DC converter (BDC). System is designed to support dual fast-charging stations: a 33kW charger for heavy EVs operating at 420V and a 10kW charger for light EVs (LEV) such as e-rickshaws operating at 120V. Both chargers utilize dual active bridge (DAB) converters, enabling bidirectional power flow, which

gives vehicle-to-grid (V2G) capability. PV system follows a double-stage configuration, employing a boost converter between PV array and voltage source converter (VSC). BDC converter maintains DC link voltage stability and charges/ discharges the battery using excess PV power in constant/variable power mode. A three-phase VSC interfaces common DC link grid and manages power exchange under grid-connected mode (GCM) and standalone mode (SAM). To enhance dynamic response and power quality, a nearest Kronecker product decomposition-based generalized hyperbolic secant robust adaptive filter (NKP-GHSRAF) is implemented for generating VSC reference currents. This filter demonstrates strong harmonic suppression, fast convergence, and resilience to noise, ensuring compliance IEEE-519 standard. VSC operates under current control in GCM and under voltage control in SAM to support stable and seamless operation. Simulation results in MATLAB/Simulink validate effectiveness of presented system in maintaining system stability, minimize total harmonic distortion (THD), and enabling smooth transitions across operational modes

[#161] A New Nine-Level Power Factor Correction Converter Triple-Output For EV Applications

RUTTALA SRINU (NATIONAL INSTITUTE OF TECHNOLOGY ANDHRAPRADESH)*

This paper proposes a novel nine-level power factor correction (PFC) converter triple-output capability designed specifically for electric vehicle (EV) charging applications. The converter topology enables high-quality input current shaping and multiple outputs, thereby addressing diverse EV charging needs. Operating in continuous conduction mode (CCM), the proposed design significantly reduces the size of inductors and capacitors. The use of a nine-level structure ensures low total harmonic distortion (THD), the input current THD measured to be only 2.9\{}%, thereby maintaining near-unity power factor and meeting stringent grid compliance standards. The efficiency of the converter is recorded as 97.2\{}% at rated load, underscoring its suitability for high-efficiency, multi-output EV power interfaces. The converter supports three distinct output power levels—300W, 500W, and 1000W—simultaneously, showcasing its versatility and scalability for multi-power-level EV charging. Simulation validation is conducted in MATLAB Simulink, confirming the accuracy of the results and the effectiveness of the control strategy. The proposed topology demonstrates a compelling combination of low THD, high efficiency, and multi-output operation, making it a strong alternative for next-generation EV charging infrastructure.

[#194] 1- and 3- Reconfigurable Highly Efficient Reduced Stage Isolated EV Charger Active Power Pulsation Buffer

Gyana Sahoo (IIT, Bombay)*; Akash Kumar Swain (Indian Institute of Technology, Bombay); Vivek Agarwal (Indian Institute of Technology, Bombay)

This paper presents a reconfigurable single-stage isolated electric vehicle (EV) charger capable of operating both 1- and 3- grid inputs. Conventional two-stage chargers suffer from higher losses, bulky DC-link capacitors, and reduced reliability. In contrast, single-stage architectures improve efficiency and compactness. However, they face challenges under 1- input due to low-frequency power pulsations, which introduce a second harmonic current component in the battery charging current. To address this, an active power buffer (APB) is integrated in the proposed topology to ensure constant current charging and mitigation of ripple-induced effects. The key novelty lies in the use of existing components of the 3- charger to implement the APB function during 1- operation, eliminating the need for additional hardware. This unified design enhances component utilization, supports flexible grid compatibility, and ensures efficient and reliable charging across a wide range of applications. Simulation results in PLECS demonstrate an efficiency of 97.2\{}% under 1- input and 97.8\{}% under 3- input, unity power factor in both cases. A detailed loss analysis is also presented to highlight the efficiency benefits of the proposed architecture.

[#202] Improved Efficiency and Dynamic Performance of Series-Series Compensated EV Wireless Charging via Partial Power Conversion

Sandhadi Naveena (Andhra University)*; Srinu naik Ramavathu (Andra University); Neelesh Yadav (Tallinn University of Technology)

Wireless Power Transfer (WPT) is emerging as a practical solution for electric vehicle (EV) charging. However conventional DC-DC converters process the entire input power, leading to lower efficiency, higher component stress, and increased cost. This paper presents a transformer-less partial power buck converter that processes only a fraction of the total power, while bypassing the rest directly to the load. The proposed topology, based on an H-bridge a switched-capacitor network, improves efficiency, reduces voltage stress, enhances power density, and lowers hardware costs. Simulation results in MATLAB/Simulink confirm effective voltage regulation and superior performance compared to traditional converters.

[#209] Power Ripple and Loss Minimization in Parallel Hybrid Wireless Charger Using Modified On-Off Keying Modulation Scheme

Rakesh Pulletikurthi (Indian Institute of Technology Madras); Deepak Ronanki (Indian Institute of Technology Madras)*

Adopting constant power (CP) charging for EVs offers superior performance, including reduced thermal stress, faster charging, and enhanced battery longevity, making it a compelling alternative to conventional constant current–constant voltage (CC–CV) charging. Effective implementation of these charging strategies necessitates a robust modulation scheme for voltage control. Among commonly used methods, phase-shifted modulation (PSM) incurs reactive power under light loads due to variable switching frequency. On–off keying (OOK), while fixed-frequency, introduces power ripple from its low-frequency control. To address this, a hysteresis-controlled OOK method is proposed for the CP charging for the parallel hybrid wireless power transfer (HWPT) system to enhance power capacity, efficiency, and misalignment tolerance in EV charging, minimizing ripple while maintaining power control. A 3.3 kW HWPT charger using the CP charging technique is modeled in MATLAB/Simulink and benchmarked against PSM and OOK. Additionally, the electric and magnetic fields of the designed couplers are analyzed using ANSYS Maxwell software; the electromagnetic fields (EMF) emissions comply IEEE C95.1-2005 and ICNIRP 2010 standards

[#338] Quantitative Comparative Analysis of Distinct Modulation Strategies for Isolated Three-phase Matrix-based AC-DC Converter

Shubham Naryani (indian institute of technology, palakkad)*; Jaydeep Saha (NUS, Singapore); Naga Brahmendra Yadav Gorla (IIT,Palakkad)

In this paper, a comprehensive comparison of distinct modulation strategies for high-frequency isolated three-phase matrix-based AC–DC converters is presented, a focus on applications such as electric vehicle (EV) fast charging. Following a brief introduction to existing research on the proposed topology, four major types of modulation strategies (along possible variations) for the single-stage three-phase matrix based AC–DC converter are discussed, accompanied by relevant theoretical formulations. Furthermore, a detailed loss model of the converter is provided, which is essential for comparative evaluation. Subsequently, a simulation-based comparative analysis is conducted for two case studies: light EV fast charging (2 kW, 100 V output) and medium EV fast charging (50 kW, 400 V output). The analysis reveals the modulation strategy that offers the best efficiency, lowest total harmonic distortion (THD), and minimal device stress for each case.

Technical Track TR06

[#102] Fault Detection in NPC Inverter using Machine Learning

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Neutral-Point Clamped (NPC) inverters are widelyused in industrial applications due to their high power quality,reduced harmonic distortion, and enhanced efficiency. However,ensuring their reliable operation remains a significant challengedue to their increased number of switches, raises the risk ofswitch faults. This paper focuses on the detection of single switchopen-circuit faults in NPC inverters using Extra Trees Classifier,an ensemble learning method, for accurate identification andclassification of faults. The proposed methodology achieved anaccuracy of 99.83\{}% demonstrating its robustness for real-worldapplications. The key advantages of this approach includesits simple architecture, resistance to overfitting, and reducedtraining time comparable performance as other existingdeep learning approaches.

[#258] AI-Enabled Early Fault Localization in Grid-Connected Hybrid Modular Multilevel Converter

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The reliability of grid-connected Hybrid Modular Multilevel Converters (H-MMCs) is crucial for stable power system operation; however, submodule (SM) failures—especially open-circuit faults in SMs pose significant diagnostic challenges. This paper presents a data-driven fault localization approach for a single-phase grid-connected H-MMC using both deep learning and machine learning techniques. By introducing faults into a single SM, capacitor voltage measurements are collected from the SMs under both normal and faulty conditions. Three classifiers—Long Short-Term Memory (LSTM) networks, Artificial Neural Networks (ANN), and Support Vector Machines (SVM)—are evaluated. The feature-based models (ANN and SVM) utilize statistical and FFT-based features extracted from 0.025-second time windows, while the LSTM model processes raw time-series data. Simulation results in MATLAB show that the LSTM model outperforms others 98.6\{}% accuracy, compared to 90\{}% for ANN and 89\{}% for SVM. The LSTM also enables early fault prediction (up to 0.25 seconds before fault onset), reduces false positives, and

demonstrates better scalability for systems multiple SMs, making it a suitable choice for real-time fault detection in H-MMCs and enhancing grid reliability.

[#270] An 11-Level Common-Ground Multilevel Quintuple-Boost Inverter for Grid-Tied PV Applications

Deepak Singh (Indian Institute of Technology Kanpur)*; Sandeep N (MNIT Jaipur); Piyush Kant (IIT Kanpur)

This article presents an enhanced high-gain common-ground-type multilevel inverter (MLI) topology tailored for photovoltaic (PV) applications, combining the advantages of quintuple voltage boosting, minimized leakage current, and reduced device count. Building upon prior 11-level inverter architectures, the proposed topology integrates an optimized switched-capacitor (SC) network to achieve high voltage gain, higher output voltage levels, and reduced voltage stress across the devices. A selective harmonics elimination modulation strategy is used to further reduce switching losses, thereby minimizing power loss. The inverter demonstrates self-voltage balancing of all SCs out the need for external sensors or feedback control, simplifying the overall system. Extensive simulation analysis on a 1 kW system validate the topology's performance under various operating modes, including unity and non-unity power factor scenarios. Compared to existing MLIs, the proposed topology achieves a superior figure of merit in terms of component count, total standing voltage, and gain, making it a compelling candidate for residential and commercial PV grid-tied systems.

[#315] Submodule Thermal Balance Technique for Medium Voltage MMC Arm Voltage Sensors

Rupam Chaki (Indian Institute of Technology Roorkee)*; Abhishek Kumar (Indian Institute of Technology Roorkee); Anubrata Dey (INDIAN INSTITUTE OF TECHNOLOGY (IIT), ROORKEE); Rupak Chakraborty (Austrian Institute of Technology Vienna)

The Modular Multilevel Converter (MMC) has become increasingly popular in medium voltage applications due to its modularity and scalability. However, traditional MMC control strategies demand a substantial number of sensors and supporting circuits, which contribute to increased hardware complexity and system cost. Moreover, thermal imbalances frequently arise between the upper and lower semiconductor devices in a submodule during normal operation. Such discrepancies are further exacerbated by ageing effects and parameter inconsistencies across different submodules, leading to non-uniform thermal stress. Persistent thermal stress concentration on specific submodules can degrade their operational lifespan and compromise the overall reliability of the MMC system. Recently,

sensor-reduced MMC control techniques have emerged for medium voltage systems, wherein all individual submodule voltage measurements are substituted a single arm voltage sensor. While this approach significantly reduces sensor count, it may intensify thermal imbalance issues, necessitating detailed investigation. In this context, the present work introduces a temperature regulation strategy for submodules in a three-phase MMC. The proposed method employs a multi-objective cost function that concurrently ensures thermal balancing and maintains submodule capacitor voltage equilibrium. Comprehensive simulation validation on a grid-connected three-phase MMC system confirm the efficacy of the proposed temperature control strategy.

[#357] RELIABILITY ASSESSMENT OF THE FAULT-TOLERANT MULTILEVEL INVERTER FEWER COMPONENTS

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Multilevel inverters, often referred to as MLIs, are essential in contemporary power conversion systems, particularly for medium- to high-power uses such as renewable energy systems, electric vehicles, and industrial drives. Nevertheless, their intricate structure, which consists of numerous power semiconductor devices, raises the chances of faults, thereby impacting both system reliability and performance. Conducting quantitative reliability analyses offers a methodical way to evaluate and improve the durability of MLIs by examining the failure probabilities of individual components as well as the entire system. This document outlines a detailed quantitative reliability evaluation of multilevel inverters utilizing mathematical modeling alongside probabilistic techniques like Failure Mode and Effect Analysis, Reliability Block Diagrams, and Monte Carlo simulations. The focus of the analysis is on crucial components like power switches, gate drivers, and passive elements, considering various operational and environmental scenarios. The proposed seven-level inverter design based on neutral point clamped and cascaded H-bridge inverters is analyzed to compare reliability metrics such as mean time to failure, system availability, and fault tolerance capabilities. The results offer valuable insights into engineering more dependable inverter designs and assist in pinpointing vulnerable components for anticipatory maintenance. This research also examines how fault-resistant designs and redundancy strategies influence the overall reliability of the system. This quantitative methodology aids in creating more durable, efficient, and fault-tolerant power electronic systems suitable for future applications.

[#410] A Model Predictive Control Scheme for Power Sharing in Open-End Winding Transformer based Microgrids

Rajeevan PP (Indian Institute of Space Science and Technology)*; Naufal N (Indian Institute of Space Science and Technology); Rajesh Joseph Abraham (Indian Institute of Space Science and Technology)

This paper presents a Model Predictive Control (MPC) method to control power in an isolated microgrid built using two voltage Source inverters (VSI) connected to an open-end winding transformer. The coordinated control of each VSI is based on the availability of power in the associated energy sources, which may be renewable sources or energy storage devices. Under different load demands and source conditions, the open-end winding transformer configuration permits both inverters to independently contribute to active (P) and reactive (Q) power sharing. Because of its dynamic reference tracking, constraint handling, and predictive capabilities, MPC is used to ensure better voltage regulation and less current ripple. MPC predicts future system behavior and optimizes control actions to meet multiple objectives simultaneously, such as keeping the transformer primary voltage constant, minimizing current ripple and sharing power in any desired ratio. Compared to traditional methods like PI or droop control, MPC can explicitly handle constraints and adapt quickly to changing conditions, providing more precise and robust power sharing between inverters in a microgrid. The strategy is validated in MATLAB SIMULINK under 2 power-sharing scenarios: equal distribution of P and Q, and unequal (e.g., 30–70\{}%) power sharing. Simulation results confirm that the approach maintains load voltage quality and optimizes inverter utilization under varying power availability conditions.

Technical Track TR07

[#234] Control of Permanent Magnet Synchronous Generator based Wind Energy Generation Solar PV Connected to 3-Phase Grid for EV charging

SANJAY KUMAR (IIT Bhilai)*; SHAILENDRA KUMAR (IIT Bhilai); BHIM SINGH (IIT Delhi)

This paper presents a hybrid renewable energy-based micro-grid charging infrastructure for electric vehicle (EV), incorporating a three-phase grid-tied solar photovoltaic array (SPVA) and a permanent magnet synchronous generator (PMSG) driven by a wind turbine. A key contribution is the development of a control scheme for grid side voltage source

converter (GSVSC) to regulate the common DC bus voltage at 400 V and ensure dynamic power sharing between sources using feed forward (FF). The system employs maximum power point tracking (MPPT) to optimize energy extraction from both solar and wind sources. Proposed control strategy includes primary and secondary control to improve grid integration, DC bus voltage regulation, and power quality under IEEE 519 standard. Field oriented control (FOC) is applied to the machine side voltage sources converter (MSVSC) for efficient PMSG operation. Proposed control approach applied to DC voltage regulation, grid power integrated charging infrastructure, and hybrid operating modes. It ensures reliability microgrid operation and enables an ample amount of control. Complete system is modeled in MATLAB/Simulink, and simulation results validate effectiveness of the proposed approach in both transient and steady state conditions under grid-connected and hybrid modes.

[#240] Decoupled Control of Parallel-Connected Dual Split-Phase Induction Motors Fundamental and Seventh Harmonic Phase Sequence

Ahna Fathima (IIST)*; Krupa Kurian (IIST); Sudharshan Kaarthik (IIST)

A novel decoupled control strategy is proposed for parallel-connected split-phase induction motors (IMs) using vector space decomposition. The system is powered by a single voltage source inverter (VSI) along two auxiliary capacitor fed inverters for decoupling. This work focuses on decoupled control fundamental voltage sequence driving one motor and seventh harmonic phase sequence voltage controlling the other motor. Vector space decomposition decomposes the system into orthogonal subspaces corresponding to different voltage sequences. The auxiliary inverters DC-link capacitors are connected to the open-end windings of the motors to apply the appropriate sequence voltages while drawing zero average power. This approach enables control of both machines out mutual interference, offering a scalable and efficient solution for high-performance multiphase drive applications. Unlike series- connected multiphase motor drives, where the stator windings carry combined load current resulting in increased losses and reduced power, the proposed method ensures that each motor operates its own current at full power capacity. The simulation is carried out in MATLAB/Simulink environment.

[#247] Fault-Tolerant Control for Dual Three-Level Inverter Fed Five-Phase OEW-PMSM Drives Under Open-Circuit Faults

Mohan Malla (National Institute of Technology Andhra Pradesh)*; Tejavathu Ramesh (National Institute of Technology Andhra Pradesh)

Permanent Magnet Synchronous Machines (PMSMs) are vital for numerous safety-critical applications that cannot afford operational disruptions. This paper introduces a robust control strategy designed to provide fault tolerance for Dual Three-Level Inverter (D-3LI) fed five-phase Open-End Winding PMSM (OEW-PMSM). The proposed scheme ensures that the five-phase OEW-PMSM can operate reliably even up to two-phase failures, out the need for additional hardware reconfigurations. This capability is especially crucial in Electric Vehicle (EV) propulsion systems, where high reliability is essential. The proposed method is based on additional current cosine components added to the remaining healthy phases using current space vector diagrams. It ensures that the torque-producing magnetic motive force (MMF) remains constant under both healthy and faulty conditions. The Proposed closed-loop control employs Direct Torque Control reduced carrier space vector modulation (DTC-RC-SVM). The simplified RC-SVM technique uses fewer carriers for generating pulses for the D-3LI. The effectiveness of the proposed fault-tolerant strategy was validated using a real-time simulator, OP5707, demonstrating that the motor maintains continuous and stable operation despite faults. The results highlight significant benefits, including reduced current and torque ripples, and improved fault-tolerant performance compared to Open-Circuit fault conditions.

[#273] Closed Loop Control of High Power BLDC Motor for Launch Vehicle Application

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—This paper discussing about the closed-loop control of Brushless DC motors for launch vehicle applications. The launch vehicle primarily comprises two types of engines: pressure-fed engines and pump-fed engines. The pressure-fed engine has specific disadvantages, including lower performance and heavier structure. It will diminish the payload capacity. Therefore, it is applicable in low-thrust applications. The conventional pump-fed engine employs a turbo pump to inject propellant into the combustion chamber. It is more complex due to the presence of turbo pumps. We can utilize an electric pump instead of a turbo pump; this is a more modern design. It is made up of a high-power BLDC motor that powers the pump. The objective of this project is to create an appropriate control strategy for an electric pump-fed engine. Three closed loop control techniques are included: field-oriented control a hysteresis current controller, field-oriented control space vector PWM, and closed loop hysteresis current control. MATLAB/SIMULINK is used to design and simulate these three approaches for high power BLDC motors.

[#274] Field Oriented Control Scheme of PMSM drive for Launch Vehicle Applications

RAYYA T C (Government Engineering College, Palakkad)*; Vinita Chellappan (Government Engineering College, Palakkad); Sangeetha V (Government Engineering College, Palakkad)

Permanent Magnet Synchronous Motors (PMSMs) offer excellent efficiency, torque density, and dynamic performance, making them ideal for precision applications, such as those used in launch vehicles. This project replaces a standard brushed DC motor a PMSM to enhance reliability and reduce maintenance costs. A closed-loop Field-Oriented Control (FOC) system is built, simulated in MATLAB/Simulink, and implemented on hardware. The cascaded control structure provides decoupled torque and flux regulation utilizing Clarke and Park transformations, MTPA operation achieved by maintaining zero d-axis current. Hardware results validate excellent torque control, reduced torque ripple, rapid dynamic response, and low current distortion under varied loads and speeds, demonstrating the viability of FOC-controlled PMSMs for aerospace applications.

[#345] Load-Side Control for a Battery-Integrated Dual-Winding Magnetic Energy Harvester

Asaf Levhar (ben gurion university)*; Riccardo Mandrioli (University of Bologna); Alon Kuperman (ben gurion university)

This paper presents a low-frequency Dual-Active-Bridge (DAB) converter unidirectional control applied to the load-side active bridge converter (ABC) in a Dual-Winding Active Magnetic Energy Harvester (D-AMEH) architecture. The proposed system enables independent control of power flow to the load— validated for power levels up to 150W—while maintaining maximum power extraction from the grid via an MPPT-controlled battery-side ABC. A closed-loop PI controller regulates the output voltage. A MATLAB-based modulation strategy generates optimal switching parameters to minimize the RMS inductor current. PSIM simulation results validate system stability under varying system conditions, demonstrating effective voltage regulation minimal current stress.

Technical Track TR08

[#36] A Data-Driven LPSP Algorithm for Optimal Design of Off-grid PV – Battery Systems

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This paper presents a dynamic algorithm for evaluating the loss of power supply probability (LPSP) in standalone photovoltaic (PV) – battery energy systems. This approach is implemented in MATLAB environment and is designed to improve the system reliability by incorporating real-time environmental data such as solar irradiation, sunshine hours and ambient temperature. Unlike traditional models based on average values, this method dynamically adjusts PV generation and battery energy based on hourly climatic inputs. The temperature correction models are applied to reflect real-time efficiency variations in PV modules and changes in usable battery energy due to ambient temperature. The battery state of charge (SoC) is continuously monitored on an hourly basis to access the energy availability and detect potential shortfalls. A structured simulation framework is developed using actual data from a selected location to represent realistic off-grid operating conditions under a constant daily load. The algorithm calculates PV energy generation, battery charge/discharge cycles and SoC variation; from which energy deficits are identified and LPSP is calculated high temporal resolution. Results are obtained such that the system maintains a high SoC and near zero LPSP for most of the year, short term reliability losses occurring during extended low-irradiance periods. The time-series analysis of SoC and LPSP in the results demonstrates the effectiveness of the model in capturing the behaviour under dynamic weather conditions. By integrating real-time and site-specific data, this study provides a more practical and accurate method for sizing and analysing off-grid PV - battery systems. The modular design of this algorithm makes it adaptable for further extension into hybrid system or artificial intelligence (AI) based forecasting framework.

[#63] A Hybrid HOGI-ANN Synchronization Technique for Fast Estimation of Grid Voltage Parameters Under Distorted Grid Conditions

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For the control and stability of modern inverter dominated modern power systems, accurate estimation of grid voltage attributes namely frequency, phase, and amplitude is

critical. This paper proposes a hybrid design that combines the strengths of a higher-order generalized integrator (HOGI) and an artificial neural network (ANN) to enhance the reliability and precision of grid voltage parameters estimation in distorted grid scenario. The HOGI structure combined synchronous reference frame phase-locked loop serves as a robust front-end estimator that tracks the parameters of fundamental component of the grid voltage. However, residual errors in amplitude and phase estimation may still persist due to dynamic grid frequency drifts. To address this issue, a supervised ANN is integrated to learn and correct those estimation errors. The ANN is trained using simulation data representing a wide range of grid disturbances. MATLAB/Simulation results demonstrate that the proposed HOGI-ANN hybrid method significantly improves estimation accuracy. In future, the proposed technique will be experimentally validated and the results would be compared conventional generalized integrators.

[#92] Ensembled deep learning algorithms for anomaly detection in batteries

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Lithium-ion batteries, which store energy, can experience thermal runaway and capacity degradation. A hybrid deep learning architecture integrates LSTM, GRU, and RNN autoencoders ensemble anomaly detection techniques such as Isolation Forest, Local Outlier Factor (LOF), and DBSCAN for fault detection. NASA's battery dataset multivariate time-series data (voltage, current, temperature, capacity) across charge discharge cycles is preprocessed by a 50-timestep sliding window to acquire degradation patterns. The dataset is partitioned into training, validation, and test splits balanced ratios to allow robust model validation. Training learns normal behavior, validation learns hyperparameter tuning, and testing covers synthetic anomalies. Five-fold cross-validation allows model robustness across different scenarios of battery aging. Feature correlation enhances performance. The LSTM-Isolation Forest ensemble achieves 90.01\{}% accuracy the integration of LSTM's temporal modeling and Isolation Forest's outlier detection on high-dimensional latent spaces. GRU-LOF achieves 86.77\{}% accuracy via local density-based detection. DBSCAN identifies outliers in low-density regions, complementing other methods via sparse data clustering, which is also investigated in this study. Dynamic thresholding reduces false positives by 40\{}% over conventional mechanisms, and real-time processing in 200 ms latency allows deployment at the edge in battery management systems (BMS). Thermal runaway in BMS for electric vehicles is avoided, and capacity fade for grid-scale energy storage is predicted, ensuring fault-tolerant aerospace operations are enhanced greater safety. The preprocessed dataset and model implementation are

shared publicly. Future work can include reinforcement learning for adaptive charging optimization using diversified battery chemistries in enhancing fault detection in energy storage applications.

[#100] Machine Learning Based Prediction and Condition Monitoring of PV Plant

Dr. N. Rama Devi (BEC)*; Mallika Kothapalli (Bapatla Engineering College); Kireeti Panchadarla (Bapatla Engineering College); Mohan bhavani Nayani (Bapatla Engineering College); Vinay Thota (Bapatla Engineering College)

The rapid expansion of data centers and industrial digitization, global energy demand has surged, necessitating a critical transition to clean and renewable energy sources, particularly solar power. Ensuring high operational efficiency in PV systems is challenging due to the variability of environmental conditions, requiring advanced forecasting and monitoring solutions. To address these challenges, this paper introduces a machine learning-based approach for predicting and monitoring the condition of rooftop photovoltaic (PV) plants. The study leverages real-time operational data collected over two years from a 50-kW solar plant at Bapatla Engineering College. An Extreme Gradient Boosting (XGBoost) regression model was employed for hourly solar power prediction, achieving a remarkable coefficient of determination (R^2) of 0.96, demonstrating excellent predictive accuracy. Additionally, thermal images obtained a FLIR thermal camera were processed using a Convolutional Neural Network (CNN) for fault detection, yielding a perfect 100\{}% classification accuracy in identifying healthy and faulty panels. This hybrid framework not only enables precise short-term energy forecasting but also ensures early fault detection, enhancing the reliability, availability, and maintenance efficiency of PV systems. The findings underline the potential of integrating machine learning techniques into PV operations as a promising avenue for intelligent energy management and sustainable power generation.

[#126] Decision Tree-Based Diagnostic Method for Series DC Arc Faults

Sarvesh Wakode (Visvesvaraya National Institute of Technology, Nagpur)*; Makarand Ballal (Visvesvaraya National Institute of Technology, Nagpur)

The increasing integration of DC sources and loadsin power systems necessitates robust protection schemes for DC circuits. One significant challenge is the detection of series arc faults, which are difficult to identify using traditional overcurrent techniques due to low fault currents. Undetected arc faults pose severe risks, including accidents and fires,

highlighting the critical need for rapid and accurate detection methods that also avoid false positives during load changes. This manuscript proposes a DC series arc fault detection method that leverages machine learning algorithms. The approach involves monitoring the current at the source end of a low-voltage transmission line and analyzing various time and time-frequency domain parameters. Data collection incorporates diverse combinations of load current, arc gap length, and fault locations. These parameters are then analyzed using different machine learning classification algorithms to distinguish between normal operation, load change conditions, and arc fault conditions System simulation and arc fault modeling are performed in MATLAB, while data analysis and machine learning applications are conducted in Python. A Raspberry Pi controller is utilized to determine the detection time.

[#412] EV Charging Demand Forecasting using Machine Learning Models: A Data-Driven Approach for Optimizing Charging Station Capacity

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The rapid proliferation of electric vehicles (EVs) necessitates efficient charging infrastructure management to ensure grid stability and optimal resource utilization. This paper proposes a machine learning (ML)-based model that predicts stochastic EV charging demand while explicitly incorporating the effects of atmospheric temperature on charging behavior. By leveraging historical data and time-series analysis, the model accurately captures spatiotemporal dependencies to forecast demand high precision, enabling proactive capacity planning. To demonstrate practical utility, we apply the predicted demand to optimize station capacity using a data-driven ML framework. Compared to existing optimization strategies, our approach achieves 18\{}% greater capacity efficiency and reduces required charging station resources by 11.06\{}% per EV class while maintaining low Loss of Load Probability (LoLP). These improvements result from the model's ability to anticipate arrivals 1-2 hours in advance, enabling dynamic resource allocation. The framework also facilitates real-time decision-making for admitting or rejecting charging requests, effectively preventing station overload.

Technical Track TR09

[#43] Performance Analysis of PV fed High Gain Converter using Adaptive Fuzzy Logic Controller

GOBIKA NIHEDHINI (NIT, PUDUCHERRY)*; THANGAVEL S (NIT PUDUCHERRY)

This study proposes a high voltage gain DC-DC converter to get high output voltage from the PV array. It is possible to get high voltage gain under low switching stress on active components of the circuit that helps in decreasing loss as well as increasing efficiency. The proposed converter is a combination of a classical boost converter, a self-lift network, and a switched capacitor voltage lift circuit. In general, the solar panel is non-linear. Hence, extracting maximum power from the panel is challenging. Therefore, this study proposes a PV-fed high gain converter controlled by Adaptive fuzzy logic MPPT algorithm. The adaptive fuzzy logic MPPT algorithm differs from the classical fuzzy logic algorithm and it provides the required switching angle to the converter as per the measured changes in the panel output. This will decrease the switching stress of the converter and increase efficiency. To analyze the performance of the converter, a MATLAB Simulink model is developed and tested at constant temperature of 25°C and variable irradiance ranges from 700 to 1000 W/m2 has acquired a high output voltage of 154.7V at a duty ratio of 0.4, respectively. The results show that this model is more reliable and simple.

[#83] A Novel Isolated Three-Port Converter based on Dual-Flyback for Solar PV Integrated E-Boat Applications

Ritika Yadav (Indian Institute Of Technology (BHU), Varanasi)*; Chinmaya K A (Indian Institute Of Technology (BHU), Varanasi)

This paper introduces a novel Three-Port Converter (TPC) based on a dual flyback topology, intended for standalone DC microgrid applications powered by rooftop photovoltaic (PV) panels in electric boats (E-boats). The proposed converter is compact and efficient, offering a significant reduction in component count compared to conventional TPCs. It consolidates the functionalities of battery charging, battery discharging, and load power supply into a single integrated system, thereby eliminating the need for multiple converters. A unified control strategy is developed to extract maximum energy from the PV array, regulate battery charging, and ensure uninterrupted power delivery to the load. Beyond conventional features such as complete control of load voltage and reduced transient current ripple, the proposed topology provides galvanic isolation between the input, battery, and load ports. Electrical isolation enhances system safety, improves fault tolerance,

and enables independent voltage domain management, which are critical in standalone and safety-sensitive applications. The converter dynamically transitions between multiple operating modes by monitoring load variations, battery state of charge (SOC), and PV availability. A front-end boost converter is integrated to facilitate maximum power point tracking (MPPT), while a unified closed-loop controller governs the overall system operation. The performance of the proposed TPC is evaluated through comprehensive modeling and simulation in MATLAB-SIMULINK.

[#189] Comparative Analysis of Voltage and Current Source Gate Driver for PV Applications

Vrundesh Pawde (VNIT, Nagpur); Ritesh Keshri (Visvesvaraya National Institute of Technology, Nagpur)*

In photovoltaic systems interfacing the utility grid, fluctuations in irradiance and temperature result in variable output from the PV array, necessitating the use of DC-DC converters for voltage regulation and maximum power point tracking. This paper presents a comparative study on the impact of Voltage Source Gate Driver (VSGD) and Current Source Gate Driver (CSGD) architectures on the efficiency of a buck converter used in PV fed grid applications. While conventional VSGDs rely on fixed voltage sources, CSGDs offer more controlled switching characteristics, potentially reducing switching losses and improving overall converter performance. This study addresses an existing gap by evaluating how gate driver topology influences converter efficiency for PV applications. A 9 kW buck converter was simulated using SIMetrix software, and the results demonstrate the performance differences between the two gate driver types. The CSGD achieved an efficiency of 96.71\{}%, significantly higher than the 87.90\{}% efficiency of the VSGD. This implies that the CSGD will significantly improve the efficiency of the PV system, which will have a substantial impact at the system level.

[#205] A Grid-Connected Transformerless Five-Level Inverter for Single-Phase PV Application

Sanket Tambe (IIT Roorkee)*; Anubrata Dey (IIT Roorkee); Snehasish Pal (HETC)

Transformerless grid-connected multilevel inverters have attracted significant attention in photovoltaic (PV) applications owing to their increased power density, improved efficiency, reduced cost, and minimal filtering requirements. Nevertheless, lack of galvanic isolation introduces a leakage current path via parasitic capacitance between PV-ground and grid. To address electromagnetic interference concerns arising from such leakage currents, this

paper proposes a transformerless five-level inverter architecture. A comparative evaluation existing transformerless topologies is presented to demonstrate optimal resource utilization. A dedicated modulation strategy is developed to ensure compliance leakage current standard VDE-0126-1-1 while effectively regulating DC link midpoint voltage through application of redundant switching states. Simulation results obtained in MATLAB and PLECS validate functionality of proposed inverter.

[#238] A Cost-Effective Non-Isolated Forward-Based Universal Micro-Converter for PV: Comparison Interleaved Single-Stage Flyback Topology

Oleksandr Husev (WUT)*; Hossein Afshari (Tallinn University of Technology); Dmitri Vinnikov (Tallinn University of Technology); Oleksandr Matiushkin (Tallinn University of Technology); Mariusz Malinowski (Warsaw University of Technology)

This paper presents a novel dual-output forward non-isolated micro-converter for Photovoltaic (PV) systems, experimentally benchmarked against a conventional cost-effective flyback isolated topology. Both TRL4 micro-converter prototypes were designed using identical components, except for their core topologies, to ensure a fair comparison. The proposed forward-based micro-inverter eliminates isolation components and integrates shared auxiliary circuits, enabling simultaneous DC and AC outputs while reducing the Bill of Materials (BOM) costs by 28\{}% for mass production. Cost savings are primarily attributed to simplified magnetics and the non-isolated architecture, making the proposed solution highly compatible photovoltaic module applications where cost is a critical barrier. Theoretical calculations and experimental waveforms are presented to validate the approach, demonstrating the feasibility and cost-effectiveness of the forward-based topology for photovoltaic module applications.

[#337] Design of a Wind Emulator for Grid-Connected Type-IV Wind Energy Conversion Systems

Abhishek Kumar (Indian Institute of Technology Roorkee)*; Rupam Chaki (Indian Institute of Technology Roorkee); Anubrata Dey (INDIAN INSTITUTE OF TECHNOLOGY (IIT), ROORKEE)

This study introduces the development and implementationof a laboratory-scale wind energy emulator utilizing a separately excited DC motor mechanically coupled a Permanent Magnet Synchronous Generator (PMSG). The system mimics wind turbine dynamics, delivering power to the grid via a three-level Neutral Point Clamped (NPC) back-to-back converter. The DC motor operates under controlled conditions to simulate varying wind

speed profiles, thereby driving the PMSG. The generated power is extracted through Maximum Power Point Tracking (MPPT) and transferred to the grid. The grid-side converter manages DC-link voltage regulation, while optimal torque control is applied at the machine-side converter to maximize energy extraction. Simulations and Experimental results validate the emulator's ability to replicate key wind turbine behaviors and demonstrate that PI-based control strategies can ensure stable operation and effective voltage control under moderate wind and load fluctuations.

Technical Track TR10

[#146] Design Optimization of Ferrite Based Axial Flux PM Machine for Low-Speed Applications

Deepak kumar (Indian Institute of Technology,Mandi)*; Himanshu Misra (Indian Institute of technology,Mandi)

This paper investigates the viability of ferrite mag-nets in low-speed Axial Flux Permanent Magnet (AFPM) ma-chines by benchmarking their performance against a dimen-sionally equivalent NdFeB-based AFPM. Both machines adopta 12-slot, 10-pole topology double-layer Fractional SlotConcentrated Windings (FSCW). The NdFeB reference designis rated at 4.5 kW and 600 RPM, representing a compact, high-torque, low-speed electric drive. Ferrite magnets, valued for theirthermal stability and cost-effectiveness, are explored as sustain-able alternatives to rare-earth magnets, particularly in low-speedapplications where their lower remanence can be offset throughgeometric optimization.A comprehensive performance evaluationincluding no-load and full-load simulations is conducted forboth machines. A multi-objective cost function is formulatedto optimize key design variables such as magnet thickness,airgap length, and yoke dimensions. The function simultaneous-lyminimizes mass, maximizes torque, and incorporates penaltyterms to enforce efficiency and flux density constraints. Over 250design variants are evaluated, and Pareto solutions are extractedto analyze trade-offs between electromagnetic performance andlightweight design. The results demonstrate that, despite inherentlimitations in magnetic strength, ferrite-based AFPM can be analternative to low speed drives.

[#162] Investigation on Low-Cost Spoke-type BLDC Motor Design Sinusoidal Back-EMF for Home Appliances

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Single-phase induction and universal motors have been widely used in home appliances. However, these motors are less efficient and reliable, and exhibit poor performance especially in part-load conditions. To overcome these concerns, this article introduces a spoke-type brushless DC (BLDC) motor for kitchen chimney and air-cooler applications. The proposed spoke-type rotor a concentrated stator winding design gives a sinusoidal phase back-EMF which improves the overall performance in terms of electric and magnetic loading capabilities. In addition, a low-cost and feasible solution is achieved the aluminum winding, even though the motor operates continuously for 78 hours and the winding temperature rise remains in the specified limits. To validate these features, a 12-slot, 8-pole spoke-type BLDC motor is investigated, and 2-D finite element (FE) analysis is carried-out using Ansys-Maxwell software for various spoke-type designs. Further, a field-oriented control (FOC) technique is implemented for precise control of the proposed spoke-type BLDC motor. Moreover, a prototype is manufactured and the performance parameters such as phase current, output torque, and output power, efficiency for various loading conditions, are experimentally measured.

[#266] Health Assessment of Solid Insulation in Power Transformers using Analytic Hierarchy Processed Weighted Scoring Technique

MULPURU GOPI (National Institute of Technology Srinagar)*; Chilaka Ranga (NIT Raipur); Kushal M Japtap (NIT, Srinagar); Teruvai Manoj (NIT AP)

Power transformer is an essential asset in power systems and is subjected to multiple stresses, such as electrical, thermal, and ambient, which can lead to sudden failures. The overall expected life largely depends on the health condition of solid (paper) insulation. In the present work, an analytical hierarchy process (AHP)- based weighted scoring technique is proposed to assess the health condition of transformer solid insulation. The three primary diagnostic attributes related to paper insulation, such as 2-furfuraldehyde (2-FAL), carbon monoxide (CO), and carbon dioxide (CO), are considered. Unlike previous approaches where attribute weights were assigned arbitrarily, data from 30 transformers collected from the Himachal Pradesh State Electricity Board are utilized for the present analysis. The proposed method integrates the AHP to generate prioritized and rational weights. Scores for each attribute are computed, and a new health index scale is introduced to

evaluate the insulation condition. The proposed approach provides a systematic and reliable method for assessing the health condition of insulation. The present work is beneficial for diagnostic experts and utility planners.

[#292] Optimal Distribution of Pole Area and Its Angular Separation on Braking Performance in Eddy Current Brakes

SRI ROSHAN (IIT(ISM) DHANBAD)*; Sethupathy Subramanian (IIT (ISM) Dhanbad)

Eddy current braking systems provide fast response,minimal wear and tear, and contactless braking. The brake actionarising from the electromagnetic force, and it can be directly-controlled by the electrical systems in the electric vehicles (EVs).Thus, they provide a direct and fast control of the brakingaction compared to mechanical braking. In this work, the focusis on arriving at an optimal distribution of pole area and theangular separation between the poles. Given the complex natureof the problem, the investigations are carried out using numericalsimulations based on the finite element method. An extensive setof simulation is performed to arrive at correlating factors. Theresults so far indicate that fewer number of poles might performbetter than more number of poles the same total polearea. When considering angular separation between the poles,the braking torque decreases as the separation between polesincreases. The findings here provide guidelines for enhancingbraking performance in eddy current brakes.

[#311] Parametric Study on the Effect of Pole Area and Radial Placement of Poles in Disk Eddy Current Braking Systems for Electric Transportation

SRI ROSHAN (IIT(ISM) DHANBAD)*; sethupathy subramanian (IIT (ISM) Dhanbad)

Eddy current brakes are well known for their ability to quickly and smoothly bring down the speed of heavy vehicles. They are reliable, have minimal wear and tear, and offer contactless braking. These features make eddy current brakes optimalfor application in heavy-duty electric transportation, includingautonomous trucks. There are several attempts in the literature to improve the braking performance of eddy current brakes. In this work, a parametric study is taken up to understand and optimize the radial positioning of the pole, as well as the optimal pole area.The study is focused on an axial-flux disk eddy current brake. The parametric study is carried out using the finite element method.The results offer practical guidelines to optimize the design of magnetic poles in disk eddy current brakes, thereby improving their braking torque and braking performance.

[#336] Investigation on Switched Reluctance Motors and out Permanent Magnets

Aprameya Karthik Susarla Ramanathan (Indian Institute of Technology Madras); Deepak Ronanki (Indian Institute of Technology Madras)*; Bijo Sebastian (Indian Institute of Technology Madras)

Switched reluctance motors (SRM) operate based on the principle of variable reluctance, which results in salient stator and rotor structures. The advantages of this type of configuration are the absence of windings on the rotor, rugged structure,adaptability to harsh environmental conditions, tolerance to faults, and simple construction. The main disadvantages of SRM are the low torque density, high torque pulsation and hence lowaverage torque. These drawbacks greatly limit the usage of SRM for applications that demand high torque density and high average torque. Hence, advanced control techniques and machine design modifications are used to improve the performance. In recent years, to address these limitations, the insertion of a permanent magnet (PM) in the structure of SRM has been proposed, which is expected to improve the performance of the machine. However, the impact of PM depends on pole configuration, pole distribution, and the position where PM is added, the positions being the stator yoke, tooth, and stator slot. Hence, the analysis of each feasible positioning of the magnet is carried out for the performance indices such as torque ripple and torque density in the ANSYS-Maxwell and Simplorer software link.

Technical Track TR11

[#73] Grid-Interactive Solar PV-BES System Sensorless PMSM Drive and MQCC-Based Power Management for Smart Water Pumping

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This paper presents a robust and intelligent control framework for a grid-integrated solar photovoltaic (PV) and battery energy storage (BES) system driving a permanent magnet synchronous motor (PMSM) for water pumping applications. A key feature of presented system is implementation of a voltage source converter (VSC) regulated by maximum quaternion correntropy criterion (MQCC) algorithm. MQCC-based control enhances

dynamic performance of VSC under highly nonlinear and uncertain operating conditions. By leveraging statistical robustness of quaternion correntropy, controller effectively suppresses impact of non-Gaussian noise, harmonics, and load disturbances, ensuring accurate and stable power flow regulation between PV source, grid, and BES. System operates under a sensorless vector control using a flux-driven model reference adaptive system (MRAS) to estimate shaft speed and position, thereby reducing system cost and improving reliability. A bidirectional DC-DC converter (BDDC) interfaces BES to common DC link, enabling energy exchange during PV shortfall or grid unavailability. Simulation studies conducted in MATLAB/Simulink under varying solar irradiance, reference speed shifts, and nonlinear load disturbance at point of common coupling (PCC) validate satisfactory performance of MQCC-controlled VSC. Results demonstrate improved system stability, rapid dynamic response, uninterrupted water pumping, and reliable support for critical loads in both islanded mode (IM) and grid-tied mode (GTM).

[#86] Power Quality Improvement in PV-Coupled Dual Multi-Level Converter System Using PI Voltage Regulator

Rupa Boddapati (Osmania University)*

This paper presents a dual converter-based topology integrated a photovoltaic (PV) array and a common DC link capacitor to enhance power quality in low-voltage distribution systems. The proposed system aims to reduce dependency on conventional energy sources by utilizing solar energy to inject active power into the grid, thereby improving energy efficiency. The architecture includes two converters: a series converter for compensating voltage sags and swells, and a shunt converter for harmonic mitigation. Both converters operate in synchronization the main supply by continuously monitoring grid conditions and applying real-time corrections. In this study, a conventional Proportional-Integral (PI) controller is employed to regulate the DC link voltage of the shunt converter. The PI controller's straightforward design and proven reliability make it suitable for basic harmonic filtering and voltage regulation tasks. MATLAB/Simulink is used to model and simulate the system under various loading conditions to validate performance. Results demonstrate that the PI-controlled system achieves acceptable levels of total harmonic distortion (THD) in accordance IEEE standards (THD < 5\{}%), and maintains voltage stability, proving its effectiveness in improving power quality in modern distribution networks.

[#306] Robust SOC Estimation Under Initial Uncertainty Using a One-Dimensional Kalman Filter

Anshul Rawat (Indian Institute of Technology); Aaqib Ahmad (Indian Institute of Technology)*; A.V Ravi Teja (Indian Institute of Technology)

Accurate estimation of the State of Charge (SOC) is essential for the safe and efficient operation of Battery Management Systems (BMS), especially in electric vehicles and energy storage systems. Traditional Coulomb counting methods rely heavily on knowing the exact initial SOC, and any error in this value can lead to large estimation errors over time. To overcome this problem, this paper combines a simple first-order electrical model of the battery a one-dimensional Kalman Filter (1D KF) for reliable, real-time SOC estimation. The method continuously updates and corrects SOC by comparing predicted terminal voltage the actual measured voltage, allowing it to handle large initial SOC errors. The algorithm is also tested added sensor noise to evaluate its performance in realistic conditions, and different sampling intervals are explored to improve tracking speed. Simulation results show that the method provides fast and accurate SOC estimation even incorrect initial values and noisy measurements, demonstrating its strong potential for implementation in low-complexity, real-time BMS environments.

[#330] Model Free Predictive Current Control Method Torque Ripple Reduction in Switched Reluctance Motor

Prabhat . (IIT ROPAR); Vishali Guleria (GHEC, Bandla, Himachal pradesh, India); A.V.Ravi Teja (IIT ROPAR)*

The Model Predictive Control (MPC) method has emerged as a promising technique for controlling Switched Reluctance Motors (SRM) drives due to its capability of handling nonlinearity while ensuring dynamic performance. However, the inherent nonlinearity of SRM and the parameter-dependent characteristics of MPC often lead to performance degradation even a slight change in motor operating conditions, making the system less efficient and robust in nature. To overcome these limitations, this article proposes a novel current-difference based Model free predictive control (MFPC) approach. This method utilizes real-time phase current information and past current error data to predict future states, which enhances prediction accuracy and robustness against parameter variations and mismatches. In addition, the proposed method significantly reduces the number of required switching states, hence reducing the computational burden and control complexity of the drive. The effectiveness of the proposed method is compared conventional MPC and MFPC methods. The proposed method is implemented on an 8/6 SRM in MATLAB/Simulink environment, and the results demonstrate superior

performance in terms of parameter tracking and torque ripple reduction.

[#331] Robust SMC Control Law for IFOC of Induction Motors Incorporating Hyperbolic Tangent Function

ROHIT RAJ (INDIAN INSTITUTE OF TECHNOLOGY ROORKEE)*; ASAD HUSSAIN (INDIAN INSTITUTE OF TECHNOLOGY ROORKEE); JYOTI RANJAN DASH (INDIAN INSTITUTE OF TECHNOLOGY ROORKEE); PRAMOD AGARWAL (INDIAN INSTITUTE OF TECHNOLOGY ROORKEE); SHARMILI DAS (INDIAN INSTITUTE OF TECHNOLOGY ROORKEE)

This paper presents the implementation of a slidingmode control (SMC) law incorporating a hyperbolic tangentswitching function as an alternative to traditional PI controllersfor indirect field oriented control (IFOC) of an induction motordrive. While PI controllers are simple and widely used, theyoften struggle parameter variations, external load disturbances, and the inherent nonlinear characteristics of induction motor systems. Moreover, PI control requires the tuning of two parameters (proportional and integral gains), which can be time consuming and sensitive to operating conditions. Unlike conventional controllers, the proposed SMC scheme requires tuning of only a single switching gain, which simplifies the overall control structure and minimizes the tuning effort. Theproposed technique significantly enhances robustness againstsystem uncertainties while effectively minimizing the chatteringeffect commonly associated conventional SMC designs. The inclusion of the smooth hyperbolic tangent function introduces a continuous control action, which reduces high-frequency oscillations, improves overall system stability, and yields superior transient and steady state performance compared to traditional methods.

[#394] Intelligent Air Conditioning System Solar PV and Load Monitoring

Gautam Raiker (Indian Institute of Science Bangalore)*; Sethu Madhavi Ramala (Indian Institute of Science Bangalore); Subba Reddy B (Indian Institute of Science Bangalore)

This paper presents the design and conceptualimplementation of an intelligent air conditioning (AC) systemintegrated a solar photovoltaic (PV) array, battery energystorage system (BESS), and grid interface. The system monitorssolar power generation, battery status, and AC load in realtime, and employs a predictive control strategy to minimizedependency on the utility grid. A Model Predictive Control(MPC) algorithm is formulated to optimize load scheduling andbattery charging based on PV availability and load forecasts, the objective of reducing AC-related grid consumption.To demonstrate practical viability, a Firebase-based IoT controlinterface is developed, enabling real-time user input and

automatedload switching.Additionally, Non-Intrusive Load Monitoring(NILM) is integrated to enable appliance-level insights, furthersupporting intelligent decision-making. Additionally, Non-Intrusive Load Monitoring (NILM) is integrated to enableappliance-level insights, further supporting intelligent decisionmaking.Preliminary results show that the system effectivelyadapts to varying solar conditions and manages grid transitionand battery utilization intelligently.

Technical Track TR12

[#50] Adaptive Directional Overcurrent Relay for Active Distribution Networks

Chirumamilla Navya (iit kharagpur)*; Gade kesava rao (iit kharagpur); Ashok kumar Pradhan (iit kharagpur)

The integration of inverter-based renewable energysources into distribution systems presents significant challengesfor conventional directional overcurrent protection schemes. Toaddress these challenges, this paper proposes an adaptive di-rectional overcurrent protection scheme. The proposed methoddynamically adjusts relay settings based on the connection statusof renewable energy sources in the distribution network. Itestimates the current direction by generating a reference signalusing differential current and post-fault current analysis. Theeffectiveness of the method is evaluated on a 33-bus distributionsystem under various fault scenarios. The results demonstratethat the proposed approach accurately identifies faults in activedistribution networks, enhancing protection reliability.Index Terms—Adaptive protection scheme(APS), Directionalovercurrent relay(DOCR), Inverter based resource(IBR)

[#82] Accurate Fault Location Calculation in a DG Penetrated System using Modified Takagi Method

Abhijith Suresh (Government Engineering College Barton hill)*; Vinod V (Government Engineering College Barton Hill); Vivek R.S. (College of Engineering Thiruvananthapuram); Sunil Kumar P.R. (Government Engineering College Idukki)

Single line to ground (SLG) fault is one of the most challenging and most frequently occurring fault in power system transmission lines and accounts for almost 85\{}% of total faults that occur. In renewable energy source integrated transmission system, accurately pointing out the fault location is tedious and a challenging task. Since the homogeneity

of the system is lost, electrical parameters from both the ends are necessary for fault calculation. Moreover, due to the presence of power electronic converters at the source end, the fault currents get modulated and relays fail to identify the fault occurrences. In this paper, using Takagi method, a new algorithm is proposed to accurately detect and exactly point out fault location in 1\{}% error. The proposed algorithm is tested under different current control strategies, fault locations and fault resistances. The results obtained shows reduced error as compared conventional methods. The three case studies conducted by varying active and reactive power injection further confirms the superiority of the proposed method.

[#95] Analysis of current limiting strategies of droop controlled grid forming inverters on delayed voltage recovery events

Upendran Mukundarajan (IIT Madras)*; Shanti Swarup K. (IIT Madras)

Due to the recent increase in renewable energy integration in the grid, both grid-following and grid-forming technologies are prevalent in inverter-based resources. Since a stability issue is associated grid-following inverters during weak grid scenarios, grid-forming inverters are mostly preferred. The literature addresses the value of grid-forming inverters in distributed energy resources for mitigating fault-induced delayed voltage recovery events.For simulating delayed voltage recovery events, the single-phase induction motor dynamic model is chosen over the composite load model as the latter introduces numerical iteration issues. The work involves modeling positive sequence droop-controlled grid-forming inverters current saturation current limiters, virtual impedance current limiters, and verification the literature. The work also analyzes the effects of the two current limiting techniques of droop-controlled grid-forming inverters on the delayed voltage recovery events in the 14 bus test system.

[#96] Low-Voltage Ride-Through Operation of a New Pumped Hydro Storage System Model Predictive Current Control

BHUMA NAGA SATYASAI VEMPALI (IIT Madras); DEEPAK RONANKI (IIT Madras)*; APPARAO DEKKA (Lakehead University)

Low-voltage ride-through (LVRT) is a critical requirement for the grid-connected systems to ensure grid stability during voltage disturbances. In this paper, the LVRT operation of a new variable-speed pumped hydro storage system (PHSS), comprising of a four-level back-to-back converter-fed wound rotor synchronous motor (WRSM), is presented. In addition, a model predictive current control (MPCC) strategy is applied for the

proposed configuration to regulate converter operation effectively during fault conditions by maintaining grid-side and machine-side requirements, such as unity power factor operation, flying capacitor voltage control, and grid/machine current control. The discrete-time models of the proposed back-to-back converter-fed PHSS configuration and their coordination during normal and LVRT operation are formulated to achieve safe operation while meeting the grid code requirements. The LVRT performance of the proposed PHSS configuration an MPCC is verified through MATLAB simulations under different voltage dip scenarios. Simulation studies demonstrate that the proposed configuration maintains an active and reactive current injection in accordance Indian grid code standards to meet the LVRT requirement.

[#138] Arc Fault Localization in Low-Voltage Distribution Systems Using Transformer-Aided DOA Estimation

Ratnakar Nutenki (IIT Kharagpur)*; Haraprasad Badajena (IIT Kharagpur); Aurobinda Routray (IIT Kharagpur); Ashok Kumar Pradhan (IIT Kharagpur)

The accurate localization of arc faults in low-voltage distribution systems is critical for the early detection and mitigation of electrical hazards. This study adopts direction-of-arrival (DOA) estimation techniques for arc-induced electromagnetic emissions using sensor arrays. While traditional subspace methods (MUSIC, Root-MUSIC, and ESPRIT) struggle the non-stationary, broadband nature of arc discharges, we propose TransMUSIC, a transformer-based framework forrobust DOA estimation under noisy conditions. TransMUSICreplaces eigendecomposition a transformer encoder thatlearns the noise subspace from temporal snapshots, followed bya neural DOA estimator for arc localization from the MUSICspectrum. Experimental results on synthetic arc discharge data,including colored noise and switching transients, show TransMUSIC significantly outperforms classical approaches in localization accuracy and low SNR robustness. The framework provides a scalable foundation for real-time arc fault localization in smart distribution systems, future implementation targeting laboratory and field measurements

[#426] Signal Construction-Based Ultra-Fast Protection Scheme for DC Feeder of an AC/DC Microgrid

Kamal Kant (Govt. Engineering College, Sheikhpura, Bihar); Salauddin Ansari (Government Polytechnic, Motihari, Bihar); Om Hari Gupta (NIT Jamashedpur, India); Krishna Murari (IIT Bhilai); Sukumar Kamalasadan (University of North Carolina at Charlotte)*

The increasing utilization of hybrid AC/DC microgridsoffers significant advantages in terms of flexibility, efficiency, and integration of renewable energy sources. However, it also

introduces new challenges, particularly in fault detection and protection—especially on the DC side. Ensuring the reliability and stability of such systems requires advanced protection techniques and precise relaying schemes. This paper proposes a signal construction-based ultra-fast protection technique for the DC feeders of a hybrid AC/DC microgrid. In this method, the present sample of a measured quantity is reconstructed using previous samples, and the estimation error during fault conditions is analyzed for fault detection. The proposed technique was implemented and tested on a hybrid microgrid modeled in the MATLAB/Simulink environment. Its performance was evaluated under both grid-connected and islanded modes of operation, demonstrating accurate and robust fault detection capabilities.

Technical Track TR13

[#181] A Comparative Study of Grid Forming Inverter Controls and Their Applications for Improving Resilience of Distribution Systems

MOHAMMED FARSHID K (IIT ROORKEE)*; HIMANSHU JAIN (IIT ROORKEE)

Grid forming inverters (GFMIs) have received significant interest in recent years as a solution for adding high levels of inverter-based resources (IBRs) in power systems. Several control techniques for parallel operation of GFMIs have also been proposed droop control, virtual impedance control, and virtual synchronous generator control receiving the most attention. However, the fundamental reasons for real and reactive power sharing achieved through these control techniques are difficult to find in literature. Further, applications of GFMIs for improving the resilience of distribution systems are limited. This paper addresses these gaps by presenting the working principles of power sharing in droop, virtual impedance, and virtual synchronous generator control techniques, and comparing their performance in improving the resilience of the IEEE 13-node feeder. Meeting the feeder's demand in the absence of the grid at the feeder head, starting an induction motor to simulate blackstart, and creating N-1 contingency of a GFMI are the three scenarios where the resilience performance of GFMIs is evaluated in this paper. Real and reactive power sharing achieved using the three control techniques under high and low X/R ratios is also discussed. Results show that while real power sharing performance of the three techniques is similar under both high and low X/R ratios, reactive power sharing performance varies significantly, virtual impedance control showing better performance. Further, GFMIs were able to meet the IEEE 13-node feeders demand under the three resilience scenarios, demonstrating their potential to improve the resilience of distribution systems.

[#190] Novel Zero-Bus Logic OR-Based Load Flow for Techno-Economic Optimization of RDNs via Coordinated DG and EVCS Placement

Anagha Rajendran K P (BITS Pilani, K K Birla Goa Campus)*; Soumyabrata Barik (IIT(ISM) Dhanbad); Sudarshan Swain (BITS Pilani, K K Birla Goa Campus)

Modern distribution networks must evolveto accommodate the growing energy demands of electricvehicles (EVs) while maintaining reliability, efficiency,and cost-effectiveness. To address these challenges, theoptimal placement and sizing of Distributed Generators(DGs), Electric Vehicle Charging Stations (EVCS), andShunt Capacitors (SCs) are crucial for enhancing bothoperational performance and economic viability in powerdistribution systems. This paper proposes a two-stageoptimization framework for the coordinated allocation ofDGs, EVCS, and SCs in a radial distribution network(RDN). In the first stage, a novel Zero-Bus Logic ORbasedForward–Backward Load Flow method is integrated a Modified Particle Swarm Optimization (MD-PSO)algorithm, guided by a fuzzy max–min decision strategy,to identify the optimal locations and sizes of DGs andSCs. In the second stage, based on the optimized valuesof these components, the ideal siting and sizing of EVCSunits are determined while accounting for the additionalload introduced by EV integration. The multi-objectiveoptimization considers real power loss, voltage deviationindex (VDI), total cost, and emissions. A comprehensivecost–benefit analysis is conducted to verify the economiceffectiveness of the proposed approach, considering investmentcosts, operation and maintenance costs. The methodologyis validated on the IEEE 33-bus system, showingimproved voltage profiles, reduced losses, and cost-effectiveDG planning in the planning horizon.

[#199] Frequency Regulation of Isolated PV-Synchronous Generator System Using Virtual Inertia Controller

Jayaprakash Allemshetty (University College of Engineering Osmania University); Mangu Bhukya (Osmania University)*

the increase in the penetration of solar photovoltaic power into the grid, the inertia of the conventional power system is being reduced. It is known that the inertia of the system which is obtained from the rotors of conventional generators regulates the frequency as acts as primary frequency controller. The power electronic converters have only DC link capacitors which provides the inertia is not sufficient the large penetration, Therefore, virtual inertia constant is included in the control algorithm to regulate the frequency in the PV-synchronous generator isolated system. The synchronous generator is controlled turbine-governor model to inject the required power to the load for continuous supply.

[#250] Multi-Source Energy Hub for EV Charging: Integration of Solar PV, Fuel Cells, and Battery Storage Grid Infrastructure

JAVED ANSARI (NTPC LTd.)*; saran Chaurasia (Department of Electrical Engineering Indian Institute of Technology Delhi); Shivam Kumar Yadav (Department of Electrical Engineering Indian Institute of Technology Delhi); Bhim Singh (Department of Electrical Engineering Indian Institute of Technology Delhi); Kuldeep Sahay (I E T Lucknow)

This paper presents a comprehensive modelling and performance of a grid-integrated multi-source renewable energy system featuring a 2.5 MW photovoltaic (PV) array seven-level multilevel inverter, 200 kW fuel cell, battery energy storage (BES), and electric vehicle (EV) charging infrastructure. System is connected to a 33 kV distribution grid through a 2.5 MVA, 33/0.415 kV transformer. Low voltage side incorporates a bidirectional converter interfacing both BES and a 500 kW EV charging station comprising ten 50 kW charging units. Detailed mathematical models are developed for each subsystem, including PV array characteristics under varying environmental conditions, seven-level inverter topology asymmetrical cascaded H-bridge configuration, fuel cell electrochemical dynamics, bidirectional power conversion, and battery storage behaviour. Integrated system model is implemented in MATLAB/Simulink environment to analyze power flow management, grid synchronization, and system stability under various operating scenarios. Simulation results validate mathematical models and demonstrate effectiveness of proposed control strategies for maintaining power quality and system reliability total harmonic distortion below 3.0\{}%.

[#430] Virtual Oscillator based Dual Loop Control for Grid Forming Modular Multilevel Converter

Hitesh Malviya (IIT Guwahati)*; Sahil Gaurav (IIT Guwahati); Chandan Kumar (IIT Guwahati)

Modular multilevel converter (MMC) is widely explored for offshore wind farms and large-scale photovoltaic plants for high voltage (HV) and medium voltage (MV) applications. The MMC comprises sub-modules (SM) connected in series and can provide higher voltage levels. The dc-ac power conversion through MMC is done through grid forming (GFM) control to regulate voltage magnitude and frequency. In the GFM mode of operation, power converters are susceptible to low-order voltage and frequency oscillations and instability. This paper presents an Andronov-Hopf oscillator (AHO) based virtual oscillator (VO) for MMC dual voltage and current control to address such issues. The modulation scheme includes rule-based SM voltage balancing to maintain SM capacitor voltages. The simulation results show the stable operation of the MMC through load variations.

[#431] Virtual Oscillator Control-Enabled Unbalanced Fault Ride-Through in Smart Transformer-Based AC Microgrids

Sahil Gaurav (Indian Institute of Technology Guwahati)*; Lakshmi Kant Rao (Indian Institute of Technology Guwahati); Chandan Kumar (Indian Institute of Technology Guwahati)

Smart transformer (ST)-based microgrids are preferred over conventional power transformer (CPT)-based microgrids because of their compact size, lighter weight, and ability to provide ancillary services. The ST LV converter works as a grid-forming converter (GFC) unit, and distributed generator (DG) converters follow the voltages set by the ST LV converter. This paper introduces a new control technique called the instantaneous symmetrical component-based Andronov-Hopf virtual oscillator (S-AHVO), which addresses the limitations of existing virtual oscillator (VO) controls that only synchronize positive-sequence voltages and fail under unbalanced conditions. The S-AHVO control allows for continuous synchronization of positive, negative, and zero-sequence voltages, enhancing microgrid operation during voltage imbalances and ensuring accurate load-sharing. It maintains the dynamic performance of existing VO controls and can be implemented out additional sensors. The paper includes a systematic design procedure, mathematical analysis, and simulation studies to validate the proposed technique.

Oral Session OP2

Technical Track TR02

[#57] Interactive Power Electronics Education Using Jupyter Notebook and the GSEIM Simulation Package

Nakul Narayanan (GEC TCR)*; Mahesh B Patil (IIT Bombay)

An innovative way for teaching/learning power electronics using Jupyter Notebooks is presented. The Notebooks are interactive, allowing users to modify circuit parameters and observe their impact on circuit behavior. The GSEIM simulation engine is interfaced the Notebooks for solving the circuit equations. Simulation examples are discussed to illustrate the capability of the Notebooks in teaching power electronics.

[#108] Constraint-Driven Control of SEPIC Converters Using Offline-Optimized MPC

Rangoli Singh (IIT BHU); Shiv Prakash (IIT BHU)*; Dhawal Dwivedi (IIT BHU); Sandip Ghosh (IIT BHU); Chinmaya K. A. (IIT BHU)

Model Predictive Control (MPC) is widely recognized for its exceptional constraint-handling and optimal control capabilities. However, its substantial computational burden poses a significant barrier to real-time implementation, especially in embedded systems limited processing resources. To address this challenge, this paper presents the application of Explicit MPC, wherein control laws are computed offline, enabling ultra-fast online execution minimal computational overhead. The proposed strategy is applied to a 100 W Single-Ended Primary Inductor Converter (SEPIC), real-time validation performed using dSPACE hardware. Experimental results demonstrate that Explicit MPC significantly outperforms conventional MPC and PI controllers in terms of transient response, steady-state precision, and disturbance rejection. Crucially, these enhancements are achieved drastically reduced computational effort, positioning Explicit MPC as a highly practical and efficient solution for real-time control in modern power electronic systems.

[#109] Active Input Power Control for SONAR Application

Joseph Mathew (IIT Madras)*; Akshat Saini (IIT Madras); Lakshminarasamma N (IIT Madras); Panchalai V N (NPOL)

Power Amplifier are the key elements in delivering the desired power to the SONAR transducers. Sensing the load parameters like output voltage and current is increasingly difficult as the inverters and the sensing circuitry are supposed to be inside the ship vessel, while the long tow cables carry the power to the SONAR transducer. For SONAR applications, the range of detection is determined by the total amount of active power supplied to the transducer. Therefore, to get configurable range of detection, there is a need to specifically control the supplied active power. An input power-based control scheme for an inverter designed for a SONAR power amplifier application is proposed in this paper. The effectiveness of the proposed control strategy has been validated through both simulations and real-time implementation. The proposed control scheme has been experimentally verified using an 800 V, 1 kW prototype.

[#115] An Improved Ultra-Gain Quadratic Boost Converter Lower Voltage Stress for Fuel Cell Applications

Ravi Kanaparthi (visvesvaraya national institute of technology)*; Jay prakash singh (visvesvaraya national institute of technology (VNIT),Nagpur); Makarand sudhakar ballal (visvesvaraya national institute of technology (VNIT), Nagpur)

This paper proposes an improved ultra-gain quadratic boost converter (AIU-QBC) lower voltage stress on the switch, which is used in fuel cell and renewable power systems. It is derived from the conventional quadratic boost converter (QBC); the front end features a QBC and is further integrated a voltage multiplier cell (VMC) as well as a switched capacitor and inductor (SCI) cell to enhance the output voltage levels and voltage gain. A series-connected capacitor (C3) decreases the voltage stress on the switch, recommends lower-rated switches, and decreases the converter cost. The input current is continuous and has low ripple content. The steady-state analysis is analysed under the continuous conduction mode (CCM). A comparison of the proposed converter the current literature has been presented. The theoretical analysis is validated by the LTspice simulation software of 200 W, 50 kHz switching frequency.

[#124] Comparative Analysis of Phase Shift Modulation Strategies of Dual Active Bridge Converter

Pavani Gupta (Indian Institute of Technology (BHU) Varanasi)*; Pragya Singh (Indian Institute of Technology (BHU) Varanasi); Dhawal Dwivedi (Indian Institute of Technology (BHU) Varanasi); Chinmaya K A (Indian Institute of Technology (BHU) Varanasi)

In recent times, the Dual Active Bridge converter has gained recognition as an efficient choice for isolated, bidirectional DC DC power conversion—ideal for EV chargers and DC microgrids. Its dual-directional power flow and galvanic isolation make it well-suited for advanced energy applications. This paper investigates DAB's steady-state performance and compares various modulation schemes to evaluate their impact on efficiency and power output. The methods studied include Single, Extended, Dual, and Triple Phase Shift strategies. Graphical plots illustrate how phase shift variations influence voltage and current characteristics. Simulations under open-loop operation confirmed strong alignment theoretical results, contributing valuable insights for future closed-loop system designs.

[#135] Non-Active Power Control of Single Phase Dual Active Bridge for improved performance

AMRITHESH KUMAR (COLLEGE OF ENGINEERING TRIVANDRUM - APJ Abdul Kalam Technological University)*; Dr. JAYAKUMAR P (COLLEGE OF ENGINEERING TRIVANDRUM - APJ Abdul Kalam Technological University)

Dual active bridges (DAB) are widely used in different power electronics applications. Controlling of the non- active power flow associated it will be highly beneficial for the effective utilization of the topology. A control strategy for controlling the inductor current and non-active power in a single-phase DAB is attempted in this paper. The triple phase shift (TPS) modulation technique is used for the control of both parameters out compromising efficiency or increasing system complexity. Also, a of non-active power flow control using single phase shift (SPS)modulation, dual phase shift (DPS) modulation, were also presented.

[#150] Performance Comparison of Single and Cascade Loop Controllers in Quadratic Boost Converter

Sumukh Surya (Bosch)*

Quadratic Boost converter (QBC) is one of themost popular high gain DC-DC converters used in renewableenergy and battery powered systems. In this paper, a singlevoltage loop and cascade control loop architectures areintegrated an ideal QBC converter operating in continousconduction mode (CCM) and performance analysis in terms ofsettling time (ts) and overshoot for different disturbances ininput, output voltage setpoints and load changes is made.Initially, QBC is mathematically modeled using volt-sec andamp-sec balance equations and the simulated using MATLAB /Simulink. Later, using FSOLVE tool in MATLAB softwaresteady state analysis is performed and the number of iterationsrequired to solve the non-linear algebraic equations is found out.Using the state space equations, the transfer functions ofperturbed output voltage to the duty cycle (Gvd) and perturbedinductor current to the duty cycle (Gid) is found out. It is foundthat both Gvd and Gid are unstable due to the presence of RightHalf Plane (RHP) zeros. Later, using the manual pole placementtechnique controllers are designed for single voltage loop andcascade loops considering a simple resistive load. It is found that the single voltage loop outperforms the cascade control in terms of ts and overshoot for various disturbances mentioned earlier. This improvement is due to higher bandwidth set in the single voltage loop control.

[#151] High-Gain Converter Optimized Voltage and Current Stress

PREETI SHARMA (PDEU)*; Kaibalya Prasad Panda (PDEU); Rajneesh Kumar (BITS); Sara Hasanpour (Islamic Azad University Ramsar)

This paper proposes a high step-up DC-DC converter incorporating two active switches, developed explicitly for sustainable energy applications. The converter achieves low input current ripple using a small-value input inductor, effectively minimizing conduction losses. A three-winding coupled inductor (TWCI), in conjunction a voltage multiplier stage, facilitates a significant voltage step-up at a reasonable duty cycle. The topology adopts a semi-trans-inverse structure, which enables high output voltage a reduced magnetic turns ratio, thereby enhancing efficiency by lowering magnetic and copper losses. Additionally, quasi-resonant operation is achieved through interaction between intermediate capacitors and the TWCI's leakage inductance, contributing to lower switching losses. Notably, the proposed converter exhibits a quadratic voltage gain characteristic, making it well-suited for applications demanding significant voltage amplification. The converter's operation are analyzed in detail, and its performance is validated through PSIM simulations. Comparative results existing converter topologies are also presented to demonstrate their effectiveness.

[#173] Design and Implementation of an Active Continuous Conduction Mode Boost Converter for Power Factor

Satyabrata Behera (National Institute of Technology, Rourkela)*; Yogesh Sharma (National Institute of Technology, Rourkela); Venkata Ramana Naik N (National Institute of Technology Warangal)

In modern industrial, commercial, and residential applications, most electronic devices require rectification of the input AC voltage for proper operation. However, the use of non-linear loads introduces significant non-linearity in input characteristics, leading to distortion in the input current waveform and deterioration of the power factor. This results in the flow of non-sinusoidal currents, which not only stresses the power grid but also adversely impacts other connected users. To address these challenges, input current harmonics must be minimized, and the current waveform must be shaped to follow the input voltage. Power Factor Correction (PFC) is a critical solution, typically implemented using either Passive or Active techniques. This paper presents the design and hardware implementation of an Active PFC system using a conventional Boost converter topology. The proposed Boost PFC approach offers continuous input current, thereby reducing Total Harmonic Distortion (THD) and significantly improving the power factor.

[#182] A Study of di/dt Effect on Spacecraft Li-ion Battery

Ananda S (U R Rao Satellite Centre)*; Yerrabothala Ram Kishore (U R Rao Satellite Centre); Jyostnarani Mahanta (U R Rao Satellite Centre); Sushma H R (U R Rao Satellite Centre); Debasmiti Mishra (U R Rao Satellite Centre); Pradeep K Peter (U R Rao Satellite Centre); Neelavathy M (U R Rao Satellite Centre)

Battery is a ubiquitous energy storage device for applications like spacecrafts. A series-parallel configured lithium-ion battery is realized. The battery behavior for fast discharging loads is studied. A typical cell is measured for resistance and reactance Electrochemical Impedance Spectroscopy (EIS). The EIS indicates an inductance of 0.7μH whereas battery test fast changing load yields 2 μH. The result indicates that the wires and bur bar arrangement in connecting cells as battery has significant contribution.

[#245] ZVS Analysis of an Interleaved Current-Fed DAB for Bipolar and Unipolar DC Grids

Lohith Pittala (University of Bologna)*; Edivan Laercio Carvalho (Tallinn University of Technology); Mattia Ricco (University of Bologna); Georgios Orfanoudakis (Hellenic Mediterranean University); Alon Kuperman (Ben Gurion University of the Negev); Riccardo Mandrioli (University of Bologna)

This paper presents a comprehensive zero-voltage switching (ZVS) analysis of an interleaved current-fed dual active bridge (CF-DAB) converter under single-phase-shift control, targeting both bipolar and unipolar DC microgrids. Compared to conventional ZVS conditions, the presence of interphase inductors, either at the input or output bridges, enables an extended ZVS region, particularly under light-load conditions. This is important to increase the overall system efficiency, since in many applications, including DC microgrids, most of the time, such a converter operates at partial load. In addition, for unipolar DC microgrids and balanced bipolar systems, the interphase inductor currents may exhibit a DC offset or remain centered around zero, while under unbalanced load conditions, a DC offset is resulted. To cover both cases, analytical ZVS boundary expressions are derived for both scenarios: and out DC offset. Experimental validation is conducted on a 300 W scaled-down prototype for the no-offset case, confirming that the CF-DAB can achieve reliable soft-switching over a wide voltage gain range appropriate inductor design.

[#278] H-Bridge and T-Type Dominance in Quad Active Bridge Converter Performance: A Comparative Analysis

Ashish Gupta (Indian Institute of Technology Kanpur)*; Piyush Kant (Indian institute of Technology Kanpur); Sunil Kumar Dube (Lucid Motors Inc.)

This paper presents a comparative study of various quad-based Dual Active Bridge (DAB) converter topologies using a consistent framework of performance indices. These indices include transformer RMS current, peak switch voltage and current stresses, turn-off voltage and current of devices, converter efficiency, and component count per module. Since voltage and current stresses significantly influence both the cost and efficiency of the converter, each topology has been designed for an identical 5 kW module rating to ensure a fair evaluation. Moreover, models are implemented in MATLAB/Simulink using the PLECS Blockset and simulated under different power rating scenarios to generate a comprehensive dataset. Analysis shows that the T-type H-bridge configuration offers superior performance across multiple indices. This paper gives an in-depth concept to the reader to select the appropriate converter topology configuration for a suitable application.

Technical Track TR03

[#155] A Novel Modulation for Modified Single-Stage Single-Phase AC to DC TAB-based converter parallel HF Compensation of Second Harmonic Power

Naresh Rana (Indian Institute of Science, Bengaluru)*; Kaushik Basu (Indian Institute of Science, Bengaluru)

Single-stage AC to isolated DC converters are gaining popularity due to their higher power density, improved efficiency, and better reliability compared to traditional two-stage systems. However, single-phase single-stage converters have a drawback—they transfer second harmonic power directly to the battery load, which can shorten battery life.To address this issue, a modified single-stage, single-phase AC to isolated DC Triple active bridge (TAB) converter is proposed. This design includes compensation for the second harmonic power ripple at the high-frequency link. A new modulation technique is introduced to ensure constant power transfer through the transformer and the DC-side full bridge, and finally to the battery load. This results in reduced conduction losses and a lower VA rating requirement for the transformer.An analytical closed-form solution

for the modulation parameters is also derived, making the control and implementation straightforward. The effectiveness of the proposed method has been validated through simulation results.

[#184] A Novel Closed-Loop Resonant Frequency Tracking for Dynamic Wireless Power Transfer Using Inverter Phase Angle Computation

parameshwari M (National Institute of Technology,Trichy)*; Arjun Bharati Kumar (National Institute of Technology,Trichy); P.Srinivasa Rao Nayak (National Institute of Technology,Trichy)

A closed-loop frequency control algorithm for dynamic wireless power transfer (DWPT) systems. This algorithmallows for robust resonant operation out extra sensingcircuitry. The method calculates the phase relationship betweenthe inverter output current and the reconstructed AC voltageusing time-series data from internal signals. Reactive behaviour is detected by computing a non-normalised discrete-time interaction term, which enables real-time switching frequency adjustment for optimal power transfer. The entire control strategy is implemented on the transmitter side, and real-time FPGA logic is used to validate it on a SiC MOSFET inverter platform. Simulation and hardware results confirm reliable frequency convergence and effective operation under varied load conditions.

[#232] New Segmented Inductive Coupler Structure Capacitive Decoupling for Interoperable Dynamic Roadway Wireless Charging Applications

Ashwaini Goswami (Indian Institute of Technology Madras); Rakesh Pulletikurthi (Indian Institute of Technology Madras); Deepak Ronanki (Indian Institute of Technology Madras)*; Subbarao Pichuka (Indian Institute of Technology Madras)

Magnetic interoperability is a critical parameter in inductive power transfer (IPT) systems, as efficient power transfer is predominantly governed by the coupling coefficient (k) between the transmitter (Tx) and receiver (Rx). IPT couplers are broadly classified as unipolar or bipolar based on the direction of magnetic flux. However, due to fundamental differences in flux orientation, magnetic interoperability between these coupler structures is inherently poor. In dynamic wireless power transfer (DWPT) applications, where lateral misalignment is common and interoperability between the couplers is essential, a bipolar coupler is typically preferred for the Tx side due to reduced power ripple and k. Despite this, unipolar Rx structures are not magnetically compatible bipolar counterparts, resulting in reduced power transfer. Also, conventional segmented doubleD (DD) structures are prone

to cross-coupling in the Tx coils due to this segmented structure, which degrades overall system efficiency. To overcome the aforementioned concerns, this paper proposes a new segmented DD transmitter structure utilising controlled current excitations to enhance magnetic interoperability. Furthermore, a modified segmented DD coupler integrated a capacitive decoupling network is introduced to mitigate the cross-coupling. A 25 kW DWPT system employing the proposed coupler design is simulated using ANSYS Maxwell for electromagnetic analysis, and the decoupling network is modelled and validated using MATLAB simulations

[#281] Implementation of Average Current Control for Load Sharing in Paralleled LCLC Resonant Converters

Shankar T (Indian Institute of Technology Madras)*; Nagesha Chitpadi (Renesas Electronics India Pvt. Ltd); Lakshminarasamma N (Indian Institute of Technology Madras)

This paper presents an average current control strategy for effectively paralleling LCLC resonant converters, targeting high-gain applications such as Electric Vehicle (EV) charging and data center power supplies. The proposed method specifically focuses on controlling the secondary tank current, offering a robust approach to achieve precise current sharing among parallel modules. This work includes the implementation and experimental validation of an averaging sensing network on the secondary side, which provides an accurate representation of the average current for feedback control. This technique significantly enhances the inherent current sharing capabilities of the paralleled converters, mitigating imbalances and improving overall system reliability and efficiency. The converter operation is verified experimentally for 400W of load power, demonstrating the effacy of average current control.

[#283] A Class E-based Power Factor Corrector Circuit Reduced DC Bus Capacitor

Pabbisetty Hari Venkata Sai Kishore (IIT Mandi)*; Amit Kumar Singha (IIT Mandi)

Driving a DC load by a grid-connected singlephase system typically results in a cascaded two-stage configuration consisting of a Power Factor Correction (PFC) circuit followed by a point-of-load (POL) DC-DC converter. This traditional two-stage architecture increases the overall component count and degrades efficiency. This paper proposes a single-stage structure based on Class E PFC, which not only eliminates the need for two-stage operation but also enables soft switching to improve efficiency. However, the PFC part continues to face persistent challenges in suppressing second harmonic components arising from source-load power imbalance. It can be mitigated using bulky electrolytic

capacitors, which are prone to reliability issues. To address this, the paper presents a hybrid solution that integrates an active power decoupling (APD) filter the Class E PFC circuit. The APD actively mitigates the 100/120 Hz ripple by injecting compensating currents. This approach significantly reduces DC-link voltage ripple, enabling the use of compact capacitors and reducing the size by at least one-tenth compared to traditional designs.

[#285] Comprehensive Performance and Magnetic Analysis of 3.3kW Inductive Wireless Power Transfer System

Ummemisbah Bhisti (Ontario Tech University)*; Joel Adubofuor (Ontario Tech University); Sheldon Williamson (Ontario Tech University)

This paper presents a comprehensive analysis of a 3.3kW Inductive Power Transfer (IPT) wireless charger system special focus on the comparison of two widely implemented compensation topologies: Series–Series (SS) and LCC–Series (LCC-S). The objectives are to find the optimum compensation strategy for mid-level Electric Vehicle (EV) applications by comparing circuit-level performance as well as heat characteristics. The study investigates consecutively three different magnetic coupling coefficients (k = 0.32, 0.265, and 0.227) in relation to their effect on the overall system efficiency, stability of power, and dynamic response of the two compensation structures under various alignment conditions.This work integrates electric optimization thermal design to develop a design approach. A thermal model is built to predict power losses in passive and switching components to enable suitable heat sink sizing under realistic working conditions. This multi-topology, multi-k analysis for selecting compensation topologies for actual WPT implementation. The methodology includes an added scalable approach that makes trade-offs between electromagnetic performance and thermal reliability for mid-power wireless EV charging.

Technical Track TR04

[#290] 3.3kW EV Charger Based on Fast switching DAB Topology: A Dual Phase Shift Approach

Avila PINTO (NITK)*; Kalpana R (NITK); Parthiban P (NITK)

This paper addresses the performance of the two-stage charging system TP(Totem-Pole)

Bridgeless PFC (Power Factor Corrector) and bidirectional Dual Active Bridge (DAB) converter GaN HEMT's (Gallium Nitride High Electron Mobility Transistors) for EV on-board charger (OBC). The switching devices of the DAB converter are controlled by double phase shift(DPS) modulation. Planar Magnetic Transformer (PMT)is implemented to enhance the performance of the DAB converter configuration of 3.3kW, and 250kHz operating frequency. The closed loop design of the converter is performed in PLECS software using digital PR \{}& PI controller for PFC \{}& DAB respectively. Thermal files of GaN devices are used to operate them at high switching frequencies and to deliver high efficiency reduced power losses. The results are compared and presented for validation.

[#296] An Efficient Bridgeless two-stage OBC using a totem-pole AC-DC converter for EV applications

SUPRABHA PADIYAR (MANIPAL INSTITUTE OF TECHNOLOGY,MANIPAL,UDUPI, KARNATAKA)*

This paper presents a bridgeless dual-stage onboard charger (OBC) for electric vehicle (EV) applications. The front-end AC-DC stage features a full-wave circuit using MOSFETs and an LC filter, which contributes to efficiency due to the exclusion of diodes. One leg of the rectifier circuit switches at high frequency, and the other switches at line frequency. The filter inductors' current and output voltage are monitored and fed back to be controlled using a tuned PI controller to generate corresponding pulses to achieve power factor correction (PFC) and voltage regulation. A push-pull converter functions as a back-end converter to charge the EV battery regulated power. The battery controller requires battery voltage and current to be monitored and fed back to incorporate CC logic through a tuned PI controller for power regulation. The proposed charger system is simulated on the MATLAB Simulink platform. Results showcase satisfactory performance and show that the input AC power quality follows the IEC 61000-3-2 standard while charging the EV battery effectively.

[#301] Analysis of a Three-Leg LLC Multiport Resonant Converter for Dual Voltage Onboard EV Charger

Swathi G V (National Institute of Technology Karnataka); Dr Dharavath Kishan (NIT K)*; Vinusha Bussa (National Institute of Technology Karnataka)

Power converters that can effectively handle broad output voltage ranges are required due to the growing demand for quicker and more flexible electric vehicle (EV) charging. To facilitate flexible charging of both 400- and 800-volt battery systems in a single

architecture, this paper proposes a novel three-leg LLC Multiport resonant converter (TL-MRC) topology for onboard EV chargers. A three-leg DC–AC inverter interfaced two high-frequency transformers and connected to separate LLC resonant tanks is used in the suggested system. Dual voltage levels are supported for different EV platforms by a pair of diode bridge rectifiers on the secondary side, which allow for series or parallel output configuration that is dynamically reconfigurable through low-side switches. The three-leg topology reduces component redundancy, maximizes transformer utilization, and facilitates modular expansion in contrast to current dual-leg or multi-transformer systems. Zero-voltage switching (ZVS), which is accomplished through resonant behavior, is used in conjunction complementary switching to analyze operational principles. Simulations verify that the converter can deliver up to 10 kW of power in both output modes high efficiency, soft switching, and improved reliability.

[#342] AC-powered Dual-Active-Bridge (DAB) EV Charger a Reduced-Current Stress Converter Using an Adaptive DC Link in Three-Phase system

RANJEET SAH (IIT BHILAI)*; SHASHANK KURM (IIT BHILAI); SHAILENDRA KUMAR (IIT BHILAI)

This article presents a converter topology for electric vehicle (EV) charging systems, comprising a three-phase Active Front-End Converter (AFEC) and a Dual Active Bridge (DAB) converter. In conventional configurations, the AFEC maintains a fixed DC link voltage, resulting in a significant voltage difference across the DAB transformer's inductor. This leads to a triangular inductor current waveform a high RMS value, causing increased conduction losses and reduced system efficiency. In the proposed architecture, the control strategy of the AFEC is enhanced to generate a variable DC link voltage that dynamically adapts to the EV battery's voltage condition. This ensures the DC link voltage remains sufficiently above the AC input voltage while aligning the voltage levels of the DAB transformer's primary and secondary windings. As a result, a load-dependent DC link is established, effectively reducing the RMS current by approximately 8\{}% to 53\{}%. This reduction minimizes current stress on the transformer, lowers conduction losses, and significantly improves overall system efficiency. Simulation results obtained using MATLAB/Simulink validate the effectiveness and performance gains of the proposed EV charging system.

[#419] Operation of Wide-Range Universal V2G Charger in DC Microgrids

Hans Anniste (Tallinn University of Technology)*; Dmitri Dmitri (Tallinn University of Technology); Andrei Blinov (Tallinn University of Technology)

This paper presents a universal wide operating range bidirectional DC-DC charger for electric vehicles (EVs), along its features that enable seamless integration into DC grids. The proposed topology provides an output voltage range of 150V to 1000V while maintaining a low component count in the power stage. Its dual-input stage design enables compatibility three DC grid configurations, including unipolar and bipolar types. Charging power and V2G operation are controlled according to a charger-specific droop curve. Additionally, the charger supports voltage balancing both during EV charging and when the EV is not connected.

[#422] Renewable-Assisted DC-Bus-Fed DWPT System Dual Transmitter and Dual Receiver for EV Charging Applications

JITHENDER TEKUMATLA (IIT Bhubaneswar); SHYAM NAVEEN KUMAR (IIT Bhubaneswar); Tummuru Narsa Reddy (IIT Bhubaneswar)*; Srinivas Bhaskar Karanki (IIT Bhubaneswar)

This paper proposes a PV-assisted and DC-busfedHigh Frequency Dual-input Interleaved (HFDII) converterbasedDynamic Wireless Power Transfer (DWPT) system forEV charging applications. In this system, the PV and DC busvoltages are connected in a cascaded manner using an HFDIIconverter. The dual transmitter and dual receiver (DTDR) basedarchitecture supports two independent wireless power channelsoperating in parallel, each powered by its own dual-source inputand converter utilizing LCC-LCC compensation. Each inverteris used to excite two transmitter coils, to generate high power fewer inverters and more power sources. The designed PVcurrent controller dynamically adjusts the duty ratio (di) of theactive switches in the interleaved converter stage based on thePV current feedback, ensuring MPPT operation. The voltagecontroller maintains the internal phase shift (Av) to regulate theload voltage.. The different case studies, such as excess, deficitpower modes, and DC bus voltage variations, are analyzed. TheIn-motion vehicle behaviour is simulated variation of mutualinductance M(t) in a trapezoidal manner. The effectiveness ofthe DWPT system has been validated through digital simulationstudies under different operating conditions.

Technical Track TR06

[#224] Discontinuous Space Vector PWM Strategies for Field-Oriented Control of Five-Phase Induction Motors Reduced Switching Losses

JEEBAN NAYAK (IIT Patna)*; Ranjan Behera (IIT Patna)

The increased demand for high-efficiency and fault-tolerant motor drives has led to the adoption of multiphase machines, specifically the five-phase induction motors (FPIMs), emerging as a promising solution for high-power drive applications such as electric vehicles and fully electric aircraft and ship propulsion systems. The two-level voltage source inverter (VSI)-fed FPIM space vector pulse width modulation (SVPWM)-based field-oriented control (FOC) offers precise torque and flux control, but the conventional SVPWM technique results in more switching transitions and thus the associated losses increase. There is always a trade-off between efficiency (switching loss) and line current ripple (torque ripple). However, in high-power and high-speed applications, the mechanical inertia of the FPIM rotor is very large and sufficient to damp out the torque ripple for smooth operation of the FPIM drive. This paper presents the implementation of discontinuous SVPWM (DSVPWM) strategies in the FOC framework for FPIMs to reduce switching losses out disturbing the dynamic performance of the drive. The proposed DSVPWM techniques reduce switching transitions in each modulation cycle. A comparative analysis between conventional SVPWM and DSVPWM is conducted using MATLAB/Simulink, evaluating switching losses and current ripple. In addition, the proposed technique is verified in hardware-in-the-loop mode the Opal-RT real-time simulator (OP5700). The results demonstrate that DSVPWM significantly reduces switching losses while maintaining waveform quality and dynamic response, making it suitable for high-efficiency drive applications.

[#244] A Compact 13-Level Step-Up Multilevel Inverter Employing Fuzzy Logic Switching Strategy

Eddu Karunakaran (NATIONAL INSTITUTE OF TECHNOLOGY KARNATAKA)*; Yellasiri Suresh (NATIONAL INSTITUTE OF TECHNOLOGY KARNATAKA)

Transformer-less inverters have gained significant popularity across a wide range of applications such as renewable energy systems, electric vehicles and medium-power industrial setups, owing to their reduced size, cost and enhanced conversion efficiency. In this context, this paper presents a novel thirteen level boost multilevel inverter (NTBMLI)

topology employing only 10 switches, 3 capacitors, 2 diodes, and a single DC source, aimed at achieving higher voltage levels minimal component count. The capacitors employed for voltage boost in the proposed NTBMLI consist of two fixed capacitors and one flying capacitor (FC). The operating modes of the proposed circuit has been presented along switching table. In contrast to conventional PWM techniques, the proposed NTBMLI utilizes a fuzzy logic PWM control switching rule-driven membership functions (MFs), offering enhanced adaptability and precise switching control. Further, the performance of the proposed NTBMLI is validated through MATLAB/Simulink to confirm its practical feasibility.

[#298] Design of P+Notch DC Link Voltage Controllers for PFC Rectifiers Operating under Prescribed Phase Margin and THD Constraints

Yoav Aminov (Ben-Gurion University of The Negev); Alon Kuperman (Ben-Gurion University of The Negev)*

It was recently demonstrated that employing a P+Notch control structure as DC link voltage regulator in grid-connected PFC rectifiers allows attaining nearly-undershoot free transient response to step-like load increase out sacrificing grid-side current total harmonic distortion (THD) at the expense of slight tolerable steady-state error of average DC link voltage. This paper establishes a set of four nonlinear design equations, based on DC link voltage phase margin and grid-side current THD as well as on upper and lower bounds of expected mains angular frequency values. Simulations results validate the proposed design methodology, demonstrating excellent matching corresponding analytical predictions.

[#304] Selective Harmonic Self-Elimination Curve Fitting Method for a Five-Level Single Phase Inverter Fixed DC voltage

Sayantika Mukherjee (IIT ROPAR)*; Pratik Kalkal (IIT ROPAR); A.V. Ravi Teja (IIT ROPAR)

In a five-level single-phase voltage source inverter, amodified selective harmonic elimination approach is proposed inthis paper that can eliminate nth harmonics and their multiples.To find the switching angles for a particular modulation index, acurve fitting technique is utilized to construct the polynomialequation the proper order as a function of modulationindex. The proposed method is tested in both simulation andhardware for elimination of triplen harmonics, and the results arepresented. This method shows that all of the triplen harmonicsare self-eliminated across a 43\{}% range of modulation index.

[#305] Graphical Analysis for Finding Multiple Solution Points of Selective Harmonic Elimination in a Two-Level Inverter

Sayantika Mukherjee (IIT ROPAR)*; Pratik Kalkal (IIT ROPAR); A.V. Ravi Teja (IIT ROPAR)

This paper exhibits multiple solution points at anypossible modulation index for eliminating some specific harmon-ics. Using just four switching angles per quarter, a two-levelinverter can give multiple solutions at any possible modulationindex that can eliminate up to 13th harmonic. A simple 3Dgraphical representation is obtained from a 4D graphical plot forbetter assessment. This paper shows how the suggested approach automatically detects the multiple solutions, i.e., multiple sets of switching angles at a specific Mi, to eliminate any specific number of selected harmonics out visually examining the graph. The each set of switching angles exhibits different harmonic properties. If one solution gives total harmonic distortion minimum then other solution may provide the minimum non-eliminating harmonics. The set of switching angles obtained from the graph is thoroughly verified the Hardware results. An FPGA-based prototype has been developed in the laboratory for this purpose.

[#344] Wave-shaper Based ANPC Active Front-end Converter for Medium Voltage Variable Frequency Induction Motor Drives

SASWAT PANDA (INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR)*; Suman Maiti (INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR)

An Active Neutral Point Clamped (ANPC) converter is popularly used in Medium Voltage (MV) drives application due to several reasons. However, this type of converter is best suited to generate up to three voltage levels, beyond which it possesses various limitations. In this work, a Wave-Shaper integrated ANPC (WS-ANPC) is proposed which can generate more number of voltage levels from an ANPC converter. The Wave-Shaper (WS) unit of ANPC is formed series connection of half-bridge submodules where the number of submodules decides the voltage levels at the output of the converter. Due to fundamental frequency switching of ANPC, and moderate frequency operation of WS, the effective switching frequency of WS-ANPC stays at the lower side, which eventually reduces overall losses of the converter. Compared to CHB or MMC, the component count (switching devices + passive elements) of WS-ANPC is far low, which can be portrayed as an attractive feature of this converter. Four-quadrant operation is a common requirement in drives, which is realized here by coupling a front-end converter WS-ANPC through a bi-directional dc-dc converter. Control methods have been developed for coordinated operation between front-end stage and motor side converter ensuring four-quadrant

operation. The proposed configuration is simulated in MATLAB/SIMULINK, and the results show satisfactory performance under various operating conditions, like speed/load change, four-quadrant operation, etc.

Technical Track TR07

[#90] A Motor Control Position Tracker for SPV Battery Fed PMaSynRM Regenerative Braking for LEV Applications

Sushant Kumar (Indian Institute of Technology, Bhilai)*; Dr. Shailendra Kumar (Indian Institute of Technology, Bhilai); Dr. Shashank Kurm (Indian Institute of Technology, Bhilai)

This paper presents an efficient control and energy management architecture for a light electric vehicle (LEV) powered by a hybrid energy source comprising rooftop-mounted solar photovoltaic (SPV) panels and a battery. To enhance energy utilization, a bidirectional DC-DC converter (BDDC) is employed, enabling regenerative braking and maintaining a stable DC link. The proposed drive employs a permanent magnet-assisted synchronous reluctance motor (PMaSynRM), offering a cost-effective alternative to traditional PMSMs by utilizing ferrite magnets while retaining high efficiency and torque performance. For torque control, the maximum torque per ampere (MTPA) technique is used to minimize stator current and reduce copper losses. A sensorless control scheme is adopted using an improved gain adaptive sliding mode observer (IGASMO) for robust back-EMF estimation reduced chattering. To further improve position and speed estimation accuracy, a dynamic harmonic extractor (DHE) is introduced to remove harmonics and DC offsets before processing through a synchronous reference frame phase-locked loop (SRF-PLL). The proposed control scheme is evaluated under the ECE-15 urban driving cycle, simulating realistic stop-and-go conditions. Results confirm improved dynamic performance, efficient energy recovery, and reliable position estimation, validating the system's applicability to urban LEV applications.

[#99] Performance Analysis of Pentacle Connected FPIM under Open Phase Fault for EV Applications

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This study examines the performance of a pentacle connected five phase induction machine under open phase fault conditions, an electric vehicle (EV) as the load. A five phase induction motor (FPIM) has been selected as the EV's propulsion unit, and its stator windings can be arranged in one of three configurations: star, pentagon, or pentacle. Among these, the pentacle connection offers superior fault tolerance compared to both star and pentagon topologies. In pentacle configured FPIM, the motor's stator winding is 1.9 times greater than the supply phase-neutral voltage. Its unique winding connection enables smoother torque production and better fault tolerance. Initially, the open-phase fault is formulated through a mathematical model using back-EMF approach. As the fault occurs, currents become unbalanced. To achieve a balanced system, a suitable fault tolerant control strategy is applied. Here, minimum derating method is chosen which involves re-defining the reference frame currents, \{}textbackslash\{}{\{}boldmath\{}{\{}$ \{}textbackslash\{}{\{}alpha \{}textbackslash\{}{\{}beta xy0 \{}$\{}. The drive's performance under a standard city driving cycle covering acceleration, deceleration, cruising, and idling is evaluated, and the post-fault performance substantiate the method's effectiveness. Finally, the proposed fault model and control scheme are validated through MATLAB/Simulink simulations.

[#128] Detection of Interturn Winding Faults in Single Phase Squirrel Cage Induction Motor by Stator Flux Monitoring System

Tushar Vilhekar (Visvesvaraya National Institute of Technology, Nagpur)*; Makarand Ballal (Visvesvaraya National Institute of Technology, Nagpur)

In the literature current and vibrations signature analysis based techniques are most commonly used for the detection faults in single phase induction motors. Since the last decade, for three phase induction motor flux based health monitoring techniques are more accurate and reliable as compared to classical techniques. Hence in this paper, novel stator flux monitoring technique is proposed for the detection of interturn winding fault in single phase induction motor. This paper deals detection of interturn faults in main and auxiliary windings. The proposed technique is also detects, the location of winding fault. The detection of location of interturn faults results in reduction maintenance cost and downtime required for maintenance. Simply monitoring RMS value the types of winding fault and its location is predicted. The advantages of the proposed technique are reduction in computation burden, low cost and less downtime, more accurate and reliable technique.

[#319] Four-Quadrant Control of CSI fed PMSM Drive for EV Applications Improved Efficiency

Ashish Kumar (Indian Institute of Technology, Roorkee); Apurv Kumar Yadav (Indian Institute of Technology, Roorkee)*

The advent of wide-bandgap (WBG) semiconductor devices, such as silicon carbide (SiC), has renewed interest in Current Source Inverter (CSI) topologies, particularly due to the high dv/dt stress limitations associated conventional Voltage Source Inverters (VSIs). This paper presents an H7type CSI architecture employing SiC-based four-quadrant (FQ) switches, developed for electric vehicle (EV) traction applications. To enhance system efficiency, a new pulse-width modulation (PWM) strategy is proposed that effectively minimizes switching transitions, thereby reducing overall switching losses. The inverter is supplied by a controlled DC current through a pre-stage converter, forming a complete current-source-based drive system. An Interior Permanent Magnet Synchronous Machine (IPMSM) is used as the traction motor, and the control strategy incorporates Maximum Torque Per Ampere (MTPA)-based Field Oriented Control (FOC) implemented a single current sensor. Simulation results validate the proposed approach by demonstrating the stable operation across both motoring and regenerative braking modes under various dynamic conditions improved inverter efficiency.

[#384] Comparative Analysis of Multi-spoke Lightning Rod to conventional ESE and Franklin Rod

Alok Verma (IIT Kanpur)*; Ramanuj Deb (IIT Kanpur)

Early Streamer Emission (ESE) Air Terminals (AT) evolved as one of the standard third-generation lightning protection devices in 1990 in Malaysia as an alternative to the then popular radioactive AT which the government put into a ban in 1988. Failure of the ESE-AT got attention in the mid 1990s and proper documentation was first archived in international conferences in 1995. Global communities like the International Council on Large Electric Systems (CIGRE) and the National Fire Protection Association (NFPA) acknowledged the failures of ESE-AT. Concentrated efforts are made in the direction of developing a new model of lightning rod apart from the existing conventional rods like the franklin rod, ESE-AT, the old radioactive AT etc. This paper proposes an eight-spoke lightning rod model based on simulation studies. The results show improved characteristics and enhanced protection ability that might be due to the multidirectional approach of the model.

[#411] Optimised Flux Reference-Based Speed Range Extension Scheme for BFI-CFI Fed Direct-Torque Controlled PMSM Drives

Rajeevan PP (Indian Institute of Space Science and Technology)*; Roshan Jabeen K (Indian Institute of Space Science and Technology)

This paper proposes an optimum flux reference-based speed range extension scheme for BFI (Battery Fed Inverter) - CFI (Capacitor Fed Inverter) fed direct torque controlled permanent magnet synchronous motor (PMSM) drives. The speed range extension of direct torque controlled PMSM drives is achieved by compensating the reactive power of the motor by using a capacitor-fed inverter, such that the voltage reserve obtained in the battery-fed inverter can be utilized for countering the increase in back EMF due to the increased speed demand. The capacitor of the capacitor-fed inverter gets charged out any pre-charging circuit, and it holds the voltage corresponding to the demanded speed. As this method of reactive power compensation is done for direct torque controlled PMSM drives, the flux reference fed to the controller is a key concern. Normally, the flux reference is the rated flux independent of the load. This paper mainly focuses on feeding the controller the optimum flux reference based on the load out compromising on the speed extension. By doing so, the capacitor voltage of the CFI can be optimized in the entire speed range. This paper also proposes a new online flux estimation scheme minimum computation. The work is simulated in the PLECS simulation platform, and experimental verification is carried out using TMS320F28069M digital controller.

Technical Track TR09

[#105] MMC-Based Grid-PV Integrated Electrolyser System for Green Hydrogen Production

Tanu Prasad (IIT Bhilai)*; Shashank Kurm (Indian Institute of Technology Bhilai); Shailendra Kumar (Indian Institute of Technology Bhilai)

This paper presents modelling and assessment of grid connected modular multilevel converter (MMC) coupled photovoltaic (PV) array and proton exchange membrane (PEM) electrolyser for green hydrogen production. The proposed system uses MMC because of its high scalability, low harmonic distortion and improved voltage quality. MMC is interface the variable DC output of the PV array and DC input requirement of electrolyser via grid. The multiple PV array provides power to the multiple electrolysers at different DC link enabling production of green hydrogen. MMC ensures efficient power

flow and follows grid compliance through the control method. A coordinated control method is designed to optimize power flow between the grid and electrolyser based on solar availability and grid demand. The dynamic response of the system is evaluated through simulation results, demonstrating stable operation under solar insolation change and electrolyser dynamic condition. The proposed system highlights the architecture of MMC based PV electrolyser system enabling green hydrogen production while supporting the grid stability and renewable energy integration.

[#231] Multipulse Rectifier Fed Reconfigurable Interleaved Buck Converter for Hydrogen Electrolysers

Anakapalli Vasu Deva (Indian Institute of Technology Madras); Deepak Ronanki (Indian Institute of Technology Madras)*

This paper proposes the design and analysis of a multipulse rectifier-fed multi-channel reconfigurable interleaved buck converter (IBC) for electrified hydrogen generation systems. Unlike existing topologies, this configuration utilizes a 24-pulse rectifier in the AC-DC stage to minimise grid current total harmonic distortion (THD) below 5\{}% out any control circuit and additional filters on the grid side. In the DC-DC stage, a multi-channel reconfigurable interleaved buck converter (IBC) is employed to reduce the electrolyser current ripple and enhance lifetime. Further, a new control scheme is implemented that selectively activates the required number of channels based on the power demand for efficient and reliable operation. A dual-loop control architecture comprising an inner current loop and outer voltage loop is implemented to regulate electrolyser voltage and current. The proposed system configuration, along the selective reconfiguration scheme based on hydrogen power generation, is validated using MATLAB simulations.

[#233] A New Transformer-Less Common Ground Micro Inverter for Single-Phase Load Applications

KOVVALI SRIMANNARAYANA (NIT AP)*; JAYARAM NAKKA (NIT AP); Arup Ratan Paul (IIT BOMBAY)

Transformer less micro inverters are widely adopted in distributed renewable energy PV systems for their compactness, high efficiency, and cost-effectiveness. However, they face critical challenges like common-mode leakage currents caused by parasitic capacitance and double-frequency power issues. This paper proposes a transformer-less common-ground (TLCGMI) micro inverter topology to address the above issues. The proposed topology inherently facilitates power decoupling capability itself, which is used to replace bulky electrolytic capacitors compact and thin-film capacitors. It employs only six switches, two

diodes, and ensures zero leakage current by neutral shorting. This paper comprehensively compares TLCGMI topologies across structure components and performance. Further, an extensive analysis and simulation/MATLAB results are provided for connecting a 35V input module a 220V utility load to confirm the topology's performance and viability.

[#313] An Optimization Approach for Sustainable Transformer Design in Photovoltaic Power Plants

Emir Yükselen (Çankaya University); Ebrahim Rahimpour (THWS)*

Transformers are critical in delivering electrical energy generated from renewable sources to the grid and end users. However, traditional transformer design methods are generally based on continuous maximum load conditions and often result in oversizing and increased costs. The increasing utilization of renewable energy sources has further highlighted these inefficiencies in modern energy systems, where energy derived from renewables is primarily converted into electrical energy for widespread public consumption. This study focuses on the optimization methods used in designing a transformer for photovoltaic power plants unique loading cycles to optimize transformer parameters, aiming for minimal material costs, a compact size, and improved efficiency. Additionally, an electromagnetic simulation analysis is performed to validate the design approach and ensure the accuracy of the applied methodology. The results highlight the potential of optimized transformer designs to improve operational efficiency and support the integration of renewable energy sources into smart grids.

[#314] Insights into JV and EQE performance metrics of solar cells

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The External Quantum Efficiency (EQE) or Incident Photo to Current Efficiency (IPCE) measurements complement current-voltage (J-V) analysis by identifying spectral mismatch losses, parasite absorption and recombination pathways. This study examines a pin silicon solar cell under varying chopper frequency and bias voltage during EQE/IPCE measurements. The J-V analysis, under illumination, reveals key performance parameters: Jsc of 31.05 mAcm-2, a Voc of 0.56 V, a FF of 62.20\{}%, and a PCE of 10.73\{}%. The low efficiency is attributed to an ideality factor n>2, indicating significant recombination

losses. High series resistance (Rs), likely due to poor contact, bulk resistance, and potential manufacturing defects, limits the performance. The voltage losses and reduced fill factor impact efficiency, while the extracted ideality factor offers insights into charge transport and recombination mechanism. Under forward bias, increased carrier injection enhances recombination, reducing quantum efficiency and the resultant integrated Jsc. Conversely, at -0.25 V, the best EQE is observed due to minimized recombination losses and enhanced carrier separation. Notably, chopper frequency variation has a negligible impact on quantum efficiency, indicating fast and efficient charge carrier dynamics in the device. However, the influence of bias voltage and chopper frequency may vary according to the type of device architecture and the material used. This study highlights the influence of bias voltage on EQE measurements, providing deeper insights into solar cell performance.

[#425] A Novel Active Power Control for Wind Energy System for Improved Frequency Regulation

Md Hasnain Arifin (UNC Charlotte); Md Shamim Hasan (UNC Charlotte); Sukumar Kamalasadan (University of North Carolina at Charlotte)*; Andrew Crosby (Siemens Energy); Krishna Murari (IIT Bhilai)

The increasing integration of renewable energy resources (RES) introduces new challenges for power system stability, primarily due to their intermittent nature and converter-driven dynamics. Among RES, wind energy systems (WES) represent a major share, making their effective control critical for maintaining frequency stability in low-inertia grids. This paper proposes an improved active power control (APC) strategy for WES by incorporating a simplified pitch control together modified machine-side and grid-side inverter control. The proposed controller is designed to enhance the frequency support capability of wind turbines and mitigate instability following disturbances. System performance is evaluated using key frequency stability metrics, including the rate of change of frequency (RoCoF) and frequency nadir. Simulation results demonstrate that the proposed APC improves frequency regulation by a minimum of 20 \{}% in Frequency Nadir compared to conventional WES control approaches, thereby enhancing the overall stability of RES-dominated power systems.

Technical Track TR11

[#87] Enhancing Power Handling using Cascaded Multilevel Inverter-Based SAPF PI Controller

Rupa Boddapati (Osmania University)*

This study introduces power electronic switches in equipment like computers, VFDs, and arc furnaces can cause power quality issues such as harmonic distortion, voltage fluctuations, and noise, leading to system failures, overheating, insulation damage, and communication disruptions. IEEE standards (IEEE Std 519-1992 and 2014) limit harmonic distortion to 5\{}%, necessitating power filters. Passive filters, though common, have limitations, such as size, instability, and resonance. Voltage source inverter-based active power filters (APFs) offer better dynamic response and flexibility. The effectiveness of APFs depends on harmonic detection and compensation control algorithms, often utilizing P-Q theory. Advanced control methods, including modified P-Q theory and Lyapunov control, enhance harmonic mitigation. Multi-level inverters, such as cascading H-bridge converters, provide improved voltage and power handling, reduced harmonics, and lower electromagnetic interference. This research employs a five-layer cascading H-bridge APF to validate P-Q theory and mitigate current harmonics, using MATLAB/Simulink for implementation.

[#104] Super Twisting Sliding Mode Control based PV fed QZSI for Grid Tied Applications

Divya S (CUSAT)*

The work presents a novel control-oriented implementation of a photovoltaic fed quasi Z-source inverter for grid connected applications, emphasizing enhanced dynamic performance and robustness through a Super-Twisting Sliding Mode Controller (STSMC). The qZSI topology offers distinct advantages such as inherent boost capability, improved shoot-through handling, and continuous input current, making it especially suitable for solar PV systems operating under fluctuating environmental conditions. To ensure maximum power extraction, a P \{}&O based MPPT algorithm dynamically adjusts the reference for the intermediate capacitor voltage. The integration of an STSMC based shoot through duty ratio controller, which ensures robust and chattering-free tracking of the capacitor voltage reference even in the presence of parameter uncertainties and external disturbances. The second-order sliding mode structure of the STSMC significantly reduces chattering,

offering smoother control action compared to conventional SMC approaches. Simulation results demonstrate fast voltage regulation, accurate MPPT tracking, and stable grid injection reduced total harmonic distortion, validating the superior dynamic and steady state performance of the proposed control approach.

[#152] Comparative Analysis of PLLs Under Frequency Variations in Electrified Aircraft Power Systems

Rama Naga Praneeth Suri (Collins Aerospace)*; Rushiraj Jani (Collins Aerospace); Rupan Sarkar (Collins Aerospace)

The development of any industry highly relies on electrification of it. Considering old oiled street lamps to rail roads and now to cars, the humankind has come leaps and bounds in electrifying things. The future of aviation industry also lies a lot in electrifying the aircraft totally or to most part of it. Additionally out compromising reliability, cost, weight and space are also to be reduced for the next generation aircraft. So a change from conventional transformer rectifier units to active front end converters are being proposed by researchers around the world. However, the change comes challenges that include electrical hurdles like power quality issues, bi-directional power flows, advanced control strategies, efficiency and reliable contingencies. One of the addition that needs to be done to mitigate the power quality issue is to add a phase locked loop (PLL). PLLs can provide phase angle information and voltage magnitude of the utility grid. This enables the aircraft systems to function even when there are any grid disturbances (in aircraft, an engine malfunction or disturbance). This paper investigates the use of different phase locked loops (PLL) for active front end converter fitted electrified aircraft concept. Basic PLLs are investigated for different frequency variations as per aerospace standards and detailed results are provided.

[#164] High-Performance FCS-MPC for a New Five-Level Voltage Source Inverter

Ali Azimi Bizaki (Lakehead University); Apparao Dekka (Lakehead University)*; Deepak Ronanki (Indian Institute of Technology Madras)

The performance of finite control-set model predictive control (FCS-MPC) for multilevel inverters (MLIs) greatly depends on the accuracy of mathematical models of the system. Particularly, the prediction accuracy is interlinked the accuracy of mathematical models, which further influence the harmonic performance and switching frequency of FCS-MPC method. In this paper, an enhanced FCS-MPC method is proposed for a new five-level voltage source inverter (5L-VSI). The new 5L-VSI is designed less flying capacitors (FCs)

and has enough redundancy switching states that allow flexible control of FC voltages and inverter currents. Moreover, the discrete-time (DT) models of 5L-VSI's currents and FC voltages are developed Heun's discretization method to implement the enhanced FCS-MPC. The proposed DT models reduce the prediction error, resulting in a low harmonic distortion and switching frequency. Simulation studies are presented to showcase the performance of the proposed enhanced FCS-MPC at steady state and transient scenarios. It is further compared the existing FCS-MPC method to demonstrate the benefits of the proposed method.

[#213] Indirect MPC Adaptive DC-Link Voltage Control for a CHB-based Shunt Active Power Filter

Christopher Curtis (Lakehead University); Joseph Fourcaudot (Lakehead University); Saige Niemi (Lakehead University); Darian Smith (Lakehead University); Apparao Dekka (Lakehead University)*; Deepak Ronanki (Indian Institute of Technology Madras)

The fixed dc-link voltage operation of a shunt active power filter (SAPF) produces high current ripple and affects its ability to compensate the grid harmonic currents under light loading conditions (i.e., operating at low modulation indices). In this paper, an indirect model predictive control (MPC) is proposed for a cascaded H-bridge (CHB) based SAPF to compensate the grid harmonic currents. In addition, an adaptive dc-link voltage control philosophy is proposed to adjust the net dc-link voltage depending on the harmonic current magnitude. Through the proposed method, CHB-based SAPF always operates at higher modulation indices, resulting in an improved performance and produces less current ripple, irrespective of the harmonic current magnitude. Also, the redundancy switching state selection algorithm is developed to equally distribute the net dc-link voltage between each H-bridge module in a CHB. Simulation studies are presented to study the performance of the proposed indirect MPC adaptive dc-link voltage control philosophy for a CHB-based SAPF current and voltage source type harmonic loads.

[#235] Generalized Model Predictive Control of Active Power Decoupling Circuits for Single-Phase Electric Vehicle Battery Chargers

Sri Sai Lahari Kommuri (Indian Institute of Technology Madras); Harish Karneddi (Indian Institute of Technology Madras); Preetha Philip (Indian Institute of Technology Madras); Deepak Ronanki (Indian Institute of Technology Madras)*

Single-phase battery chargers are widely used as onboard chargers for overnight electric vehicle (EV) charging, where compactness and reliability are critical. However, these chargers commonly suffer from second-order harmonic currents and low-frequency voltage

ripple, necessitating the use of large dc-link electrolytic capacitors. Although these capacitors offer high capacitance per volume, however, they are more fragile and deteriorate aging. Therefore, passive and active decoupling circuits (APDs) are needed to minimize the capacitance and improve the reliability of the charger. Although several solutions have been proposed, a comprehensive and unified approach to APDs remains limited. This paper presents a generalized model predictive control (MPC) framework for APD circuits that reduces energy storage requirements. The proposed framework is uniformly applied across multiple APD topologies, such as buck, boost, buck-boost, and DC-split capacitor. The performance of the proposed approach is applied to these APD circuits and simulated in MATLAB Simulink. The findings underscore the potential of integrated APD design and advanced control strategies to eliminate bulky passive components, paving the way for reliable and compact single-phase ac–dc converter systems.

Technical Track TR12

[#122] Stability Analysis of Smart Grids Under Dynamic Load-Altering Attacks

AnilKumar Badavath (Department of Electrical Engineering, Indian Institute of Technology Roorkee)*

This paper investigates dynamic load-altering attacks (DLAAs) in power systems, focusing on their impact on system stability through manipulating load dynamics. We derive a state-space model that integrates power flow equations, generator dynamics, and attack mechanisms modeled as proportional controllers using grid frequency feedback. The analysis reveals how attack parameters can alter system poles, potentially destabilizing the power system. Additionally, we perform structural and graph-theoretic vulnerability analyses, simulate DLAAs on a 9-bus system a defense strategy, and evaluate their effects on frequency and load stability. The simulation includes an initial 1-second normal operation phase to establish a baseline, followed by the attack phase. The results highlight the vulnerability of power systems to such cyberattacks, the effectiveness of mitigation strategies, and the importance of topological analysis in identifying critical buses.

[#131] Cyber Modelling of Advanced Metering Infrastructure using ns-3

Issa Mlimbila (IIT Roorkee); Amit Kumar (IIT Roorkee)*; Himanshu Jain (IIT Roorkee)

Digital communication is an integral component of smart grids. However, while physical flow of electricity has been modelled for decades, it is difficult to find instances of simulating the communications network for smart grid planning and operations. This paper addresses this gap by presenting an approach to simulate advanced metering infrastructure (AMI) communication model, firewalls and cyberattacks in ns-3. The performance of the developed model is evaluated on several IEEE 13, 34, 37, and 123 node feeder models. Based on the simulations performed using these models it was observed that propagation delay and system loss are key parameters that influence the performance of the AMI system. Results for the impact of cyberattacks and out the consideration of firewalls show significant differences in the time taken for data to be received at the desired locations. Further, the wall-clock time required to simulate the IEEE 123-node feeder is around 19 seconds, compared to 3 seconds it takes to simulate the IEEE 13-node feeder. Although the amount of data exchanged increased 11 folds, the time taken increased only 6 times, demonstrating the scalability potential of ns-3 for modelling AMI systems.

[#159] A Flexible and Robust Framework for Intrusion Detection in DER Cyber-Physical Security

Suresh Mogilicharla (Indian Institute of Technology Roorkee)*; Manoj Tripathy (Indian Institute of Technology Roorkee); Mital Kanabar (GE Grid Solutions)

The rapid integration of distributed energy resources (DERs) into modern power systems has revolutionized energy generation and distribution, giving rise to a highly dynamic and interconnected cyber-physical ecosystem. However, this evolution has also expanded the attack surface, making DERs increasingly vulnerable to sophisticated cyber-physical threats. Traditional intrusion detection systems (IDSs) primarily focus on algorithmic performance, often neglecting the architectural flexibility required to address the heterogeneity and flexibility demands of DER environments. This work proposes a flexible and robust intrusion detection framework specifically tailored for DER cyber-physical security. The framework integrates both cyber and physical data streams, enabling comprehensive anomaly detection and enhanced situational awareness. It adopts a defense-in-depth strategy to strengthen resilience and employs a microservices-based design using open-source tools to ensure flexibility and ease of integration existing energy infrastructure. The proposed framework is deployed on a Raspberry Pi and validated in a microgrid lab environment. Evaluation results demonstrate its effectiveness in identifying diverse cyber-physical attack scenarios, while highlighting the benefits of architectural adaptability and operational robustness.

[#294] Cyber-Physical Security Testbed for Wide Area Damping Control utilizing IEEE C37.118 Protocol: Assessing Denial-of-Service Attack Impact

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Wide-area measurement systems (WAMS) play a crucial role in enhancing power grid situational awareness. WAMS facilitates the transmission of critical measurements and control signals for wide-area damping controllers (WADC). However, significant vulnerability exists in communication networks which exposes the system to malicious cyberattacks such as false data injection (FDI), denial-of-service (DoS) attacks, and signal spoofing attacks. This paper presents a comprehensive real-time cyber-physical testbed for WADC, designed to emulate realistic operational environments and investigate the impact of cyber attacks. The proposed testbed adheres to IEEE C37.118 protocol standards for synchrophasor communication and integrates industry-standard hardware components, including the OPAL-RT real-time simulator, phasor measurement unit (PMU), real-time automation controller (RTAC), global positioning system (GPS) synchronization clock, and network switches. Software components such as network simulator-3 (NS-3) and open phasor data concentrator (OpenPDC) are also incorporated. The impact of the DoS attack on critical applications like WADC system performance is analyzed. The proposed testbed serves as a valuable research platform for comprehensive vulnerability assessments and cyberattack evaluations in the wide-area monitoring, protection, and control (WAMPAC) applications. It provides a robust environment to investigate potential threats, develop effective detection mechanisms, and explore defense strategies against a diverse range of cyberattacks.

[#308] A New Hybrid Model-Driven Machine Learning Approach for Cyber Attack Detection in Autonomous E-Mobility Systems

Praneeth Reddy Pesala (Indian Institute of Technology Madras); Durga Prasad Pilli (Indian Institute of Technology Madras); Deepak Ronanki (Indian Institute of Technology Madras)*; Apparao Dekka (Lakehead University)

Autonomous e-mobility systems have advanced functionalities such as remote monitoring and control, self-drivingcapability, vehicle-to-vehicle (V2V), and vehicle-to-infrastructure(V2I) communication. These features enhance the operational autonomy, efficiency, and responsiveness of electric vehicles (EVs).However, these capabilities also introduce cyber security

vulnerabilities, particularly false data injection (FDI) attacks that cancompromise the integrity and reliability of critical componentssuch as the electric drive system (EDS). This paper proposes ahybrid cyberattack detection framework that combines model-derived features a machine learning (ML) classifier todetect FDI-type data integrity attacks in the EDS. The proposedmethod is validated through MATLAB-Simulink under various conditions. FDI-type attack scenarios. Furthermore, the key performance characteristics of the proposed method are comprehensively analyzed and demonstrated. The results show that the proposed approach achieves high detection accuracy and robustness, thereby enhancing the cybersecurity resilience of autonomous e-mobility systems

[#424] Impact of BESS Placement on Fault Current Contribution in Bulk Power Systems

Md Shamim Hasan (UNC Charlotte); Md Hasnain Arifin (UNC Charlotte); Sukumar Kamalasadan (University of North Carolina at Charlotte)*; Michael Smith (UNC Charlotte); Andrew Crosby (Siemens Energy); Krishna Murari (IIT Bhilai)

The growing penetration of renewable energy sources (RESs) such as solar PV and wind in bulk power systems is accelerating the adoption of Battery Energy Storage Systems (BESS) to mitigate intermittency and support grid stability. As inverter-based resources (IBRs), BESS units provide fast active and reactive power support but typically exhibit low short-circuit current (SCC) contribution during faults, posing challenges for system stability and protection sensitivity. This paper presents a detailed analysis of BESS SCC contribution under both Grid-Following (GFL) and Grid-Forming (GFM) control architectures, emphasis on the influence of connection topology in high-voltage transmission networks. Simulation studies reveal that BESS placement and connection method significantly impact SCC magnitude. In particular, connecting the BESS via a dedicated transformer at strong grid nodes can increase the peak fault current contribution by up to \{}$1.5\{}$ times compared to connecting via a common transformer, while GFM control can further enhance SCC capability relative to GFL operation. The results provide practical guidelines for location and mode-dependent BESS deployment to improve bulk system fault response and ensure reliable protection coordination.

Technical Track TR13

[#106] Dynamic performance analysis of BESS in grid disturbance scenarios

SOMA SEKHAR POLA (cpri)*

Abstract: This paper investigates the dynamic response and stabilizing capabilities of Battery Energy Storage Systems (BESS) in a power grid under various disturbance scenarios using PSS®E simulations. Four cases are examined: sudden loss of a large hydro generator, abrupt load shedding, a three-phase fault, and generator loss under different BESS droop settings. The study demonstrates that BESS plays a critical role in maintaining grid stability by supporting both frequency and voltage regulation. In the case of generation loss, BESS discharges active power to limit frequency decline, while during load shedding, it absorbs power to mitigate over-frequency conditions. Under fault conditions, BESS helps reduce rotor angle oscillations and enhances voltage recovery. Additionally, the impact of droop settings on frequency control performance is analysed, highlighting the need for optimal parameter tuning. The results confirm that BESS significantly enhances system resilience and dynamic performance, making it a valuable asset in modern power systems high renewable penetration. Keywords—Battery energy storage system, Droop control, Voltage and frequency disturbances

[#200] A Study of Cybersecurity and Resilience of Dual Active Bridge Converters in Electric Vehicle Fast Charging Systems

V S R Varaprasad Oruganti (Ontario Tech University)*; Kushan Tharuka Lulbadda (Ontario Tech University); Tarlochan Sidhu (Ontario Tech University); Sheldon Williamson (Ontario Tech University)

The rapid adoption of electric vehicles (EVs) and the growing need for high-power EV fast charging infrastructure have led to the widespread use of advanced power electronic topologies, notably the Dual Active Bridge (DAB) converter. While DAB-based DC fast chargers offer efficient bidirectional power flow and galvanic isolation, their integration into cyber-physical systems introduces new cybersecurity vulnerabilities. Existing research primarily focuses on communication protocols (e.g., OCPP, ISO 15118), backend networks, or electric vehicle supply equipment (EVSE) firmware security, often overlooking the converter level where actual power processing occurs as a critical attack surface. This study investigates recent literature on cyber threats specific to DAB-based fast charging systems and reviews resilience strategies that enhance system robustness. Emphasis

is placed on the unique attack surfaces of the DAB architecture, potential impacts on the control loop and power flow, and the integration of real-time monitoring, secure communication protocols, and anomaly detection systems. The objective is to provide the EV charging and smart grid research communities an understanding of emerging threats at the converter and its controller level and to propose recommendations for designing cyber-resilient EV fast charging systems. The paper concludes by outlining key research gaps and offering directions for future work to support the development of secure and reliable DAB-based EV fast charging infrastructure.

[#252] Modal Analysis of Offshore Wind Farms Integrated via Hybrid HVDC–MVDC Transmission Systems

Priya Singh (Victoria University of Wellington)*; Ramesh Rayudu (Victoria University of Wellington); Praveen Kumar (Oak Ridge National Laboratory)

Extensive simulation tools and numerous studies have deepened the understanding of inter-area oscillations in power systems, revealing their underlying sources and guiding the development of effective damping strategies. However, the integration of offshore wind farms using hybrid transmission technologies, specifically HVDC and MVDC-based systems introduces new dynamic challenges. This paper investigates the dynamic impact of integrating two distinct offshore wind farms, each employing different transmission technologies, into an onshore transmission network. The study explores the oscillatory modes introduced by the interaction between these systems, which differ from traditional power system oscillations. A detailed modal analysis is conducted to classify these modes based on frequency ranges, eigenvalue analysis used to identify critical modes and assess their influence on grid stability. Furthermore, participation analysis is employed to understand the contribution of each system component to the oscillatory modes, helping to identify the key sources of instability. The findings highlight the importance of understanding these system-specific oscillations and emphasize the need for effective control strategies to ensure the stable integration of offshore wind farms hybrid transmission technologies into the grid.

[#263] Single Loop dq control scheme based islanded distributed micro-grid system

Vaishnavvignesh G Iyer (IIT Guwahati)*; Cilaveni Satish Chandra (IIT Guwahati); Ravindranath Adda (IIT Guwahati); Sreenath J G (IIT Guwahati); Praveen Tripathy (IIT Guwahati)

Distributed microgrids have become popular afterthe emergence of cost-effective PV/battery-

based inverter systems. Multiple inverters connected in parallel to supply localloads have become a common occurrence. A major task in suchsystems is to maintain the load sharing between the parallelconnected inverter systems. Droop control has been presented in literature to ensure proportional sharing of the loads between the parallel inverter systems. In islanded distributed systems, specifically, the inner loop becomes complicated due to the cascaded control structure required for LC filtered systems. This article aims to simplify this by implementing droop control single-loop dq control scheme. The proposed droop control structure is tested and validated through simulation in an islanded mode two-inverter distributed microgrid system. Further, the proposed scheme is tested against the cascade dq-based droop control to test their dynamic performance.

[#269] Decentralized Virtual Inertia-Damping Supervisory Secondary Control for DC Microgrids

Abualkasim Bakeer (Tallinn University of Technology); Andrii Chub (Tallinn University of Technology)*

DC microgrids have recently gained significant attention due to their efficiency, ease of power flow management, and seamless integration renewable energy sources (RESs) compared to traditional AC microgrids. However, their low inherent inertia poses stability challenges under disturbances such as load changes or generation variability, since they depend mainly on power electronics converters. This paper proposes a decentralized virtual inertia-damping control (DVIDC) to enhance the stability of DC microgrids. The case-study DC microgrid is an islanded grid, which consists of two battery energy storage systems (BESSs), each equipped the proposed DVIDC control, operating alongside photovoltaic (PV) generation and DC loads. A droop control mechanism is integrated into each DVIDC unit to enable power sharing among distributed energy storage units. To eliminate steady-state error on the main DC bus voltage, a supervised secondary (SS) controller based on integral action is added to the BESS control loop. Using PLECS software, the proposed approach demonstrates reduction in the DC bus voltage deviations by more than 37\{}textbackslash\{}{\{}}\{}% and eliminates the steady-state error under various disturbance conditions compared to the conventional droop control implementation.

[#271] Adaptive Damping Control for Mitigating Sub-Synchronous Resonance in Wind Energy Systems Under Variable Operating Conditions

Jawaharlal Bhukya (VNIT)*; Rishav Dutta (Visvesvaraya National Institute of Technology (VNIT))

The integration of Doubly-Fed Induction Generator-based wind power plants (DFIG-WPPs) into series-compensated transmission networks introduces Sub-Synchronous Resonance (SSR), threatening system stability under varying wind and grid conditions. This paper proposes a PI-based supplementary damping controller embedded in the inner loop of the Rotor Side Converter (RSC) and the outer loop of the Grid Side Converter (GSC). The control allocation leverages the RSC's high torque controllability and the GSC's enhanced observability of voltage dynamics, enabling fast and robust mitigation of sub-synchronous oscillations. A modified IEEE First Benchmark Model is employed to assess controller performance under stepwise series compensation changes (5\{}%, 20\{}%, 60\{}%) across multiple wind speeds. Simulation results demonstrate effective damping of SSR-induced electromagnetic torque oscillations, validating the controller's efficacy under diverse operating conditions. The proposed strategy offers a practical and converter-level solution for improving the resilience of DFIG-WPPs in weak or dynamically evolving grid environments.

Oral Session OP3

Technical Track TR02

[#40] Square-Wave-Fed Hybrid Symmetrical Cockcroft-Walton Voltage Multiplier for High-Voltage DC Power Supplies

Sara Baldisserri (University of Bologna); riccardo mandrioli (University of Bologna)*; Lohith Kumar Pittala (Universita di Bologna); Gabriele Neretti (University of Bologna); Vincenzo Cirimele (University of Bologna); Mattia Ricco (University of Bologna); Andrea Cristofolini (University of Bologna)

Voltage multiplier (VM) circuits are widely used in high-voltage DC power supplies. Traditional VMs are typically fed by sinusoidal voltages delivered by a resonant converter coupled a high-voltage step-up transformer. This paper explores a transformerless solution, achieved by feeding the VM directly the inverter's square wave output voltage. Although requiring additional multiplier stages to get the same output voltage level, this configuration simplifies the implementation and allows the elimination of the first smoothing capacitor, making it potentially suitable for size- and weight-sensitive applications. The performance and feasibility of the square-wave-fed hybrid VM are evaluated, and it is demonstrated that lower output voltage ripple and drop and better system dynamic response are achieved this approach.

[#289] Active Voltage Balancing Strategy for a Three-Level Flying Capacitor Boost Converter

Cilaveni Satish Chandra (IIT Guwahati)*; Vaishnavvignesh G Iyer (IIT Guwahati); Ravindranath Adda (IIT Guwahati); Praveen Tripathy (IIT Guwahati); Sreenath J G (IIT Guwahati)

Multilevel flying capacitor (FC) DC-DC converters have benefits such as reduced inductor size, lower voltage stress, and high power density. However, incorporating a FC poses the challenge of maintaining the FC voltage at the desired value. Therefore, additional methods such as passive and active balancing, must be employed to maintain the FC voltage as desired. Passive balancing methods rely on external components to equalize the FC voltage, whereas in active balancing control based techniques are used to maintain the FC voltage. In this paper, a three-level FC boost converter (3-L FCBC) is operated in open-loop to verify the working and it is observed that the FC voltage is not maintained at half of the output capacitor voltage. Next, the active flying capacitor voltage balancing (AFCVB) strategy is implemented to maintain the FC voltage at half of the value of output capacitor voltage. Finally, a lab-scale prototype of the 3-L FCBC is developed and experimental results in open-loop and AFCVB are presented.

[#293] A 1.5 µW Quiescent Power, 85% Peak Efficiency, Digitally Controlled Energy Subsystem for Powering the IoT Node

Siddhant Nagar (IIT Dharwad)*; Saroj Mondal (IIT Dharwad)

This paper presents an energy-efficient design ofa microwatt (µW) RF energy harvesting system that scavengesenergy from nearby GSM and/or WiFi bands and operates ina wide range of available input power, from -24 to 0 dBm. Thesystem comprises a primary and/or a secondary DC-DC con-verter(s) to ensure MPPT, power the loads, and charge/dischargethe supercapacitor when there is a surplus of energy. To ensuremaximum power transfer efficiency, the switching converter'sactive and passive components are sized optimally, employ adigital control logic for maximum power point tracking, andensure zero-current switching. The design has been implementedin 0.18 µm CMOS technology node that operates for a wide rangeof input voltages from 0.2 to 0.6 V and supplies regulated outputvoltages of 1.1 and 0.9 V for the available input power rangebetween 7-30 µW. The peak efficiency of the system is 85\{}%, andthe control circuitry consumes 1.5 µW of quiescent power.

[#320] An Active Current Source Gate Driver-Based Voltage Balancing of Series-Connected Devices

Siddhartha Suyal (Indian Institute of Technology, Roorkee); Apurv Kumar Yadav (Indian Institute of Technology, Roorkee)*

WBG devices show superior performance and better loss profile compared to Si counterparts; however, their lower voltage blocking capability limits their use in high-voltage applications. Multiple devices can be connected in series to overcome this challenge, but that results in unequal voltage distribution even among well-matched devices. This paper proposes a current injection technique for a current-based gate driver (CSD) that reduces the steady-state voltage unbalance of the series-connected devices. The proposed technique uses an external capacitor connected across the drain of the device to inject an appropriate additional amount of current into the gate, depending on the slew rate experienced by the device. The simulation technique was tested in LTSpice using the GaN device model for GS61008 and the SiC device model for C2M0040120D, using a DC link of 100V and 500V, respectively. The results obtained were found to be in accordance the theoretical analysis.

[#389] Z-Source Inverter a Floating Capacitor Cascade for Enhanced Performance and Single Supply Operation

Remya K P (Govt. Engineering College,Thrissur, India affilaited to APJ Abdul Kalam Technological University)*; Dr Jaison Mathew (GEC Thrissur); Vasuda K V (Government Engineering College Thrissur)

This paper presents a single source multilevelinverter topology based on conventional two-level Z-SourceInverter (ZSI) and floating capacitor cascaded inverter.The topology is particularly suited for loads like openend winding induction motor where the end connectionsof the load are available. By cascading the ZSI the floating capacitor configuration, the topology attainsvoltage boosting capability, thereby reducing the sourcevoltage requirement and improving the cost effectiveness.The floating capacitor is charge balanced at a voltage of0.5Vi using redundant switching states, where Vi is the peakmagnitude of the dc rail voltage of the Z-source inverter.The topology can generate four-level effective pole voltageswhich helps to improve the power quality. The paperalso analyzes the operation of the proposed topology foroperation at high and low power factors. It is observed thatunder low power factor operation, if the inductance in theimpedance network is low, discontinuous conduction mode(DCM) sets in leading to poor performance. This issue canbe mitigated either by using an active switch in series the source to ensure continuous conduction or by employinga high-value inductance in the impedance network. Theopera-

tion of the proposed system is analyzed using a 2.2 kWRL load in MATLAB/Simulink. Applications like electricvehicles and ship propulsion systems are expected to benefitfrom this topology.

[#403] A Reconfigurable Series / Parallel IGBT-Based DC Solid-State Circuit Breaker

Prasanth Pithanisetty (Indian Institute of Technology Dharwad); Daniel Dsa (Indian Institute of Technology Dharwad)*; Moein Ghadrdan (Institute for Technical Physics, Karlsruhe Institute of Technology); Satish Naik Banavath (Indian Institute of Technology Dharwad); Giovanni De Carne (Institute for Technical Physics, Karlsruhe Institute of Technology); Edivan Laercio Carvalho (Tallinn University of Technology)

Modern Direct-Current (DC) power systems operate increased load demand as well as high system voltage. Ultrafast and reliable fault isolation is a necessity in DC systems due to the absence of natural current zero-crossing points and the rapid rise of fault current. This paper presents a modular reconfigurable DC Solid-State Circuit Breaker (SSCB) using two IGBT switches that can operate either in series, offering high voltage blocking capability, or in parallel for high current carrying capability. The key challenges in such a reconfigurable SSCB, such as unequal static and dynamic voltage balancing in series connection of switches and current sharing imbalance in parallel operation, are discussed. A hardware prototype of the reconfigurable SSCB rated at 400V, 25A is experimentally validated against overcurrent events, considering the series as well as parallel connection of IGBTs.

[#404] Current Stress Minimization of DAB using Particle Swarm Optimization

Rajarshi Saha (IIT Guwahati)*; Shabari Nath (IIT Guwahati)

Dual active bridge (DAB) converter is a DC-DC converter which provides galvanic isolation and bidirectional power flow. Power transmission is controlled in DAB based on the phase-shifted modulation strategies. Dual phase-shift (DPS) is a phase-shifted modulation technique where both the internal (phase-shift between either-side switches in DAB) and external phase-shifts (phase-shift between the primary and secondary winding voltages of DAB) are given in DAB. So, DPS has two-degree of freedom of power flow, and it has good capability of power flow control. Multiple combinations of phase-shift ratios are possible for a given power transfer in DAB. Due to many combinations of phase-shift ratios, different values of transformer peak current in DAB are possible. High peak current increases the switching stress and affects the system's efficiency. So, minimization of the

peak current is necessary in DAB. In the existing literature, minimization of peak current considering all the possible conditions of phase-shift ratios is not performed. This paper identifies all the transformer peak current conditions and minimizes the transformer peak current using particle swarm optimization (PSO) in all conditions of phase-shift ratios in DPS of DAB. The values of optimized peak current and the corresponding optimized conditions are verified, and comparisons of the optimized peak current other peak currents for different combinations of phase-shift ratios are done in MATLAB/Simulink.

[#406] Active SoC Equalization for Multi-Module Li-ion Battery Packs Using Bidirectional Flyback and Buck-Boost Converters

Kiran Sai Dasari (Indian Institute of Science, Bangalore)*; Subba Reddy Bassapa (Indian Institute of Science, Bangalore); Subba Rao Mopidevi (Vignan's Foundation for Science Technology and Research, Guntur)

The adoption of electric vehicles (EVs), renewable energy systems, and grid-scale storage has underscored the critical role of lithium-ion battery management and the need for effective cell balancing. This study aims to develop a hybrid, scalable active balancing architecture that ensures uniform State of Charge (SoC) across both cells and modules in large battery packs, enhancing efficiency, safety, and lifespan. The proposed approach integrates Bidirectional Flyback Converters (BFCs) in an Input-Series Output-Parallel (ISOP) configuration for intra-module cell-to-cell (C2C) balancing, and a Bidirectional Buck-Boost converter (BBC) for inter-module (M2M) balancing, coordinated through average SoC-based control algorithm. Simulation models of 7S1P, 14S1P, and 14S2P battery configurations were developed to assess system performance under varying SoC imbalance scenarios. Results demonstrate the system's effectiveness: for a 7S1P module, a 25\{}% initial SoC deviation was reduced to less than 1\{}% in 85 seconds; in a 14S1P setup, module-level SoC differences of up to 14.9\{}% were equalized in 70 seconds. Furthermore, for a 14S2P battery pack, simultaneous intra- and inter-module balancing achieved convergence in 68–70 seconds, maintaining uniformity across 28 cells under varied imbalance conditions. The architecture enables rapid convergence, low control complexity, bidirectional energy flow, and modular scalability, making it a strong candidate for integration into next-generation Battery Management Systems in electric mobility and energy storage applications.

[#408] Comparative Study on Thermal Aspects of Boost and Quadratic Boost Converter Using PLECS

Sumukh Surya (Bosch); Gopal Krishna (NITK)*; Venkatesha Perumal (NITK)

Due to the evolution of electric vehicles (EVs), DC – DC converters have gained utmost importance. Ensuring efficient power transfer by maintaining optimum junction temperature (Tj) in MOSFETs is extremely crucial. In this paper, thermal performances of ideal boost (BC) and quadratic boost converters (QBC) are compared by estimating the Tj of MOSFETs for room (RT) and high temperatures (HT) operating conditions and simulated using PLECS software. The converters are designed to operate in Continous Conduction Mode (CCM), simulated using appropriate step size and a first order thermal model is built in PLECS software to estimate the MOSFET heat sink temperature for the converters. These converters are designed for low power rating, and it is observed that the heat sink temperature for QBC is always higher by at least 5°C in all operating conditions.

[#421] Modeling and Control of a Five-Port DC-DC Converter for Integrating Renewable Energy Sources

Ravi Ranjan (IIT Bhubaneswar); Pragya Nand Singh (IIT Bhubaneswar); Srinivas Karanki (IIT Bhubaneswar)*

This article presents a novel five-port converter(5PC) architecture that integrates multiple energy sources usinga minimal number of components. A comprehensive mathematical model and control strategy are developed for one of the operating modes in which the biomass source is inactive. The 5PC interfaces solar PV, wind energy, a battery storage system (BSS), and biomass, while delivering a regulated output to the load. Maximum power point tracking (MPPT) is implemented for both the solar and wind sources to maximize energy extraction. In addition, the 5PC maintains a constant output voltage of 380 V. The converter is capable of adapting to varying energy source availability through mode-based operation.

[#423] Constant Output Power LED Driver for Line Voltage Variation High Power Factor

Manjunath Hegde (IIT Dharwad)*; Aditi Sardesai (IIT Dharwad); Nagaveni S (IIT Dharwad)

Light-emitting diodes (LEDs) have emerged as the preferred technology for modern lighting applications due to their high luminous efficacy, long operational lifetime, and environmental benefits. However, the performance and reliability of LED systems are critically dependent on their driver circuits. Conventional AC–DC drivers generally employ bulky inductors and electrolytic capacitors for current regulation and power factor correction, which not only increase system cost and size but also compromise long-term

reliability owing to component degradation. To overcome these limitations, this work presents an inductor-less AC LED driver architecture that ensures constant power delivery improved efficiency and power quality. The proposed design utilizes a diode bridge rectifier to generate pulsating DC, directly driving the LED load and associated driver circuitry, voltage references are produced by a dynamic voltage reference generator circuit and these voltage references regulates the current across five sequential LED strings through MOSFET op-amp control, enabling stage-wise activation even under varying input conditions. Simulation and prototype evaluation demonstrate \{}4̃ W output power an efficiency exceeding 87\{}% and a power factor of 0.98, validating the design as a compact, cost-effective, and reliable solution for solid-state lighting.

[#435] Development and Realization of a Compact, Rugged and Highly Efficient Power Supply Unit for non-linear Pulsating Loads in High Power Radar applications

NANDAKUMAR S (DRDO)*; Aabir S Wani (DRDO); Dr Vijay Hiralal Bhosale (DRDO); Suchith Rajagopal (DRDO)

Operation of a RADAR system requires DC powerranging from as low as few kWs to the order of MWs to providebias supply to a large number of Transmit Receive Modules(TRMs) for its desired operation. Since, the readily availablepower is usually AC (available via State Electricity Boards orDiesel Generator Sets), it needs to be converted to DC at desiredvoltage level via power electronic converters maintaining powerquality parameters for reliable operation. This is accomplished byPower Supply Unit (PSU) which is based on DC-DC convertertechnology fed by a diode based rectifier source at its inputterminals. The developmental stages of this PSU along various challenges faced during realization and testing stagesis presented in this paper. This PSU is modular in nature toallow easy scaling as per load power requirements. This PSUalso offers various functional and protection features. It is apart of transportable system; thus complies various MIL-STDs and can be operated in a wide range of temperatureand altitude as per JSS55555 specifications. This PSU has beenrealized through industry partner, qualified as per MIL-STDsand has been successfully tested for its desired performance.

Technical Track TR04

[#177] Moving Discretized Control Set Model Predictive Control of Single-Stage GaN-Based Onboard Electric Vehicle Battery Charger

Rajasekhar Attada (IIT Madras); Deepak Ronanki (Indian Institute of Technology Madras)*; Rajesh Sura (Dynolt Technologies Private Limited)

This paper proposes a single-stage onboard charger (OBC) that consists integration of a dual-active-bridge (DAB) a direct matrix converter, using monolithic GaN-based devices for electric vehicle battery charging applications. So far, in single-stage AC to DC converter using a direct-matrix converter has been realized a common-source back-to-back MOSFET configuration. This configuration results in parasitics and highconduction loss due to the presence of two series-connecteddevices. In contrast, the monolithic GaN-based bidirectionalswitch offers superior performance, having a high breakdownvoltage capability and reduced conduction losses, enabling ahigher power density for OBCs. In addition, a moving discretizedcontrol set model predictive control (MDCS-MPC) is used toaccomplish the desired load regulation and maintain unity powerfactor (UPF) at the grid interface. To validate the proposedcontrol method, a 3.3 kW single-stage OBC is simulated on the PLECS platform, and constant-current (CC) and constant-voltage (CV) charging control techniques are implemented usingan MDCS-MPC

[#179] A Multifunctional LLC Resonant-Tank Based Topology for Compact and Efficient Integration of OBC and APM Architectures in Electric Vehicles

Tarun Panwar (Indian Institute of Technology, Mandi)*; Ashutosh Rai (Indian Institute of Technology, Mandi); Venkata Ratnam Vakacharla (Indian Institute of Technology, Mandi)

A dual-output, unified, multifunctional charger topology utilizing a three-winding transformer an LLC resonant tank is proposed, which integrates an on-board charger (OBC) the auxiliary power module (APM). Traditionally, these two components have been designed and implemented separately, which not only adds to the overall cost, weight, and complexity of the vehicle's power electronic systems but also results in under-utilization of space and components. The proposed topology used eight switches, a three-winding transformer, and LLC resonant tank for achieving simultaneous charging of HV and LV batteries. Additionally, the integration is tailored to charge LV battery via HV battery as and when required. The performance of the suggested converter is examined using MATLAB simulations, showing effective voltage control between 200–350 V for the 2

kW HV battery port and 38–60 V for the 0.5 kW LV battery port. The results validate the functionality, flexibility, and practicality of the integrated charger in a unified framework.

[#180] Current-Fed Converter based CLC Compensated Parallel Transmitters for Inductive Wireless Charging of Multiple Electric Vehicles

Rajat Kumar (IIT Mandi)*; Abhishek Singhal (IIT Mandi); Venkata Ratanam Vakacharla (IIT Mandi)

This paper presents a current-fed converter based inductive wireless power transfer (IWPT) system using a CLCcompensated parallel transmitter array for simultaneous and independent charging of multiple electric vehicles (EVs). The system is powered by a single full-bridge current source inverter and a shared common primary CLC compensation network, reducing component count and improving scalability. CLC compensation provides a constant output voltage across the multiple ports, which remains unaffected by the number of connected electric vehicles (EVs), as well as variations in coupling or the load resistance of the EVs. Each transmitter coil is connected in series a capacitor and switch, enabling selective activation based on EV presence while preserving resonance. This allows flexible isolation of particular transmitter coil not paired a receiving coil using relay switches, while other coils continue supplying power to their respective EVs out interruption. The proposed design achieves constant output current and zero input phase angle at resonance, independent of load variation, coil coupling, or number of connected EVs. Simulation results for a threechannel configuration demonstrate that output voltage and power delivery remain unaffected by other channels, confirming full electrical decoupling. This proposed architecture is wellsuited for dynamic and scalable EV charging scenarios such as parking lots and public charging stations.

[#183] SiC Device Based Four Phase Interleaved Bidirectional DC-DC Converter Control Wide Voltage Range Operation for EV Drive Validation

Murali Manohar S R (CDAC)*; Arun Raj Kumar K P (CDAC); Sreedevi M L (CDAC); Jinuraj K G (CDAC); Renji V Chacko (CDAC)

The need for advanced test platforms capable of evaluating complete EV systems, drivetrain components, and electric vehicle supply equipment (EVSE) is growing as a result of the rapid developments in electric vehicle (EV) and EV subsystem. Traditional testing methods involving physical energy storage systems are often expensive and time-consuming. This paper presents the design, control, hardware implementation, and testing

of a wide voltage range, four-phase interleaved bidirectional DC-DC converter based on silicon carbide (SiC) devices for use in a battery emulator platform. The emulator system, comprising an Active Front End (AFE), Dual Active Bridge, and Bidirectional DC-DC Converter (BDC), enables programmable power delivery for EV subsystem testing. The BDC multi-phase interleaved topology offers benefits including high power density, reduced current ripple, improved thermal distribution and scalability. Integration of SiC devices enhances switching performance, efficiency, and thermal capability, supporting high dynamic load conditions encountered in EV operations. The converter supports bidirectional power flow for both buck (discharge) and boost (charge) modes, enabling realistic simulation of EV scenarios of acceleration, cruising and regenerative braking. This system facilitates cost-effective, high-fidelity testing and validation of EV subsystems under real-world driving profiles. For the bidirectional converter, three nominal voltage levels - 48 V, 72 V, 360 V - are chosen as output voltage requirements in accordance battery voltage levels of two-, three- and four-wheeler electric vehicles respectively. This converter is designed for a power capability of 30kW. Synchronous buck and synchronous boost mode operations are tested power level of 16 kW and current capacity up to 100A.

[#253] A Seven Level Inverter Topology Reduced Switch Count for Electric Vehicle Applications

Hari Krishna Bhukya (National Institute of Technology Karnataka); Dr Dharavath Kishan (NIT K)*; UdayKumar R Yaragatti (National Institute of Technology Karnataka)

In recent years, substantial progress has been made in electric vehicle (EV) technology, necessitating the development of effective and efficient solutions in the field of power electronics. The development of multilevel inverters is the viable solution for electric vehicle power systems, lower harmonic distortions, enhanced quality power , and the ability to transfer power in both directions. The present research is a revolutionary multisource multilevel inverter (MLI) fewer switches. Suggested multilevel inverter utilizes two asymmetric battery sources. This distinctive arrangement facilitates the production of a more refined seven-level AC voltage output quality is improved decreased harmonic content, addressing the rigorous power quality necessity of the electric vehicle requirement. This research provides comprehensive simulation outputs derived from MATLAB-Simulink, regarding the suggested MLI's performance under varied operational situations and load circumstances. The simulation results involve examining the output voltage waveforms and conducting a harmonic analysis.

[#262] Design of an Optimized Two-Leg DC–DC Dual-Mode Converter for Wireless EV Charging

Lalitha Pai B (National Institute of Technology Karnataka); Dr Dharavath Kishan (NIT K)*; Kalpana R (National Institute of Technology Karnataka)

Wireless power transfer (WPT) for electric vehicle (EV) battery charging has gaining more attention due to its convenience, safe, and flexible against environmental hazards. Resonant inductive power transfer (RIPT) based wireless charging is the most common method for EV battery charging applications. The current EV models are designed 800 or 400V battery pack voltages. This paper proposes a Dual–Mode resonant DC–DC Converter for RIPT system. The proposed converter can be operated as a Voltage Doubler for 800V battery packs or a Current Doubler for 400V packs. This converter topology employs series/series resonant networks on both sides of the RIPT system. To check the compatibility of the proposed converter, the MATLAB/Simulink simulations are carried out. The simulation results demonstrate the effective power delivery at a 7.2 kW load an peak efficiency of 98.05\{}% for both modes. The proposed topology can further be used for a wide range of EV battery charging applications by controlling the primary side inverter duty cycle or switching frequency.

Technical Track TR05

[#111] A Comprehensive Review of the Battery Thermal Management Systems for Fast-Charging of Lithium-ion Batteries

Chandan Chetri (Ontario Tech University)*; Alvin Huynh (Ontario Tech University); Latha Anekal (Ontario Tech University); Sheldon Williamson (Ontario Tech University)

The growing demand for electric vehicles (EVs) and portable electronic devices has increased the need for rapid charging of lithium-ion batteries (LIBs). However, fast charging generates excessive heat, necessitating efficient battery thermal management systems (BTMSs). BTMS plays a critical role in maintaining battery safety, performance, and longevity by regulating thermal loads and addressing issues like thermal runaway, electrode degradation, and overheating during fast charging and discharging. This paper presents a comprehensive review of existing BTMS strategies, categorized into active and passive systems. Active systems include air, liquid, and thermoelectric cooling, while passive systems include phase change materials (PCM) and heat pipes. Furthermore,

hybrid BTMSs combining both approaches are discussed. This study evaluates the BTMSs based on thermal performance, efficiency, and suitability for fast-charging applications.

[#119] Hybrid Cell Balancing of Lithium-Ion Batteries Using Multi-Inductor Active Topology Combined Passive Dissipation Control

dheeraj gurijala (student)*

Cells in a battery pack frequently have different State of Charge (SoC) values due to manufacturing irregularities and environmental factors, which can reduce system efficiency and pose safety risks. A Battery Management System (BMS) monitors and controls individual cell behavior to ensure reliable performance and extend battery life. Cell balancing, which equalizes the charge levels of all cells to minimize SoC discrepancies, is a crucial function of the BMS, improving the battery's overall efficiency, longevity, and protection against conditions like overcharging and deep discharging. This paper presents a hybrid cell balancing strategy that combines active and passive balancing methods, automatically switching between the two in response to real-time SoC variations. In this system, active balancing transfers charge across cells using a configuration of three inductors, while passive balancing uses resistors to dissipate excess energy from cells higher charge levels. This hybrid approach merges the strengths of both techniques, providing a cost-effective and efficient solution well-suited for applications where both performance and affordability are essential.

[#186] Impact of Faulty Cell Position on Terminal Voltage Characteristics in Li-Ion Battery Packs

TWINKLE PATTNAIK (VNIT, Nagpur)*; Devang P. Kubitkar (VNIT, Nagpur); Ragiree Shusmashalini (VNIT, Nagpur); Makarand S. Ballal (VNIT, Nagpur)

Conventional battery monitoring systems often rely on thermal scanning across all regions of a lithium-ion battery (LIB) pack, which increases system complexity, slows down detection, and demands significant computational resources. Such approaches are inefficient for large-scale applications where fast and reliable fault localization is critical. This paper presents a novel method for early fault detection at the strip level using electrical diagnostics. By analyzing voltage deviations across parallel-connected strips, the proposed technique accurately identifies the faulty strip out requiring exhaustive thermal imaging. This targeted approach reduces hardware requirements, minimizes data processing, and enables faster fault localization. Simulation results demonstrate the effectiveness of this method in enhancing the speed and efficiency of fault detection in high-capacity battery

systems.

[#261] Performance Evaluation of Passive Cell Balancing Topologies for Parallel-Connected Lithium-Ion Battery Packs Using Minimum SoC Strategy

Kiran Sai Dasari (Indian Institute of Science, Bangalore)*; Shreyansh Kulshrestha (Institute of Technology, Nirma University); Bhavesh Rameshbhai Vagh (Indian Institute of Science, Bangalore); Ritk Bind (Madan Mohan Malaviya University of Technology); Subba Reddy Basappa (Indian Institute of Science, Bangalore)

State of Charge (SoC) imbalance in lithium-ion battery packs leads to reduced efficiency, increased energy loss, and long-term degradation especially in parallel-connected series string (nSmP) configurations. This paper presents a comparative analysis of two passive cell balancing topologies: the conventional switched shunt resistor method and a modified two-diode, two-contactor topology. A minimum SoC-based balancing strategy is implemented for both cases to regulate cell-level SoC variation. The study is conducted on a 7S2P lithium-ion battery pack model in MATLAB/Simulink under varying charge and discharge C-rates (0.2C to 3C) for both identical (ID) and non-identical (NID) cell scenarios, reflecting practical variations due to aging and manufacturing variances. Under ID cell conditions, the modified topology achieves up to 57\{}% faster balancing, 35\{}% lower energy loss, and 4\{}% higher efficiency compared to the conventional method. Under NID cell conditions, it continues to outperform by 42\{}% faster balancing, 32\{}% lower energy loss, and maintains system efficiency above 90\{}%. These improvements are attributed to enhanced string-level isolation and directional current control, which minimize inter-string circulation and power dissipation. The results validate the viability of the proposed topology's effectiveness and robustness, making it a scalable and energy-efficient solution for battery management in electric vehicles and stationary storage systems.

[#359] A Single Stage Battery Balancing and Power Transfer using M2L2M Configuration via Integrated DC-DC Topology for Electric Vehicles

Namrata Narayan (Indian Institute of Technology, Mandi)*; Parmar Pratikkumar (Indian Institute of Technology, Mandi); Moumita Das (Indian Institute of Technology, Mandi)

This paper presents a novel real-time battery management and power transfer strategy based on a proposed DC-DC converter topology for electric vehicle (EV) applications. The system adopts a module-to-load-to-module (M2L2M) configuration, enabling simultaneous

SOC-based battery balancing and high voltage output regulation for efficient intra-pack energy redistribution and traction power delivery. Four 48V, 72Ah lithium-ion batteries are connected in series, each interfaced a synchronous boost converter using SiC switches operating at 100kHz, collectively delivering 1kW. A control algorithm manages SOC balancing and voltage regulation. Simulation results validate the proposed topology's high efficiency, fast balancing, and stable power transfer under varying load and SOC conditions. Future work will examine thermal effects on balancing and power performance.

Technical Track TR07

[#178] An Improved ASMO-Based Sensorless Control Strategy for PMSM Using Grey Wolf Optimization

Jinlun Liu (Xi'an Jiaotong-Liverpool University); Mianzhi Wu (University of Pennsylvania)*; Zirui Zhuang (Xi'an Jiaotong-Liverpool University); Hanyu Zhang (Xi'an Jiaotong-Liverpool University)

The sliding mode observer (SMO) is widely employed in sensorless control of permanent magnet synchronous motor (PMSM), but the performance of PMSM is limited by sensitivity of the observer to the variation of system parameters and inaccuracies in position estimation caused by the variation of BEMF speed. To address these problems, this article proposes one Grey Wolf Optimization (GWO)-based method to optimize the gains of the adaptive sliding mode observer, enhancing the performance of the sensorless control of PMSM. After GWO optimization, the error between estimated speed and reference speed of the rotor was reduced, and the adaptability of ASMO has improved. This paper introduces the mathematical models of traditional and adaptive SMO, and elaborates in detail the design of GWO, which is used to optimize the gain of ASMO. The optimization results of GWO were described in detail through simulation and experiments, verifying the effectiveness of this method.

[#193] Regenerative Braking in Open Ended Induction Machine for EV Applications

Arindam Paul (NIT AGARTALA); Moumita Paul (NIT AGARTALA); Diptanu Majumder (NIT AGARTALA); Prabir Kasari (NIT AGARTALA); Bikram Das (NIT Agartala)*; Anindita Jamatia (NIT AGARTALA); Soham Chakraborty (IISc Bangalore); ABANISHWAR CHAKRABARTI (NIT AGARTALA); Sujit K. Biswas (St. Thomas College of Engg. & Tech.)

This paper proposes a new scheme for supercapacitor based regenerative braking for an electric vehicle (EV) using an open end induction motor (OEIM), out utilizing DC-DC converters. In the proposed scheme, the OEIM is connected between two 3-phase inverters at its two ends, one inverter dc side connected to a battery source while the other inverter dc side is connected to a supercapacitor. During normal running condition, the battery supplies the active power and the reactive power is supplied by the supercapacitor. However, during regenerative braking, the supercapacitor as well as the battery absorbs the braking current, maximizing energy recovery. During rapid acceleration the motor receives active power from both the inverters, which improves performance of the EV. The proposed regenerative braking strategy enhances performance and efficiency of the EV drive.

[#275] Design and Analysis of Push-Pull Converter for Microsatellite and PMDC Motor Drive Systems

VANDANA V (Government Engineering college Sreekrishnapuram, Palakkad.)*; Vinita Chellappan (Government Engineering College Palakkad)

The design, simulation, and hardware implementation of a high-efficiency push-pull DC-DC converter for PMDC motor drives and microsatellite electric propulsion are presented in this study. The converter steps up a variable input to a regulated high-voltage output for Hall thrusters, and in the motor drive setup, it is integrated a chopper for current-fed control. Closed-loop PID control ensures stable voltage and current regulation. MATLAB Simulink simulations validate performance, including efficiency and loss analysis. Hardware uses compact, cost-effective components analysis of conduction, switching, diode, and transformer losses. Results confirm high efficiency and stable operation across varying input and load conditions, highlighting the converter's suitability for both aerospace and industrial use.

[#324] Performance Analysis of Shaft Voltage in 3L-NPC fed Induction Motor using LS-PWM and PS-PWM a Carbon brush

Asad Hussain (Indian Institute of Technology, Roorkee)*; Jyoti Ranjan Dash (Indian Institute of Technology, Roorkee); Rohit Raj (Indian Institute of Technology, Roorkee); Pramod Agarwal (Indian Institute of Technology, Roorkee)

In this paper, the effects on shaft voltage generated in an Induction motor driven by a three-phase, three-level NPC inverter operating under high switching frequency PWM techniques are presented. High-frequency switching operation generates common-mode voltage, leading to shaft voltage build-up and potentially damaging the bearings in the long

run due to the circulating bearing current. Phase-shifted carrier PWM and Level-shifted carrier PWM are analyzed to determine their impact on the shaft voltage characteristics. Later, a sealed brass holder containing a carbon brush from Helwig Carbon Products is placed on the motor body perpendicular to the shaft as a mitigation tool. The results provide insights into the influence of high switching frequency PWM techniques and carbon brush on shaft voltage levels. Simulation and experimental-based studies are analyzed to assess how much shaft voltage is mitigated. Thus, protective measures against bearing failure can be developed to enhance the reliability and longevity of induction motor drive systems.

[#325] Recuperation and Reuse of Regenerative Braking Energy in EOT Cranes

Rakesh Jha (Aartech Solonics Ltd. Bhopal)*

Electric overhead traveling (EOT) cranes are integral to material handling in industries such as manufacturing, transportation, and construction, where the movement of heavy loads demands substantial operating energy. Operation of cranes results in significant energy loss during braking, contributing to inefficiency. This work demonstrates the application of supercapacitor-based energy storage systems to capture and reuse regenerative braking energy in EOT cranes. By connecting a supercapacitor bank to the DC link through a bidirectional DC-DC converter, the system effectively stores braking energy and repurposes it for subsequent lifting operations thus reducing the stress on the electrical grid and reducing maximum demand. Field results demonstrate that this approach reduces net energy consumption by approximately 14\{}%, lowers emissions, and alleviates peak demand stress on local power infrastructure. Overall, the integration of regenerative braking energy recovery enhances operational efficiency and supports sustainability in industrial crane operations.

[#348] Split Core CT Based Energy Harvester for Powering Transmission Line Inspection Robots

Souvik Das (Indian Institute Of Technology Dharwad)*; Abhijit Kshirsagar (Indian Institute of Technology Dharwad); Pratyasa Bhui (Indian Institute of Technology Dharwad)

Conventional transmission line inspection methodsface several limitations, leading to the increasing adoption ofautonomous crawling robots. However, powering these systemsremains a challenge. Onboard batteries have limited operationallife, while solar-based solutions depend on variable weatherconditions. This work proposes a design approach for

a split-core current transformer (CT) based energy harvester to meetthe power demands of such inspection robots. The harvesteroperates on inductive principles by utilizing the magnetic fieldsurrounding live transmission lines and transferring the energyto the CT's secondary side. A systematic methodology for coredesign and magnetic material selection is presented. The proposed configuration is validated through finite element analysis (FEA), showing a maximum extractable power of 72 W for a 300 A, 50 Hz transmission line current.

Technical Track TR08

[#165] Machine Learning-Based Prediction of Load Current THD in SVPWM Inverters: A Comparative Study

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The work presented here investigates and compares several machine learning approaches to estimate the Total Harmonic Distortion (THD\{}_I) in the load current generated by Space Vector Pulse Width Modulated (SVPWM) inverters. A dataset was generated using a two-level, three-phase inverter a fixed RL load by varying switching frequency and modulation index in MATLAB/Simulink. The study evaluates six regression models: Random Forest (RF), Gradient Boosting (GB), Support Vector Regression (SVR), k-Nearest Neighbors (KNN), Multilayer Perceptron (MLP), and Gaussian Process Regression (GPR). All models utilize z-score normalization for features and min-max normalization for the current THD target. Performance is assessed using Mean Absolute Error(MAE),Root Mean Square Error(RMSE), and Coefficient of Determination R\{}$\hat{2}$ on validation and test sets. Gradient Boosting yields the most accurate predictions. The results show that machine learning can effectively replace simulation for fast and accurate THD estimation in load current of inverter systems.

[#188] Residual-Aware Reinforcement Learning Bayesian Transformer for Probabilistic Anomaly Detection in PV Systems

Apoorva Choumal (Delhi Technological University)*

Reliable detection of operational anomalies in large-scale photovoltaic plants is constrained by unavailability of labeled data data, non-stationary ramp-rate dynamics and the need for well-calibrated uncertainty estimates. This research proposes an end-to-end framework that couples residual-driven anomaly exploration uncertainty-aware prediction to deliver probabilistic anomaly score. A ten-year, 5-min resolution dataset from 1058 kW Dessert garden, Australia solar farm is used to access the proposed framework. A deterministic gradient-boosting model is trained to estimate the photovoltaic power output using meteorological parameters. The normalized prediction residual is employed as an unsupervised anomaly indicator. Further, the anomaly identification is formulated as a sequential decision-making problem which is addressed using reinforcement learning agent. At each step the agent selects the time window and sensor subset expected to maximize residual deviation, thereby concentrating computational effort on high-risk segments. The selected sequences are processed by a Transformer encode whose decoder is trained as a Bayesian neural network via Monte-Carlo dropout. The model outputs both point prediction and epistemic variance, enabling closed-form computation of anomaly probability from prediction errors. The proposed framework attains a CRPS of 0.041 70\{}% faster inference and apporx. 33\{}% lower false alarm rate hence providing an accurate and operational viable methodology for real-time PV anomaly detection.

[#215] Levenberg–Marquardt Trained Neural Network Based Fault Current Limitation in Synchronverter

Lipun Naik (Indian institute of technology dharwad)*; Animesh Sahoo (Indian institute of technology dharwad); Mayukha Pal (ABB Ability Innovation Center, Asea Brown Boveri Company, Hyderabad)

Synchronverters are grid-forming inverters that mimic synchronous generator behavior in modern power systems. A key challenge in their control is limiting fault current during grid disturbances, which can otherwise damage equipment or destabilize the grid. Conventional virtual impedance methods often lack adaptability and real-time responsiveness under fast-changing conditions. This paper proposes a neural network-based approach to regulate inverter voltage references for effective fault current suppression. A shallow feedforward neural network is trained using the Levenberg–Marquardt algorithm to map system states to control actions high precision and robustness. The network dynamically adjusts dq-axis voltage references in response to overcurrent conditions, keeping current levels in safe operational limits out requiring fixed parameter tuning. Simulation results show high prediction accuracy, rapid response, and strong generalization across varied operating scenarios. Compared to traditional control strategies, the proposed method offers a flexible, data-driven alternative that significantly enhances protection, adaptability, and dynamic stability in synchronverter-based systems.

[#237] Modified Mantis Shrimp Optimization-Based Global MPPT for Photovoltaic Systems under Partial Shading Conditions

Nisha Poothullil (Government Engineering College Thrissur)*; Abdul Saleem (Government Engineering College Thrissur)

Maximum Power Point Tracking (MPPT) techniques are essential for enhancing the efficiency of photovoltaic (PV) systems under dynamically changing environmental conditions. Under partial shading conditions (PSC), PV arrays exhibit multiple peaks in their power-voltage (P-V) characteristics, posing a significant challenge for conventional MPPT techniques to accurately locate the Global Maximum Power Point (GMPP). Although traditional techniques perform well under uniform irradiance, they often fail under PSC due to their inability to distinguish between local and global peaks. To address this, a Modified Mantis Shrimp Optimization (MSO) algorithm is proposed in this paper. Inspired by the predatory and sensory behavior of mantis shrimps, the algorithm effectively balances exploration and exploitation to ensure rapid and accurate tracking of the GMPP. The proposed MSO algorithm demonstrates improved tracking performance faster convergence, reduced power oscillations, and high efficiency. MATLABbased simulation results under various partial shading scenarios validate the superiority of the proposed approach when compared conventional P\{}&O, PSO, and GWO techniques. The MSO algorithm exhibits excellent dynamic response and robustness, making it a promising solution for real-time MPPT in PV systems under non-uniform irradiance.

[#351] Fusion-Based Approach For Accurate SoH Estimation of Lithium-Ion Batteries: A Review

AMITAVA DUTTA (Volvo India Ltd.)*; Makarand S Ballal (Visvesvaraya National Institute of Technology)

The Accurate estimation of State of Health (SoH) and Remaining Useful Life (RUL) of lithium-ion batteries is critical for enhancing the safety, reliability, and performance of battery-powered systems, particularly in EVs and ESS applications. This study proposes a fusion-based diagnostics approach that integrates Coulomb Counting (CC) Capacitance-based analysis to overcome the individual limitations of each method. CC, while widely adopted due to its simplicity and real-time applicability, on the other hand, Capacitance-based analysis provides valuable insights into electrochemical aging by correlating capacity fade impedance or voltage response metrics. By combining these two techniques which compliments each-other, this proposed fusion methodology offers to enhance accuracy, robustness, and early fault detection capability across various aging modes and operating conditions. This review will examine the theoretical approach by understanding the

foundations and comparative performance metrics of state-of-the-art fusion models. It also highlights key challenges, such as temperature dependencies, measurement noise, and data-driven model generalization, while outlining future research directions for real-time onboard implementation. The study aims to provide a consolidated knowledge base for researchers and engineers seeking advanced SoH and RUL estimation techniques for lithium-ion batteries.

[#360] Optimal Demand Side Management using Multi-Objective Particle Swarm Optimization

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In this study, a probable programming model is proposed to optimize the peak demand-supply gap of a smart grid, in a short term to minimize operating costs and emissions renewable sources. In order to achieve an accurate model, the use of a probability density function to predict the wind speed and solar irradiance is proposed. On the other hand, in order to resolve the power produced from the wind and the solar renewable uncertainty of sources, the use of demand response programs the participation of residential, commercial and industrial consumers is proposed. The multi-objective particle swarm optimization method is utilized to solve this problem. In order to validate the proposed model, it is employed on a sample smart micro-grid, and the obtained numerical results clearly indicate the impact of demand side management on reducing the effect of uncertainty induced by the predicted power generation using wind turbines and solar cells.

Technical Track TR10

[#192] A Novel Dual-Field Modulated Generator Using Dual Halbach Arrays for Wind Energy Applications

Neel Shrivastava (IIT Goa')*; Sashidhar Sampathirao (IIT Goa); Bidyadhar Subudhi (IIT Goa)

This paper presents a dual-field-modulated generator featuring dual armatures and dual Halbach magnet arrays, designed for enhancing energy capture in direct-drive wind energy systems. The topology comprises concentric inner and outer armatures, a rotating hollow cylindrical rotor placed between them. This rotor has permanent magnets embedded

along both its inner and outer circumferential surfaces, enabling magnetic interaction both armatures. The Halbach magnetization enhances the air-gap flux density and effectively isolates the magnetic interaction between the two armatures. To facilitate field modulation, ferromagnetic segments are strategically placed on both sides of the rotor, enhancing induced voltage in each armature. The electromagnetic behavior of the machine is analysed through finite element simulations (FEA), demonstrating superior voltage output, power density, and torque generation.

[#112] Design of High-Speed Outer Rotor Permanent Magnet Synchronous Motor (PMSM) for Blower Application Novel Asymmetrical Stator Structure

ATHIRA VENUGOPAL (VSSC)*; SHINOY K.S (VSSC); BABY SEBASTIAN (VSSC)

In this paper, a novel Permanent MagnetSynchronous Motor (PMSM) an asymmetrical statorstructure is proposed for blower applications. ConventionalPermanent Magnet DC (BLDC) motors are commonly used inblower systems; however, they suffer from high torque rippleand supply voltage harmonic distortion due to the symmetricalnature of the stator design. The proposed asymmetrical statorconfiguration reduces harmonics, leading to improvedperformance and smooth operation in blower systems. Theconcentrated winding configuration improves the overallefficiency. The paper covers the design, electromagneticsimulations using FEM, hardware implementation, andcomparison between simulation results and actual hardwareperformance. Furthermore, flow trial results on blower arealso demonstrated validating the proposed design.

[#129] Digital Twin Approach for the Stator Winding Interturn fault Detection of Alternator

Makarand Ballal (Visvesvaraya National Institute of Technology, Nagpur)*; Bhimrao Umare (Visvesvaraya National Institute of Technology, Nagpur); Hiralal Suryawanshi (Visvesvaraya National Institute of Technology, Nagpur); Rashmi Singh (Visvesvaraya National Institute of Technology, Nagpur); Tushar Vilhekar (Visvesvaraya National Institute of Technology, Nagpur); Kanchan Ghule (Visvesvaraya National Institute of Technology, Nagpur)

This paper concentrates on the approach of digital twins in alternator stator winding Interturn fault (SWIF) diagnosis. Initially, a mathematical relationship was established linking alternator dynamics to stator winding interturn faults (SWIF). A real-time 10 kW, 400 V, 50 Hz alternator model was developed in MATLAB Simulink, a corresponding Digital Twin in ANSYS Maxwell finite element tool is considered for analysis. The

models were dynamically coupled via co-simulation. SWIF detection was achieved through negative sequence current analysis, ANSYS simulations confirming change in magnetic flux distribution proportional to fault severity. This integrated Digital Twin approach demonstrated accurate, real-time fault detection and severity assessment, validating its effectiveness for predictive maintenance and operational reliability enhancement in alternator systems under varying conditions.

[#187] A Boundary-Based Numerical Method to Calculate Inductance Matrix for Non-Circular Planar Spiral Winding

Parvathy Saji (National Institute of Technology Calicut)*; Muhammed Nashrah K (National Institute of Technology Calicut); Ashiq Muhammed PE (National Institute of Technology Calicut)

Air-core planar inductors play a pivotal role in contemporary electronic systems, particularly in high-frequency and space-constrained applications. By leveraging printed circuit board technology, these inductors mitigate challenges associated iron losses and magnetic saturation commonly observed in traditional materials. Non-circular coil geometries, such as rectangular and square spiral windings, have gained prominence due to their layout flexibility and favourable electromagnetic properties, making them well-suited for applications including wireless power transfer (WPT), radio frequency integrated circuits (RFICs), and near-field communication (NFC). This paper addresses the limitations of existing analytical and numerical methods to calculate the inductance of square and rectangular spiral windings, which either overlook the mutual inductance effects of adjacent turns or involve significant computational challenges. A computationally efficient, non-FEM filament-based approach is proposed to accurately compute turn-by-turn inductance matrices while incorporating the effect of adjacent turns. The proposed method is validated through experimental results for rectangular windings and 3D finite element simulations for square spiral windings.

[#350] Correlation between apparently dissimilar inductor geometries through Geometric Inversion: An approach using Mathematical Topology

Gourab Banerjee (IIEST Shibpur)*; Mainak Sengupta (IIEST Shibpur)

This paper presents inverse geometric transformation between two existing core geometries, torus and pot. The approach is hitherto unknown in the field of magnetics, as is displayed by lack of even a single relevant published work. The point-to-point mapping respect

to an inversion sphere is presented to establish a plausible relation between apparently dissimilar inductor core geometries. In the process, an appropriate Mobius transformation is applied to obtain the inverted twins of the existing geometries. It is found that, as a result, core and window areas in both the geometries swap their locations and hence roles. To validate the proposition, experiments are carried out using commercially available toroidal and pot core inductors, which may be correlated through inversion and then the closest standard available cores are chosen. Finite element simulations are carried out to validate the design specifications. The inductors are tested in a lab-developed GaN-based Synchronous Buck Converter, rated 48 V to 24 V, 60 W (switching at 200 kHz). The different performance indices have been correlated the proposed inversive transformation. The proposed study can be extended, among other objectives, to obtain innovative core geometries from existing ones and can be used for improved performance depending on application requirements.

[#409] A Comparative Study of Flux-Reversal and Flux-Switching Permanent Magnet Machines

Shivendra Singh (Indian Institute of Technology Dharwad); Aravind G (Indian Institute of Technology Dharwad); Amarkumar Kushwaha (Indian Institute of Technology Dharwad)*

Flux Reversal Synchronous Machines (FRSM) and Flux Switching Synchronous Machines (FSSM) are two permanent-magnet machine topologies distinguished by their rotor and stator magnet arrangements and slot-pole combinations. This work presents a fair, direct comparison of a 6-slot/14-pole FRSM and a 12-slot/14-pole FSSM. Both geometries were optimized to maximize average torque while maintaining identical electrical loading, permanent-magnet volume, and outer diameter. The design variables included the width of the rotor-tooth, the width of the stator-tooth, and the split ratio. Performance parameters such as electromagnetic torque, back-EMF quality, and cogging torque for each optimized topology are assessed using finite-element analysis. The results indicate that, at the same split ratio, the FRSM delivers 19\{}% higher average torque than the FSSM. However, the harmonic content in the back-emf and dynamic torque-ripple are higher in FRSM in comparison to that of FSSM. Moreover, the FRSM requires lower current density to achieve equivalent torque, implying reduced copper losses and improved thermal performance. These findings demonstrate that, under equivalent design constraints, the FRSM topology offers superior torque density and electromagnetic efficiency, making it a compelling choice for high-performance PMSM applications.

Technical Track TR11

[#236] Model Predictive Control of Small-Scale Electrified Hydrogen Electrolyzer Systems

Jairam Kinjarapu (Indian Institute of Technology Madras); Deepak Ronanki (Indian Institute of Technology Madras)*; Rajesh Sura (Indian Institute of Technology Madras)

Hydrogen electrolyzers have gained significant attention for the green energy transition. This study explores the application of predictive control in an electrified hydrogen electrolyzer system to achieve better dynamic and steady-state performance while ensuring grid compliance through proper isolation. This article presents a converter topology for hydrogen electrolysis applications based on an active front-end (AFE) rectifier and a dual active bridge (DAB) converter. To meet the grid power quality requirements, an AFE rectifier finite control set model predictive control (FCS-MPC) is utilized, ensuring unity power factor (UPF) operation. For the DAB converter, a moving discretized control set model predictive control (MDCS-MPC) strategy is used to enhance the dynamic performance reduced computational burden. In a PI-based control, cascading of dual loops used in the AFE rectifier and voltage loop of the DAB converter leads to slow dynamic response and complex modeling. The proposed dual predictive control method ensures a stable voltage low current ripple on the electrolyzer side and better dynamic performance while ensuring the power quality of the grid.

[#243] Research on Optimizing Controller Configuration for CRAFT Power Supply System Based on Particle Swarm Optimization and BP Neural Networks in Nuclear Fusion

Ling Zhang (Institute of Plasma Physics, Hefei Institutes of Physical Science, Chinese Academy of Sciences)*

the development of nuclear fusion technology, high power supply control system plays crucial role in nuclear fusion. Proportional-Integral-Derivative (PID) controller is widely used in power supply control field. In this research, a novel hybrid algorithm optimizing controller parameters is proposed using a combination of Particle Swarm Optimization (PSO) and Back propagation (BP) neural networks. The proposed method aims to enhance the performance of the control system by fine-tuning the PID parameters for a given transfer function. The approach includes training the neural network on randomly generated PID parameters and then using PSO to optimize these parameters in an iterative manner. The results demonstrate the effectiveness of the proposed method in

achieving optimal PID parameters for system control and realizing a more sensitive, stable and efficient response in the high power operation. This research is not only of great significance to the advancement of nuclear fusion technology, but also provides useful reference for the application of adaptive optimization methods in other fields.

[#264] Single-Loop Output Power Control Strategy for Standalone Single-Phase Inverters in Dynamic Load Applications

Akshat Saini (IIT Madras)*; Ajit Kumar Upadhiya (IIT Madras); Lakshminarasamma N (IIT Madras)

This paper presents a simplified output power control strategy for a single-phase inverter, suited for applications where the load parameters are unknown or varying. Various control strategies based on both stationary and synchronous reference frames exist. Typically, widely adopted methods rely on synchronous reference frame transformations; however, in single-phase applications, the addition of orthogonal imaginary components introduces delays into the system, degrading the dynamic response and making it slower and oscillatory. Whereas, stationary frame-based approaches struggle to eliminate steady-state error while tracking sinusoidal references. The proposed method provides a simplified approach by computing the averaged output power and employing a tuned PI controller to track the reference load power zero steady-state error. The key advantages of this approach lie in its simplicity, easier tuning, reduced computational burden, and ease of practical implementation, making it particularly suitable for high-frequency, power-sensitive applications. Simulation results validate the effectiveness of the proposed method in maintaining required power delivery.

[#288] An Enhanced Multi Complex Coefficient Filter based Control for a Grid Connected Solar Energy Conversion System

Dr. NIRMAL MUKUNDAN (Khalifa University)*; Ahmed Al Durra (Khalifa University); Mohamed El Moursi (Khalifa University); Tarek El Fouly (Khalifa University)

Industrial prosumers are increasingly adopting solar energy conversion systems and integrating them the grid to enhance reliability. However, the quality of power exchange the grid is often compromised due to poor grid voltage conditions and distortions introduced by local nonlinear loads. This paper presents a grid integration control strategy capable of operating reliably under weak grid conditions and in the presence of nonlinear, imbalanced, and reactive local loads. The ability of a multi-complex coefficient filter (MCCF) to extract the positive-sequence voltage component is typically affected by high-order harmonics, DC offsets, frequency deviations, and amplitude or phase imbalances. To

address these challenges, an enhanced MCCF structure is proposed by incorporating a second-order generalized integrator (SOGI)-based prefilter. The bandwidth of the SOGI controller is carefully tuned for optimal filtering, and the corner frequencies derived from the MCCF are fed back to the SOGI to maintain accurate grid synchronization. The use of SOGI supports the fast dynamic response of the enhanced MCCF, owing to its computational simplicity. The improved SOGI used for active load current component extraction enables the SECS to effectively meet both the reactive power and harmonic compensation requirements of local loads. The proposed single-stage system eliminates the need for a separate DC-DC converter by integrating maximum power point tracking (MPPT) and grid synchronization control in a single voltage source converter. The system performances are validated through a MATLAB/Simulink model under varying source, load, and PV-side conditions.

[#356] State Estimation and Validation of Nonlinear System Dynamics of a SPV Array Integration Different BESS using Error Metrics

Annavarapu Ankamma Naidu (Indian Institute of Technology Roorkee)*

The recent surge in renewable penetration andadvanced technologies in power electronics have made DCmicrogrids more feasible. In this article, the development andvalidation of the nonlinear system dynamics of the integration ofthe solar photovoltaic(SPV) array a hybrid energy storagesystem(HESS) using error metrics. Implemented a nonlinearstate estimation technique to study the behavior or estimate thestates of the test system. The control algorithms implementedwere: 1) A perturb \{}& observe(P\{}&O) MPPT algorithm to trackthe maximum power point(MPP). 2) Combination of state-of-charge (SOC)-based battery reference current generation andinner current control algorithm a simple PI controllerto give switching pulses to bidirectional converters to meetload demands under solar irradiance uncertainties. Validationof the developed test system model, state estimation, and controlalgorithms implemented in MATLAB/Simulink.

[#437] DPS-Based Control Strategy of DAB Converter Reduced-Order Modelling for Off-board EV Charging Applications

Shashi kumar Kondoju (VNIT Nagpur)*; Pradyumn Chaturvedi (VNIT Nagpur); Aman Sheikh (VNIT Nagpur)

The demand for effective, high-power, isolated fast-charging systems has increased due to the growing popularity of electric vehicles (EVs). Because they facilitate quick battery replacement and can facilitate bidirectional operation for vehicle-to-grid (V2G)

applications, off-board chargers are essential in this situation. Due to its high efficiency, galvanic isolation, and soft-switching ability, the Dual Active Bridge (DAB) converter is a promising option for these kinds of systems. A reduced-order modelling and control approach for a DAB converter running on Dual Phase-Shift (DPS) modulation is presented in this paper. Regarding the secondary phase-shift variable, a simplified small-signal model is developed. D\{}_2 which is chosen as the primary control input, and the complementary variable D\{}_1 is established offline by applying an optimization criterion to minimize device stress and inductor peak current. PI controllers are made for constant current (CC) and constant voltage (CV) modes of operation based on reduced-order transfer functions. This allows for a seamless transition between the various charging profile stages. By comparing frequency responses to the switching model, simulation studies verify the accuracy of the reduced-order model. They also show strong CC and CV regulation under various load conditions and decreased current stress. Future work will concentrate on hardware validation using battery-based loads and experimental verification of CC–CV transitions. The suggested approach offers a systematic framework for control-oriented modelling of DPS-operated DAB converters in EV fast-charging applications.

Technical Track TR12

[#277] Coordination of RT-DFIG-DC based WECS to Interconnect a DC Microgrid and an AC Grid

SOHAM CHAKRABORTY (NETAJI SUBHASH ENGINEERING COLLEGE)*; Abanishwar Chakrabarti (NIT Agartala); Dipten Maiti (Jadavpur University); Sujit Biswas (St. Thomas College of Engineering and Technology)

This paper proposes a coordinated control strategy for a rotor tied doubly fed induction generator-direct current (RT-DFIG-DC) based wind energy conversion (WECS) scheme that can also be utilized to interconnect a DC microgrid and AC utility grid for bidirectional power exchange through the asynchronous machine. The rotor terminals of the asynchronous machine are directly connected to the AC grid via slip rings at the point of common coupling (PCC). The stator terminals are interfaced to the DC microgrid using a 3-phase 2-level stator side converter (SSC) that handles the slip power of the RT-DFIG-DC. This scheme permits the harnessed wind power to be directed either to the DC microgrid or to the AC grid. The DC microgrid can also exchange bidirectional active power the AC grid the help of interfacing DC/DC converter coupled to the SSC. The DC microgrid has PV power and a backup hybrid battery energy storage system (BESS) comprising of a smaller Lithium-ion supercapacitor based storage higher power density (to supply power during abrupt changes in load demand) and a larger lead acid BESS

higher energy density. The coordinated control scheme, analysis and simulated results exhibit effectiveness of proposed scheme in interconnection of DC microgrid to an AC grid through a RT-DFIG-DC based WECS.

[#328] Multi-Objective Optimization Study of Hybrid Renewable Energy System

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This research proposes an optimal, feasible and economical Hybrid Renewable Energy System (HRES). An analysis is conducted in terms of Net Present Cost (NPC), Levelized Cost of Energy (LCOE), Capital cost and Annual Generation, results shows that hybrid system outperforms the standalone Solar or Wind energy system. The proposed system supplements the 1960MW Hydro Power Plant at Koyna Dam, in Maharashtra. 1\{}% (approximately 20 MW, precisely 19.6 MW) of dam capacity is considered as a fixed output power requirement. The systems designed to meet this demand are 27MW and 62MW for standalone solar and wind respectively, the hybrid system combines 13.6MW solar and 32MWwind capacities to achieve the same 20MW output.

[#428] A Coordinated DC Protection Strategy Based on Prioritized DC-link Capacitor Fault Interruption

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Protection in DC networks has long been a complex challenge, often cited as a major barrier to their widespread adoption. The presence of large DC-link capacitors and the replacement of traditional transformers power electronic converters result in fault current rise rates that are tens to hundreds of thousands of times higher than those in AC grids. Despite this, conventional DC protection strategies are largely borrowed from the traditional AC protection mindset. In DC networks, the inclusion of large link capacitors renders the series insertion of circuit breakers, similar to AC network practices, suboptimal, as breakers must interrupt fault currents injected from multiple sources varying escalation time constants. Existing DC circuit breaker technologies also inherently struggle to simultaneously achieve high interruption speed, minimal size, and low cost. Consequently, relying on a single circuit breaker does not necessarily offer an optimal solution in terms of speed, cost, or compactness. This paper investigates fault propagation mechanisms in common power electronic converter topologies and proposes a novel protection strategy

based on the coordinated use of multiple circuit breakers varying interruption speeds. The proposed approaches are also validated through comprehensive simulations.

[#432] MPC-Enabled DAB Converter Control to Reduce LVdc Ripple in SST-Based Microgrid

Somnath Meikap (IIT Guwahati)*; Hitesh Malviya (IIT Guwahati); Sahil Gaurav (IIT Guwahati); Chandan Kumar (IIT Guwahati)

Solid-state transformers (SSTs) are increasingly adopted for microgrids due to their bidirectional power flow capability, availability of medium-voltage dc and low-voltage dc (LVdc) links, and ancillary services. However, linear and nonlinear loads at the low-voltage ac (LVac) side introduce ripple into the LVdc link. These ripples cause capacitor stress, efficiency degradation, and deterioration of power quality. The conventional ripple mitigation techniques, such as oversized dc-link capacitors, passive/active filters, etc. increase the cost and volume of system, and limits the dynamic performance. Another conventional approaches include direct compensation that impose large current stress on the dc-dc converter. To address these challenges, this paper proposes a model predictive control (MPC) strategy for the dual active bridge (DAB) converter to actively suppress LVdc ripple in SST-based microgrids. The proposed method leverages a moving discretized control set MPC (MDCS-MPC) framework that explicitly models both second-harmonic and higher-order ripple dynamics, adaptively adjusting the DAB phase-shift angle for effective ripple mitigation. Simulation results demonstrate reduced LVdc voltage ripple, and improved voltage regulation under nonlinear load conditions.

[#433] Series Arc and Hotspot Detection in Power Distribution Box for Industrial Loads

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The Industrial Power Distribution Box (PDB) distributes incoming power supply to diverse loads, including resistive, inductive, and power electronics-based devices, each distinct current profiles. Circuit breakers allocate these loads to designated sections of the distribution unit. Mechanical vibrations in industrial environments can loosen connections at breaker poles, causing series arc faults. Such faults often remain undetected due to their low intensity, leading to progressive damage, potential shutdowns, and significant economic losses. Early detection of arc faults is therefore critical. While literature addresses series faults, it lacks effective methods for detecting arc faults in application-specific circuits diverse loads. This work proposes a hybrid detection solution for resistive

and power electronic (i.e., DC and RF generator) loads using current measurement and harmonic component analysis, complemented by a temperature-based detection algorithm for localized secondary protection. Arc current and temperature data are logged wirelessly, eliminating communication cabling and reducing space requirements. The proposed method is validated in semiconductor industry PDBs varied loads, demonstrating high detection accuracy under different noise levels.

[#436] Study on Effects of Voltage Harmonics of Electric Propulsion based Ship Power System on Long Range Radar Power System

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Long Range Radars require few Mega Watts of Power for detection and tracking of targets at hundreds or sometimes few thousands of kilometers away. Ship based Long Range Radars require compact and highly reliable Power System of higher capacity. Power Systems of Ship Based Radars are designed to allow minimum leakage current. Low leakage currents are maintained by using circuit topologies which offer lower leakage currents and components having higher Insulation Resistance (IR). Electric Propulsion Systems inject harmonics into the Ship Power System which in turn effects the performance and reliability of Radar Power System. In this paper details of the voltage harmonics study carried out on Radar Power System are presented.

Technical Track TR13

[#171] Elastic Synergy: A Hybrid Price–Incentive Framework for Peak-Shaving and Utility Profit Optimization in Residential Demand Response

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This paper presents a hybrid demand response (DR) framework that integrates price-based and incentive-based mechanisms to manage residential electricity loads.The approach utilizes dynamic pricing via a price-elasticity matrix and timevarying monetary incentives via incentive elasticities to balance peak demand reduction and utility profit maximization simultaneously. Three DR schemes are formulated—an Incentive based model, a Real time price based model , and a Hybrid model—each governed by distinct demand reduction

rules. The hybrid model combines both elasticity types, computing demand reduction as a weighted sum of price- and incentive-driven responses. Numerical simulations involving 10 residential users over a 24-hour period, using realistic consumption profiles and calibrated parameters, demonstrate that the hybrid strategy consistently outperforms single-mode approaches. Notably, the hybrid model achieves deeper peak shaving, smoother load redistribution, and more than 200\{}% higher utility profit compared to the Incentive driven DR scheme and 90\{}% higher utility profit compared to the Price Driven DR scheme. The results indicate that jointly leveraging price signals and targeted incentives provides a more effective and economically beneficial DR solution.

[#197] Least-cost generation expansion modeling for Karnataka for FY 2031–32

Shristy Srivastava (CSTEP); Ammu Jacob (CSTEP)*

Karnataka's rapidly growing economy has led to a surge in energy demand, making it one of the top ten energy-consuming states in India. Therefore, it becomes essential to conduct a least-cost generation expansion planning study for the state's power sector. This paper provides a simulation-based comprehensive least-cost planning methodology for Karnataka, which includes capacity expansion modeling to identify the optimal capacity additions required and production cost modeling to analyze the optimal dispatch the new plant additions planned by the financial year (FY) 2031–32. The least-cost planning aligns Karnataka's energy requirements by FY 2031–32 and decarbonization goals, thereby serving the policy mandates. In this study, FY 2023–24 and FY 2031–32 were considered the base and target years, respectively. Constraints such as the renewable purchase obligation (RPO) target, reserves, spinning reserves, reliability requirements, and resource availability for Karnataka were applied. Results obtained from the simulation indicate that a versatile mix of non-fossil fuel energy sources can meet the FY 2031–32 energy demand for Karnataka. Moreover, although some investment is required for the new plant additions, a significant decrease in the state's average power purchase cost for FY 2031–32 is observed. This study would be useful in providing actionable insights to policymakers, utilities, energy planners, and stakeholders, enabling the planning for a resilient and economically efficient power sector for Karnataka by FY 2031–32.

[#257] Optimization and Techno-economic Assessment of Hydrogen-Electrolyzer-Fuel Cell Hybrid Grid Connected EV Charging Stations Different Dynamic Loads and Control Techniques

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The planet-wide shift to clean and sustainable energy systems has now become increasingly urgent to meet the immediate environmental challenges. This work offers a thorough optimization and techno-economic analysis of an EV HybridCharging Station powered by a combination of hydrogen-basedelectrolyzers and fuel cells and renewable energy sources fordynamic load conditions. Optimization is done in HOMER Pro major techno-economic criteria such as LCOE, NPC, andrenewable fraction. Several configurations, including solar photovoltaic (PV) panels, solar hydrogen storage, and fuel cell technologies, were analysed in order to assess the system performance in the presence of different EV charging and auxiliary load profiles. The techno-economic analysis demonstrates a reduction in NPC from \{}$25.5M to \{}$25.0M and a decrease in LCOE from \{}$0.120/kWh to \{}$0.110/kWh, offering substantial cost savings. The optimized system was also simulated in DIgSILENT PowerFactory to investigate its dynamic behavior and the potential benefits of advanced control methodologies such as Voltage-IQ Droop Control, Load Shedding Logic and Fuel Cell Response Coordination.The findings reveal that the incorporation of hydrogen systemscan improve energy resilience and alleviate grid pressures aswell as the operational emissions. Furthermore, advanced control methods resulted in system stability enhancement under peak load and fault situations.

[#267] Optimal Placement of Electric Vehicle Charging Station Using Machine Learning and Optimization Techniques

GANESH MATTAPARTHI (NIT TIRUCHIRAPPALLI)*

The rapid adoption of electric vehicles necessitates the strategic deployment of Electric Vehicle Charging Stations (EVCS) to ensure accessibility, efficiency, and user convenience.

Given the complexity of siting decisions involving spatial, infrastructural, and demand-side variables, machine learning has emerged as a critical tool for deriving data-driven insights to support informed infrastructure planning. This paper presents a framework for optimal EVCS siting that integrates machine learning-based suitability prediction two distinct metaheuristic approaches: Particle Swarm Optimization and a GIS-enhanced Firefly Algorithm, to evaluate trade-offs between fitness optimization and spatial feasibility. Emphasis is placed on evaluating the balance between maximizing fitness outcomes and adhering to spatial feasibility constraints. The results reveal distinct trade-offs in site selection patterns, convergence behaviour, and deployment practicality. Spatial visualizations further aid in interpreting the effectiveness of each approach. The proposedapproach lays a foundation for adaptive, spatially integrated,and scalable planning models suited for an evolving smart cityecosystems.

[#302] Seamless Integration of Passive Islanding Detection and Low Voltage Ride-Through in Synchronverter-Based Grid-Forming Inverters

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Traditional islanding detection and low-voltage ride-through (LVRT) mechanisms are often implemented separately in inverter control. This separation can lead to conflicts during grid disturbances, especially in systems high-penetration inverter-based resources (IBRs). Grid-forming inverters (GFMIs), particularly synchronverter-based designs, must satisfy both anti-islanding and LVRT requirements under evolving grid codes. This paper proposes a seamless integration strategy that combines passive islanding detection LVRT functionality for synchronverter-based GFMIs. A coordinated control framework is developed, unifying a novel passive islanding detection function (IDF) a voltage-dependent LVRT response. The IDF uses virtual flux, inverter current, and virtual speed deviation as key indicators. A time-domain logic discerns between unintentional islanding and grid faults in 30 ms. This ensures the timely activation of either disconnection or LVRT protocols. Simulation results on an IEEE 1547-based test system validate the approach. The method maintains voltage support during fault-induced sags and detects islanding reliably in 80 ms. It significantly reduces the non-detection zone (NDZ) out compromising system stability. The proposed strategy provides an efficient and practical solution for next-generation synchronverter-controlled GFM Inverters.

[#321] Integration of Battery Energy Storage System into Secondary Reserve Anciliary Services: Insights from Delhi's Kilokri BESS Testing

Adarsh Nagarajan (BSES Rajdhani Power Limited)*; Avinash Kumar (BSES Rajdhani Power Limited); Murthi Thandavarayan (BSES Rajdhani Power Limited); Abhishek Ranjan (BSES Rajdhani Power Limited)

Battery Energy Storage Systems (BESS) represent a transformative technology for enhancing grid stability through rapid frequency response ancillary services. This paper presents a comprehensive summary of India's first Automatic Generation Control (AGC)-integrated BESS deployment and testing conducted at Kilokari Substation, New Delhi. Aimed at addressing the challenges posed by the increasing shares of intermittent renewable energy sources, the Kilokari project evaluates the operational performance of a 20 MW / 40 MWh system providing Secondary Reserve Ancillary Services (SRAS). Detailed field measurements highlight BESS capabilities, such as rapid response times, robustness to communication disturbances, and alignment global ancillary service performance benchmarks. However, the analysis also identifies significant regulatory barriers in the existing Central Electricity Regulatory Commission (CERC) Ancillary Services Regulations, particularly concerning compensation frameworks that disadvantage BESS operators during SRAS-down events. Drawing insights from established international regulatory practices in the United States, this paper recommends targeted regulatory amendments to facilitate effective and economically viable integration of BESS into India's ancillary service market.

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